

Continuity and Differentiability

Question1

The number of points in the interval $[2, 4]$, at which the function $f(x) = [x^2 - x - \frac{1}{2}]$, where $[\cdot]$ denotes the greatest integer function, is discontinuous, is _____.

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Answer: 10

Solution:

For the greatest integer function

$$f(x) = [x^2 - x - \frac{1}{2}],$$

the function is discontinuous at those points where

$$x^2 - x - \frac{1}{2}$$

takes an integer value.

So, let

$$x^2 - x - \frac{1}{2} = n,$$

where n is an integer.

We need to find how many such points lie in the interval $[2, 4]$.

First, consider

$$g(x) = x^2 - x - \frac{1}{2}.$$

Now check its values at the endpoints:

$$\text{At } x = 2,$$

$$g(2) = 4 - 2 - \frac{1}{2} = \frac{3}{2}.$$

$$\text{At } x = 4,$$

$$g(4) = 16 - 4 - \frac{1}{2} = \frac{23}{2} = 11.5.$$

Also,

$$g'(x) = 2x - 1.$$

For $x \in [2, 4]$, we have $2x - 1 > 0$, so $g(x)$ is strictly increasing on $[2, 4]$.

Hence, as x goes from 2 to 4, $g(x)$ takes every value from $\frac{3}{2}$ to $\frac{23}{2}$ exactly once.

Now count the integers in the interval

$$[\frac{3}{2}, \frac{23}{2}] = [1.5, 11.5].$$

These integers are

$$2, 3, 4, 5, 6, 7, 8, 9, 10, 11.$$

There are 10 integers.

Since each integer value corresponds to exactly one $x \in [2, 4]$, the function $f(x)$ is discontinuous at exactly 10 points.

Therefore, the required number of points is

$$\boxed{10}.$$

Question2

The number of points, at which the function

$f(x) = \max \{6x, 2 + 3x^2\} + |x - 1| \cos \left| x^2 - \frac{1}{4} \right|, x \in (-\pi, \pi)$, is not differentiable, is

_____ .

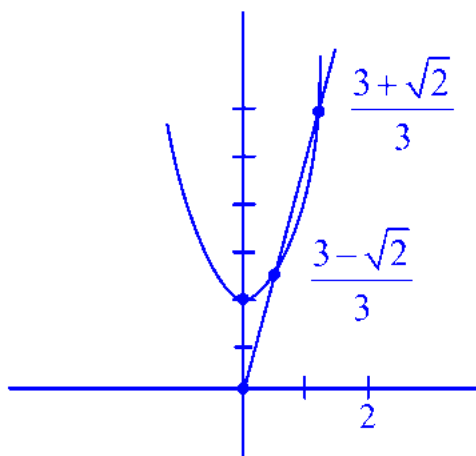
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Answer: 3

Solution:

For the part $\max \{6x, 2 + 3x^2\}$, the function can change its “active” expression only at those points where the two inside expressions become equal.

$$2 + 3x^2 = 6x \Rightarrow 3x^2 - 6x + 2 = 0$$



$$x = \frac{6 \pm \sqrt{12}}{6} = \frac{3 \pm \sqrt{2}}{3}$$

At these two values of x , the max function switches between $6x$ and $2 + 3x^2$. Also, the derivatives of the two expressions are different there (since $\frac{d}{dx}(6x) = 6$ and $\frac{d}{dx}(2 + 3x^2) = 6x$), so $\max \{6x, 2 + 3x^2\}$ is not differentiable at both points.

Now look at the second part $|x - 1| \cos \left| x^2 - \frac{1}{4} \right|$. The factor $\cos \left| x^2 - \frac{1}{4} \right|$ is continuous for all x , and at $x = 1$ it becomes $\cos \left| 1 - \frac{1}{4} \right| = \cos \left(\frac{3}{4} \right) \neq 0$.

Because $\cos \left| x^2 - \frac{1}{4} \right|$ is nonzero at $x = 1$, multiplying it by $|x - 1|$ does not remove the “corner” of $|x - 1|$. So $|x - 1| \cos \left| x^2 - \frac{1}{4} \right|$ is not differentiable at $x = 1$.

Total number of points where $f(x)$ is nonDifferentiable is 3

Question3

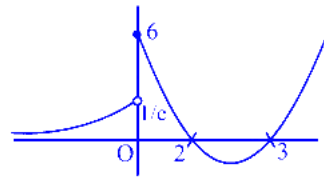
Let $f(x) = \begin{cases} e^{x-1} & , x < 0 \\ x^2 - 5x + 6 & , x \geq 0 \end{cases}$ and $g(x) = f(|x|) + |f(x)|$. If the number of points where g is not continuous and is not differentiable are α and β respectively, then $\alpha + \beta$ is equal to _____

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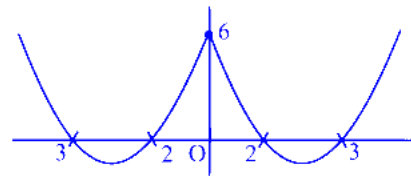
Answer: 4

Solution:

Graph of $f(x)$

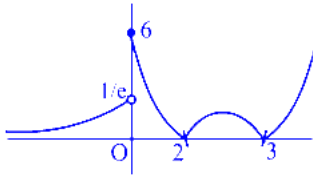


Now graph of $f(|x|)$



$f(|x|)$ is continuous function and it is non diff. at $x = 0$

Graph of $|f(x)|$



First look at $|f(x)|$.

For $x < 0$, we have $f(x) = e^{x-1}$, which is always positive, so $|f(x)| = e^{x-1}$. This part is continuous and differentiable for all $x < 0$.

For $x \geq 0$, $f(x) = x^2 - 5x + 6 = (x - 2)(x - 3)$. This becomes 0 at $x = 2$ and $x = 3$, and it changes sign between 2 and 3. So when we take absolute value, $|f(x)|$ gets a "sharp corner" at $x = 2$ and $x = 3$, hence it is not differentiable at $x = 2, 3$.

At $x = 0$, check continuity of $|f(x)|$: left value tends to $|e^{-1}| = e^{-1}$, but right value is $|f(0)| = |6| = 6$. Since these are not equal, $|f(x)|$ is discontinuous at $x = 0$ (and therefore also not differentiable at $x = 0$).

Now consider $g(x) = f(|x|) + |f(x)|$.

Here $f(|x|)$ always uses the $x \geq 0$ definition because $|x| \geq 0$. So $f(|x|)$ is continuous for all x , but it is not differentiable at $x = 0$ because of $|x|$.

So, discontinuity in g can only come from $|f(x)|$, which happens at $x = 0$ only. Also, g will be non-differentiable where either term is non-differentiable, i.e., at $x = 0, 2, 3$.

$|f(x)|$ is discontinuous at $x = 0$

$|f(x)|$ is non-diff. at $x = 0, 2, 3$

$g(x) = f(|x|) + |f(x)|$

$g(x)$ will be discontinuous at $x = 0$

$g(x)$ will be non diff at $x = 0, 2, 3$

$\alpha = 1, \beta = 3$

$\alpha + \beta = 1 + 3 = 4$

Question4

Let $f(x) = \begin{cases} x^3 + 8; & x < 0, \\ x^2 - 4; & x \geq 0, \end{cases}$ and $g(x) = \begin{cases} (x - 8)^{1/3}; & x < 0, \\ (x + 4)^{1/2}; & x \geq 0. \end{cases}$

Then the number of points, where the function $g \circ f$ is discontinuous, is ____ .

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Answer: 3

Solution:

We need to find the points where the composite function $g \circ f$ is discontinuous.

That means we first find

$$(g \circ f)(x) = g(f(x)).$$

So we must see, for each value of x , which branch of g will be used.

1. Given functions

$$f(x) = \begin{cases} x^3 + 8, & x < 0, \\ x^2 - 4, & x \geq 0 \end{cases}$$

and

$$g(x) = \begin{cases} (x-8)^{1/3}, & x < 0, \\ (x+4)^{1/2}, & x \geq 0. \end{cases}$$

Now,

$$(g \circ f)(x) = g(f(x)).$$

The definition of g depends on whether its input is negative or non-negative. So we must check the sign of $f(x)$.

2. Find where $f(x) < 0$ and where $f(x) \geq 0$

Case 1: $x < 0$

Then

$$f(x) = x^3 + 8.$$

We check when this is negative:

$$x^3 + 8 < 0 \implies x^3 < -8 \implies x < -2.$$

So for $x < 0$:

- if $x < -2$, then $f(x) < 0$,
- if $-2 \leq x < 0$, then $f(x) \geq 0$.

Case 2: $x \geq 0$

Then

$$f(x) = x^2 - 4.$$

We check when this is negative:

$$x^2 - 4 < 0 \implies x^2 < 4 \implies -2 < x < 2$$

But here $x \geq 0$, so this becomes

$$0 \leq x < 2.$$

Thus for $x \geq 0$:

- if $0 \leq x < 2$, then $f(x) < 0$,
- if $x \geq 2$, then $f(x) \geq 0$.

3. Write $(g \circ f)(x)$ piecewise

Now use the correct branch of g .

(i) When $x < -2$

Here $f(x) < 0$, so use

$$g(y) = (y-8)^{1/3}.$$

Thus

$$(g \circ f)(x) = ((x^3 + 8) - 8)^{1/3} = (x^3)^{1/3} = x.$$

So for $x < -2$,

$$(g \circ f)(x) = x.$$

(ii) When $-2 \leq x < 0$

Here $f(x) \geq 0$, so use

$$g(y) = \sqrt{y+4}.$$

Thus

$$(g \circ f)(x) = \sqrt{(x^2 - 4) + 4} = \sqrt{x^2} = |x|.$$

So for $-2 \leq x < 0$,

$$(g \circ f)(x) = \sqrt{x^3 + 12}.$$

(iii) When $0 \leq x < 2$

Here $f(x) < 0$, so use

$$g(y) = (y - 8)^{1/3}.$$

Thus

$$(g \circ f)(x) = ((x^2 - 4) - 8)^{1/3} = (x^2 - 12)^{1/3}.$$

So for $0 \leq x < 2$,

$$(g \circ f)(x) = (x^2 - 12)^{1/3}.$$

(iv) When $x \geq 2$

Here $f(x) \geq 0$, so use

$$g(y) = \sqrt{y + 4}.$$

Thus

$$(g \circ f)(x) = \sqrt{(x^2 - 4) + 4} = \sqrt{x^2} = x$$

since $x \geq 2$.

So for $x \geq 2$,

$$(g \circ f)(x) = x.$$

4. Check continuity at possible boundary points

Each expression is continuous in its own interval. So we only check the joining points:

$$x = -2, \quad x = 0, \quad x = 2.$$

At $x = -2$

Left-hand limit:

for $x < -2$, $(g \circ f)(x) = x$, so

$$\lim_{x \rightarrow -2^-} (g \circ f)(x) = -2.$$

Right-hand limit:

for $-2 \leq x < 0$, $(g \circ f)(x) = \sqrt{x^3 + 12}$, so

$$\lim_{x \rightarrow -2^+} (g \circ f)(x) = \sqrt{(-2)^3 + 12} = \sqrt{4} = 2.$$

Since

$$-2 \neq 2,$$

the function is discontinuous at $x = -2$.

At $x = 0$

Left-hand limit:

$$\lim_{x \rightarrow 0^-} (g \circ f)(x) = \sqrt{0^3 + 12} = \sqrt{12}.$$

Right-hand limit:

$$\lim_{x \rightarrow 0^+} (g \circ f)(x) = (0^2 - 12)^{1/3} = (-12)^{1/3}.$$

These are not equal, so discontinuous at $x = 0$.

At $x = 2$

Left-hand limit:

$$\lim_{x \rightarrow 2^-} (g \circ f)(x) = (2^2 - 12)^{1/3} = (-8)^{1/3} = -2.$$

Right-hand limit:

$$\lim_{x \rightarrow 2^+} (g \circ f)(x) = 2.$$

These are not equal, so discontinuous at $x = 2$.

5. Conclusion

The function $g \circ f$ is discontinuous at

$$x = -2, \quad x = 0, \quad x = 2.$$

Hence, the number of points of discontinuity is

3.

Question5

Let $f : \mathbf{R} \rightarrow (0, \infty)$ be a twice differentiable function such that $f(3) = 18$, $f'(3) = 0$ and $f''(3) = 4$.

Then $\lim_{x \rightarrow 1} \left(\log_e \left(\frac{f(2+x)}{f(3)} \right)^{\frac{18}{(x-1)^2}} \right)$ is equal to :

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Options:

A.

9

B.

18

C.

1

D.

2

Answer: D

Solution:

$$\ln \left(\lim_{x \rightarrow 1} \left(\frac{f(x+2)}{f(3)} \right)^{\frac{18}{(x-1)^2}} \right) (1^\infty)$$

$$\ln \left(e^{\lim_{x \rightarrow 1} \frac{18}{(x-1)^2} \left(\frac{f(x+2)}{f(3)} - 1 \right)} \right)$$

$$\ln \left(e^{\lim_{x \rightarrow 1} \frac{f'(x+2)}{2(X-1)}} \right)$$

$$\ln \left(e^{\lim_{x \rightarrow 1} \frac{f''(x+2)}{2}} \right)$$

$$\lim_{x \rightarrow 1} \frac{f''(3)}{2} = \frac{4}{2} = 2$$

Question6

Let $[\cdot]$ denote the greatest integer function, and let $f(x) = \min \left\{ \sqrt{2}x, x^2 \right\}$.

Let $S = \{x \in (-2, 2) : \text{the function } g(x) = |x| [x^2] \text{ is discontinuous at } x\}$.

Then $\sum_{x \in S} f(x)$ equals

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Options:

A.

$$2 - \sqrt{2}$$

B.

$$2\sqrt{6} - 3\sqrt{2}$$

C.

$$1 - \sqrt{2}$$

D.

$$\sqrt{6} - 2\sqrt{2}$$

Answer: C

Solution:

$$g(x) = |x| \lfloor x^2 \rfloor$$

Points of discontinuity of $g(x)$ in $(-2, 2)$ are $(\pm 1, \pm\sqrt{2}, \pm\sqrt{3})$

$$\therefore S = \{-1, 1, -\sqrt{2}, \sqrt{2}, -\sqrt{3}, \sqrt{3}\}$$

$$\therefore f(x) = \min \{ \sqrt{2x}, x^2 \}$$

$$\therefore \sum_{x \in S} f(x) = -\sqrt{2} + 1 - 2 + 2 - \sqrt{6} + \sqrt{6} = 1 - \sqrt{2}$$

Question 7

If $\lim_{x \rightarrow 0} \frac{e^{(a-1)x} + 2 \cos bx + (c-2)e^{-x}}{x \cos x - \log_e(1+x)} = 2$, then $a^2 + b^2 + c^2$ is equal to :

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Options:

A.

$$3$$

B.

$$5$$

C.

$$9$$

D.

$$7$$

Answer: D

Solution:

$$\lim_{x \rightarrow 0} \left[1 + \frac{(a-1)}{1!}x + \frac{(a-1)^2}{2!}x^2 + \dots \infty \right] + 2 \left[1 - \frac{(bx)^2}{2!} + \dots \infty \right] + (c-2) \left[1 - \frac{x^1}{1!} + \frac{x^2}{2!} - \dots \infty \right]$$

$$\frac{x \left(1 - \frac{x^2}{2!} + \dots \infty \right) - \left(x - \frac{x^2}{2} + \dots \infty \right)}{(c+1) + (a-c+1)x + \left[\frac{(a-1)^2}{2} - b^2 + \frac{(c-2)}{2} \right] x^2 + \dots \infty} = 2$$

$$c+1=0 \Rightarrow c=-1$$

$$a-c+1=0 \Rightarrow a+1+1=0 \Rightarrow a=-2$$

$$\left(\frac{(a-1)^2}{2} - b^2 + \frac{(c-2)}{2} \right)$$

$$\left(\frac{1}{2} \right)$$

$$\Rightarrow 9 - 2b^2 - 3 = 2 \Rightarrow 2b^2 = 4 \Rightarrow b^2 = 2$$

$$\therefore a^2 + b^2 + c^2 = 4 + 2 + 1 = 7$$

Question8

Let $f(x) = \begin{cases} \frac{ax^2+2ax+3}{4x^2+4x-3} & , x \neq -\frac{3}{2}, \frac{1}{2} \\ b & , x = -\frac{3}{2}, \frac{1}{2} \end{cases}$ be continuous at $x = -\frac{3}{2}$. If $f \circ f(x) = \frac{7}{5}$, then x is equal to:

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Options:

A.

1.4

B.

2

C.

1

D.

0

Answer: C

Solution:

$$\lim_{x \rightarrow -\frac{3}{2}} \frac{ax^2+2ax+3}{4x^2+4x-3} = b$$

$$ax^2+2ax+3=0 \quad \text{at } x = -\frac{3}{2}$$

$$\Rightarrow \frac{9a}{4} - 3a + 3 = 0$$

$$\Rightarrow \frac{3a}{4} = 3$$

$$\Rightarrow a = 4$$

$$\lim_{x \rightarrow -\frac{3}{2}} \frac{4x^2+8ax+3}{4x^2+4x-3}$$

$$\begin{aligned}
&= \lim_{x \rightarrow \frac{-3}{2}} \frac{(2x+1)(2x+3)}{(2x-1)(2x+3)} \\
&= \frac{-2}{-4} = \frac{1}{2} = b \\
\therefore f(x) &= \frac{2x+1}{2x-1} \\
f(f(x)) &= \frac{7}{5} \\
\therefore \frac{2f(x)+1}{2f(x)-1} &= \frac{7}{5} \\
\Rightarrow f(x) &= 3 \\
\frac{2x+1}{2x-1} &= 3 \\
\Rightarrow x &= 1
\end{aligned}$$

Question9

If $f(x) = \begin{cases} \frac{a|x|+x^2-2(\sin|x|)(\cos|x|)}{x} & , x \neq 0 \\ b & , x = 0 \end{cases}$

is continuous at $x = 0$, then $a + b$ is equal to :

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Options:

A.

1

B.

2

C.

0

D.

4

Answer: B

Solution:

Refunctioning the given function.

$$f(x) = \begin{cases} \frac{ax+x^2-\sin 2x}{x} & , x > 0 \\ \frac{-ax+x^2+\sin 2x}{x} & , x < 0 \\ b & , x = 0 \end{cases}$$

$$RHL = \lim_{x \rightarrow 0^+} \left(a + x - 2 \frac{\sin 2x}{2x} \right) = a - 2$$

$$LHL = \lim_{x \rightarrow 0^-} \left(-a + x + 2 \frac{\sin 2x}{2x} \right) = -a + 2$$

For continuity:

$$LHL = RHL \Rightarrow a - 2 = -a + 2 \Rightarrow a = 2$$

$$b = f(0) = RHL = 2 - 2 = 0$$

$$a + b = 2 + 0 = 2$$

Question10

If the function $f(x) = \frac{e^x(e^{\tan x - x} - 1) + \log_e(\sec x + \tan x) - x}{\tan x - x}$ is continuous at $x = 0$, then the value of $f(0)$ is equal to

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Options:

A.

$$\frac{2}{3}$$

B.

$$\frac{1}{2}$$

C.

$$2$$

D.

$$\frac{3}{2}$$

Answer: D

Solution:

Given that $f(x)$ is continuous at $x = 0$ means $\lim_{x \rightarrow 0} f(x) = f(0)$

$$f(0) = \lim_{x \rightarrow 0} \frac{e^x(e^{\tan x - x} - 1) + \log_e(\sec x + \tan x) - x}{\tan x - x}$$

$$f(0) = \lim_{x \rightarrow 0} \frac{e^x(e^{\tan x - x} - 1)}{\tan x - x} + \lim_{x \rightarrow 0} \frac{\log_e(\sec x + \tan x) - x}{\tan x - x}$$

Using standard limit $\frac{e^{f(x)} - 1}{f(x)} = 1$ if $f(x) \rightarrow 0$

Let $f(x) = \tan x - x$, then $\lim_{x \rightarrow 0} \frac{e^{\tan x - x} - 1}{\tan x - x} = 1$

Second limit is $\frac{0}{0}$ form use L'Hopital Rule

$$\begin{aligned} f(0) &= 1 + \lim_{x \rightarrow 0} \frac{\frac{d}{dx}(\log(\sec x + \tan x) - x)}{\frac{d}{dx}(\tan x - x)} \\ &= 1 + \lim_{x \rightarrow 0} \frac{\frac{1}{\sec x + \tan x}(\sec x \tan x + \sec^2 x) - 1}{\sec^2 x - 1} \\ &= 1 + \lim_{x \rightarrow 0} \frac{\sec x \tan x + \sec^2 x - \sec x - \tan x}{(\sec x + \tan x)(\sec^2 x - 1)} \\ &= 1 + \lim_{x \rightarrow 0} \frac{\tan x(\sec x - 1) + \sec x(\sec x - 1)}{(\sec x + \tan x)(\sec^2 x - 1)} \\ &= 1 + \lim_{x \rightarrow 0} \frac{(\sec x - 1)(\sec x + \tan x)}{(\sec x + \tan x)(\sec^2 x - 1)} \\ &= 1 + \lim_{x \rightarrow 0} \frac{1}{\sec x + 1} \\ \Rightarrow f(0) &= 1 + \frac{1}{2} = \frac{3}{2} \end{aligned}$$

Question 11

Let $\alpha, \beta \in \mathbb{R}$ be such that the function $f(x) = \begin{cases} 2\alpha(x^2 - 2) + 2\beta x & , x < 1 \\ (\alpha + 3)x + (\alpha - \beta) & , x \geq 1 \end{cases}$ be differentiable at all $x \in \mathbb{R}$. Then $34(\alpha + \beta)$ is equal to

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Options:

A.

48

B.

84

C.

36

D.

24

Answer: A

Solution:

$$f(x) = \begin{cases} 2\alpha(x^2 - 2) + 2\beta x, & x < 1 \\ (\alpha + 3)x + (\alpha - \beta), & x \geq 1 \end{cases}$$

Given $f(x)$ is differentiable at all $x \in \mathbb{R}$.

This means $f(x)$ is continuous and differentiable at $x = 1$.

1. Continuity at $x = 1$

$$\text{L.H.L} = \text{R.H.L} = f(1)$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x)$$

$$\lim_{x \rightarrow 1^-} [2\alpha(x^2 - 2) + 2\beta x] = \lim_{x \rightarrow 1^+} [(\alpha + 3)x + (\alpha - \beta)]$$

$$2\alpha(1 - 2) + 2\beta(1) = \alpha + 3 + \alpha - \beta$$

$$-2\alpha + 2\beta = 2\alpha - \beta + 3$$

$$3\beta - 4\alpha = 3 - (1)$$

2. Differentiability at $x = 1$

$$f'(x) = \begin{cases} 2\alpha(2x) + 2\beta, & x < 1 \\ \alpha + 3, & x > 1 \end{cases}$$

$$\text{L.H.D} = \text{R.H.D}$$

$$f'(1^-) = f'(1^+)$$

$$2\alpha(2 \cdot 1) + 2\beta = \alpha + 3$$

$$3\alpha + 2\beta = 3 - (2)$$

Solving equation (1) and (2)

$$\text{eqn (1)} \times 2 - \text{eqn (2)} \times 3$$

$$-8\alpha - 9\alpha = -3 \implies \alpha = \frac{3}{17}$$

Substitute $\alpha = \frac{3}{17}$ in equation (1):

$$3\beta - 4 \times \frac{3}{17} = 3$$

$$3\beta = 3 + \frac{12}{17} = \frac{63}{17}$$

$$\beta = \frac{21}{17}$$

$$\text{Value of } 34(\alpha + \beta) = 34\left(\frac{3}{17} + \frac{21}{17}\right) = 6 + 42 = 48$$

Question 12

Let $[t]$ denote the greatest integer less than or equal to t . If the function

$$f(x) = \begin{cases} b^2 \sin\left(\frac{\pi}{2} \left[\frac{\pi}{2}(\cos x + \sin x) \cos x\right]\right), & x < 0 \\ \frac{\sin x - \frac{1}{2} \sin 2x}{x^3}, & x > 0 \\ a, & x = 0 \end{cases}$$

is continuous at $x = 0$, then $a^2 + b^2$ is equal to :

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Options:

A.

$$\frac{1}{2}$$

B.

$$\frac{5}{8}$$

C.

$$\frac{3}{4}$$

D.

$$\frac{9}{16}$$

Answer: C

Solution:

It is given that $f(x)$ is continuous at $x = 0$

This mean at $x = 0$

$$\text{L.H.L} = \text{R.H.L} = f(0) = a$$

Calculating L.H.L using the definition of $f(x)$ for $x < 0$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} b^2 \sin \left(\frac{\pi}{2} \left[\frac{\pi}{2} (\cos x + \sin x) \cos x \right] \right) = a$$

$$\Rightarrow \lim_{x \rightarrow 0^-} b^2 \sin \left(\frac{\pi}{2} \left[\frac{\pi}{2} (\cos^2 x + \frac{\sin 2x}{2}) \right] \right) = a$$

$$\cos^2(0^-) = 1^- \text{Less than } 1$$

$$\sin 2(0^-) = 0^- \text{Less than } 0$$

$$\Rightarrow a = b^2 \sin \left(\frac{\pi}{2} \left[\frac{3.14}{2} (1^- + 0^-) \right] \right)$$

$$\Rightarrow a = b^2 \sin \left(\frac{\pi}{2} [1.57(1^-)] \right)$$

$$\Rightarrow a = b^2 \sin \left(\frac{\pi}{2} [1.57^-] \right), [1.57^-] = 1\{[t] \rightarrow \text{G.I.F}\}$$

$$\Rightarrow a = b^2 \sin \left(\frac{\pi}{2} \times 1 \right) = b^2$$

Calculating R.H.L using the function definition for $x > 0$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{\sin x - \frac{1}{2} \sin 2x}{x^3} = a$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \frac{\sin x - \frac{1}{2} \times 2 \sin x \cos x}{x^3} = a$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \frac{\sin x (1 - \cos x)}{x \cdot x^2} = a$$

$$\text{Using limits result } \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \text{ and } \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

Putting these result

$$\text{Using limits result } \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \text{ and } \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

Putting these result

$$a = 1 \cdot \frac{1}{2} = \frac{1}{2}$$

$$\text{So } a^2 + b^2 = a^2 + a \{b^2 = a \text{ from } L. H. L\}$$

$$= \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

Question 13

Let $y = y(x)$ be a differentiable function in the interval $(0, \infty)$ such that $y(1) = 2$, and $\lim_{t \rightarrow x} \left(\frac{t^2 y(x) - x^2 y(t)}{x - t} \right) = 3$ for each $x > 0$. Then $2y(2)$ is equal to :

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Options:

A.

27

B.

18

C.

23

D.

12

Answer: C

Solution:

$$\lim_{t \rightarrow x} \frac{t^2 y(x) - x^2 y(t)}{x - t} = 3$$

Substituting

$$t \rightarrow x + h \Rightarrow h \rightarrow 0$$

$$\lim_{h \rightarrow 0} \frac{(x+h)^2 y(x) - x^2 y(x+h)}{-h} = 3$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{x^2 y(x) + h^2 y(x) + 2xhy(x) - x^2 y(x+h)}{-h} = 3$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{-x^2[y(x+h) - y(x)] + y(x)(h^2 + 2xh)}{-h} = 3$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{x^2[y(x+h) - y(x)]}{h} - \lim_{h \rightarrow 0} \frac{y(x)(h+2x)h}{h} = 3$$

Use

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$x^2 y'(x) - 2xy(x) = 3$$

Equation (1) is differential equation can be written by replacing

$$y'(x) = \frac{dy}{dx} \quad \text{and} \quad y(x) = y$$

$$\Rightarrow x^2 \frac{dy}{dx} - 2xy = 3$$

$$\Rightarrow \frac{dy}{dx} - \frac{2}{x}y = \frac{3}{x^2} \text{ (Linear differential equation)}$$

Integrating factor

$$\text{I.F.} = e^{\int -\frac{2}{x} dx} = e^{-2 \ln x} = e^{\ln\left(\frac{1}{x^2}\right)} = \frac{1}{x^2}$$

$\therefore$ Solution of linear differential equation

$$y \cdot (\text{I.F.}) = \int (\text{I.F.}) \cdot \frac{3}{x^2} dx$$

$$\Rightarrow y \cdot \frac{1}{x^2} = \int \frac{1}{x^2} \cdot \frac{3}{x^2} dx = \int \frac{3}{x^4} dx$$

$$\Rightarrow \frac{y}{x^2} = 3 \times \frac{x^{-4+1}}{-4+1} = -\frac{1}{x^3} + C$$

$$\Rightarrow y = -\frac{1}{x} + Cx^2 \text{ (General solution)}$$

Apply initial condition

$$y(1) = 2$$

$$\Rightarrow 2 = -\frac{1}{1} + C(1) \Rightarrow C = 3$$

$$\text{So particular solution is : } y(x) = -\frac{1}{x} + 3x^2$$

$$\therefore y(2) = -\frac{1}{2} + 3(2)^2 = 12 - \frac{1}{2} = \frac{23}{2}$$

$$\Rightarrow 2y(2) = 23$$

Question14

The value of

$$\lim_{x \rightarrow 0} \frac{\log_e (\sec(ex) \cdot \sec(e^2x) \cdot \dots \cdot \sec(e^{10}x))}{e^2 - e^{2 \cos x}}$$

is equal to

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Options:

A.

$$\frac{(e^{10}-1)}{2e^9(e^2-1)}$$

B.

$$\frac{(e^{20}-1)}{2e^2(e^2-1)}$$

C.

$$\frac{(e^{10}-1)}{2(e^2-1)}$$

D.

$$\frac{(e^{20}-1)}{2(e^2-1)}$$

Answer: D

Solution:

$$\text{Let } L = \lim_{x \rightarrow 0} \frac{\ln(\sec(ex) \cdot \sec(e^2x) \cdot \dots \cdot \sec(e^{10}x))}{e^2 - e^{2 \cos x}}$$

Simplify : we use $\ln(abc \dots) = \ln(a) + \ln(b) + \ln(c) \dots$ and take e^2 common in denominator

$$L = \lim_{x \rightarrow 0} \frac{\ln \sec(ex) + \ln \sec(e^2x) + \dots + \ln \sec(e^{10}x)}{e^2[1 - e^{2 \cos x - 2}]} \dots\dots\dots(1)$$

Now understanding dominating term

- If $A \rightarrow 1 \Rightarrow \ln A \approx A - 1$
- as $x \rightarrow 0 \Rightarrow \sec(ex) \rightarrow 1 \Rightarrow \ln \sec(ex) \approx \sec(ex) - 1$
- so generalized $x \rightarrow 0 \Rightarrow \ln \sec(e^kx) \approx \sec(e^kx) - 1$
- If $A \rightarrow 0 \Rightarrow e^A - 1 \approx A$
- so in denominator as $x \rightarrow 0 \Rightarrow 2(\cos x - 1) = 0$
- so, $e^{2(\cos x - 1)} - 1 \approx 2(\cos x - 1)$

Substituting this dominating term in equation (1)

$$L = \lim_{x \rightarrow 0} \frac{(\sec(ex) - 1) + (\sec(e^2x) - 1) + \dots + (\sec(e^{10}x) - 1)}{-e^2 \cdot 2(\cos x - 1)}$$

$$= \lim_{x \rightarrow 0} \frac{\left(\frac{1}{\cos(ex)} - 1\right) + \left(\frac{1}{\cos(e^2x)} - 1\right) + \dots + \left(\frac{1}{\cos(e^{10}x)} - 1\right)}{2e^2(1 - \cos x)} = \lim_{x \rightarrow 0} \frac{\frac{1 - \cos(ex)}{\cos(ex)} + \frac{1 - \cos(e^2x)}{\cos(e^2x)} + \dots + \frac{1 - \cos(e^{10}x)}{\cos(e^{10}x)}}{2e^2(1 - \cos x)}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos(ex)}{\cos(ex) \cdot 2e^2(1 - \cos x)} + \lim_{x \rightarrow 0} \frac{1 - \cos(e^2x)}{\cos(e^2x) \cdot 2e^2(1 - \cos x)} + \dots + \frac{1 - \cos(e^{10}x)}{\cos(e^{10}x) \cdot 2e^2(1 - \cos x)} \dots \dots \dots (2)$$

Again using dominating term

$$\text{at } x \rightarrow 0, \cos(e^k x) = 1, k = \{1, 2, \dots, 10\}$$

$$L = \lim_{x \rightarrow 0} \frac{1 - \cos(ex)}{2e^2(1 - \cos x)} + \lim_{x \rightarrow 0} \frac{1 - \cos(e^2x)}{2e^2(1 - \cos x)} + \dots + \frac{1 - \cos(e^{10}x)}{2e^2(1 - \cos x)}$$

Understanding dominating term again

$$\text{If } A \rightarrow 0 \Rightarrow 1 - \cos A \approx \frac{A^2}{2} \text{ So } 1 - \cos(e^k x) \approx \frac{(e^k x)^2}{2}, k \in \{1, 2, \dots, 10\}$$

Substitute this value in equation (2) :

$$L = \frac{\lim_{x \rightarrow 0} \frac{(ex)^2}{2}}{2e^2 \cdot \frac{x^2}{2}} + \frac{\lim_{x \rightarrow 0} \frac{(e^2x)^2}{2}}{2e^2 \cdot \frac{x^2}{2}} + \dots + \frac{\lim_{x \rightarrow 0} \frac{(e^{10}x)^2}{2}}{2e^2 \cdot \frac{x^2}{2}}$$

x^2 and $\frac{1}{2}$ get cancelled so L becomes :

$$L = \frac{e^2}{2e^2} + \frac{(e^2)^2}{2e^2} + \dots + \frac{(e^{10})^2}{2e^2}$$

$$\text{take } \frac{1}{2e^2} \text{ common } L = \frac{1}{2e^2} [e^2 + e^4 + e^6 + \dots + e^{20}]$$

$e^2 + e^4 + \dots + e^{20}$ is a G.P. whose first term is e^2 common ratio is e^2 so sum of G.P. is $a \frac{(r^n - 1)}{r - 1}$ where a is first term and r is common ratio.

$$L = \frac{1}{2e^2} \times e^2 \left[\frac{(e^2)^{10} - 1}{e^2 - 1} \right] L = \frac{e^{20} - 1}{2(e^2 - 1)}$$

Question15

Let $f(x) = \lim_{\theta \rightarrow 0} \left(\frac{\cos \pi x - x^{\left(\frac{2}{\theta}\right)} \sin(x-1)}{1 + x^{\left(\frac{2}{\theta}\right)} (x-1)} \right)$, $x \in \mathbb{R}$. Consider the following two statements :

(I) $f(x)$ is discontinuous at $x = 1$.

(II) $f(x)$ is continuous at $x = -1$.

Then,

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Options:

A.

Neither (I) nor (II) is True

B.

Only (II) is True

C.

Only (I) is True

D.

Both (I) and (II) are True

Answer: A

Solution:

The function is defined as :

$$f(x) = \lim_{\theta \rightarrow 0} \left(\frac{\cos \pi x - x^{\left(\frac{2}{\theta}\right)} \sin(x-1)}{1 + x^{\left(\frac{2}{\theta}\right)} (x-1)} \right)$$

Let $n = \frac{2}{\theta}$. As $\theta \rightarrow 0, n \rightarrow \infty$ (specifically, let's consider θ approaching 0 such that the power grows). The behavior of the function depends on the value of x because of the term x^n .

- **Case 1 :** $|x| < 1$ As $n \rightarrow \infty, x^n \rightarrow 0$.

$$f(x) = \frac{\cos \pi x - 0}{1+0} = \cos \pi x$$

- **Case 2 :** $|x| > 1$ As $n \rightarrow \infty, x^n \rightarrow \infty$. We divide the numerator and denominator by x^n :

$$f(x) = \lim_{n \rightarrow \infty} \frac{\frac{\cos \pi x}{x^n} - \sin(x-1)}{\frac{1}{x^n} + (x-1)} = \frac{0 - \sin(x-1)}{0 + (x-1)} = \frac{-\sin(x-1)}{x-1}$$

- **Case 3 :** $x = 1$ If $x = 1$, then $\cos(\pi) = -1$ and $\sin(0) = 0$.

$$f(1) = \frac{-1 - (1^n \cdot 0)}{1 + (1^n \cdot 0)} = -1$$

- **Case 4 :** $x = -1$ If $x = -1$, then $\cos(-\pi) = -1$ and $\sin(-2) = -\sin(2)$. Since $n = 2/\theta$ is even, $(-1)^n = 1$.

$$f(-1) = \frac{-1 - (1 \cdot \sin(-2))}{1 + (1 \cdot -2)} = \frac{-1 + \sin 2}{-1} = 1 - \sin 2$$

So $f(x)$ become

$$f(x) = \begin{cases} \frac{-\sin(x-1)}{x-1} & \text{if } x \in (-\infty, -1) \cup (1, \infty) \\ \cos \pi x & \text{if } x \in (-1, 1) \\ 0 & \text{if } x = 1 \\ 1 - \sin 2 & \text{if } x = -1 \end{cases}$$

Analyze the Statements

Statement (I) : $f(x)$ is discontinuous at $x = 1$.

- Value at $x = 1$: $f(1) = -1$.
- Left-hand limit ($x \rightarrow 1^-$) : Uses the $|x| < 1$ definition: $\lim_{x \rightarrow 1^-} \cos \pi x = \cos \pi = -1$.
- Right-hand limit ($x \rightarrow 1^+$) : Uses the $|x| > 1$ definition: $\lim_{x \rightarrow 1^+} \frac{-\sin(x-1)}{x-1}$. Using the standard limit $\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$, this equals -1. Since $LHL = RHL = f(1)$, the function is continuous at $x = 1$.

Conclusion : Statement (I) is False.

Statement (II) : $f(x)$ is continuous at $x = -1$.

- Value at $x = -1$: $f(-1) = 1 - \sin 2$.
- Left-hand limit ($x \rightarrow -1^-$) : Uses $|x| > 1$ definition : $\frac{-\sin(-1-1)}{-1-1} = \frac{-\sin(-2)}{-2} = \frac{\sin 2}{-2}$.
- Right-hand limit ($x \rightarrow -1^+$) : Uses $|x| < 1$ definition : $\cos(-\pi) = -1$.

Since the limits do not match, the function is discontinuous at $x = -1$.

Conclusion : Statement (II) is False.

Question 16

If $\lim_{x \rightarrow 2} \frac{\sin(x^3 - 5x^2 + ax + b)}{(\sqrt{x-1} - 1) \log_e(x-1)} = m$, then $a + b + m$ is equal to :

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Options:

A.

5

B.

6

C.

8

D.

10

Answer: B

Solution:

$\therefore$ Denominator = 0 at $x = 2$

$\Rightarrow$ Numerator = 0 at $x = 2$

$$2^3 - 5(2)^2 + a(2) + b = 0$$

$$\Rightarrow 2a + b = 12 \quad \dots (1)$$

$$\therefore m = \lim_{x \rightarrow 2} \frac{\frac{\sin(x^3 - 5x^2 + ax + b)}{x^3 - 5x^2 + ax + b} \cdot (x^3 - 5x^2 + ax + b)}{\left(\frac{x-1-1}{\sqrt{x-1}+1}\right) \cdot \frac{\log_e(1+(x-2))}{x-2} \cdot (x-2)}$$

$$\Rightarrow m = \lim_{x \rightarrow 2} 2 \left(\frac{x^3 - 5x^2 + ax + b}{(x-2)^3} \right)$$

$$\Rightarrow m = \lim_{x \rightarrow 2} 2 \left(\frac{3x^2 - 10x + a}{2(x-2)} \right)$$

$\therefore$ Denominator = 0 at $x = 2$

$\Rightarrow$ Numerator = 0 at $x = 2$

$$\Rightarrow 3(2)^2 - 10(2) + a = 0 \rightarrow a = 8$$

put $a = 8$ in (1), we get $b = -4$

$$\therefore m = \lim_{x \rightarrow 2} 2 \left(\frac{6x-10}{2} \right) \rightarrow m = 2$$

$$\therefore a + b + m = 6$$

Question17

The product of all possible values of α , for which

$$\lim_{x \rightarrow 0} \left(\frac{1 - \cos(\alpha x) \cos((\alpha+1)x) \cos((\alpha+2)x)}{\sin^2((\alpha+1)x)} \right) = 2, \text{ is :}$$

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Options:

A.

-2

B.

1

C.

-1

D.

$\frac{5}{4}$

Answer: C

Solution:

$$\lim_{x \rightarrow 0} \frac{1 - \cos \alpha x \cdot \cos((\alpha+1)x) \cos((\alpha+2)x)}{(\alpha+1)^2 x^2}$$

By using L.H. Rule we get

$$\Rightarrow \frac{1}{2}(\alpha+1)^2 [\alpha^2 + (\alpha+1)^2 + (\alpha+2)^2] = 2$$

$$\alpha^2 + (\alpha+1)^2 + (\alpha+2)^2 = 4(\alpha+1)^2$$

$$\alpha^2 + (\alpha+2)^2 = 3(\alpha+1)^2$$

$$\Rightarrow \alpha^2 + 2\alpha - 1 = 0$$

$$\text{Product} = -1$$

Question 18

Let $f(x)$ and $g(x)$ be twice differentiable functions satisfying $f''(x) = g''(x)$ for all $x \in \mathbf{R}$, $f'(1) = 2g'(1) = 4$ and $g(2) = 3f(2) = 9$. Then $f(25) - g(25)$ is equal to :

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Options:

A.

20

B.

40

C.

-20

D.

-40

Answer: B

Solution:

Let us define a new function $h(x)$ taking the difference between $f(x)$ and $g(x)$.

$$h(x) = f(x) - g(x)$$

Differentiating this function twice with respect to x , we get:

$$h''(x) = f''(x) - g''(x)$$

We are given that $f''(x) = g''(x)$ for all $x \in \mathbf{R}$. Therefore, their difference is zero:

$$h''(x) = 0$$

Now, let us integrate both sides with respect to x to find $h'(x)$:

$$h'(x) = \int 0 \, dx = c_1$$

where c_1 is a constant of integration.

We are given the values of the first derivatives at $x = 1$. Let us simplify them:

$$f'(1) = 4$$

$$2g'(1) = 4 \implies g'(1) = 2$$

Using these values, we can find the exact value of $h'(1)$:

$$h'(1) = f'(1) - g'(1) = 4 - 2 = 2$$

Since $h'(x) = c_1$ for all values of x , it follows that:

$$c_1 = 2$$

Substituting this back, we get our first derivative equation:

$$h'(x) = 2$$

Next, we need the original function $h(x)$. We find it by integrating $h'(x)$ with respect to x :

$$h(x) = \int 2 \, dx = 2x + c_2$$

where c_2 is another constant of integration.

We are given the values of the functions at $x = 2$. Let us simplify these values as well:

$$g(2) = 9$$

$$3f(2) = 9 \implies f(2) = 3$$

Using these values, we can find the value of $h(2)$:

$$h(2) = f(2) - g(2) = 3 - 9 = -6$$

Now, substitute $x = 2$ and $h(2) = -6$ into our integrated equation for $h(x)$:

$$h(2) = 2(2) + c_2$$

$$-6 = 4 + c_2$$

$$c_2 = -6 - 4$$

$$c_2 = -10$$

Substituting the value of c_2 back into the equation for $h(x)$, we have the completed function rule:

$$h(x) = 2x - 10$$

The question asks for the value of $f(25) - g(25)$, which is represented directly by $h(25)$. Let us substitute $x = 25$ into our function $h(x)$:

$$h(25) = 2(25) - 10$$

$$h(25) = 50 - 10$$

$$h(25) = 40$$

Therefore, $f(25) - g(25) = 40$.

The correct answer is Option B.

Question19

Let $f(x) = \lim_{y \rightarrow 0} \frac{(1 - \cos(xy)) \tan(xy)}{y^3}$. Then the number of solutions of the equation $f(x) = \sin x$, $x \in \mathbb{R}$ is :

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Options:

A.

0

B.

2

C.

3

D.

1

Answer: C

Solution:

$$f(x) = \lim_{y \rightarrow 0} \frac{1 - \cos(xy)}{(xy)^2} \cdot \frac{\tan xy}{xy} \frac{(xy)^3}{y^3}$$

$$f(x) = \frac{x^3}{2}$$

$$f(x) = \sin x$$

$$\frac{x^3}{2} = \sin x$$

3 points of intersection

Question20

The value of $\lim_{x \rightarrow 0} \left(\frac{x^2 \sin^2 x}{x^2 - \sin^2 x} \right)$ is :

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Options:

A.

2

B.

3

C.

4

D.

6

Answer: B

Solution:

$$\lim_{x \rightarrow 0} \frac{x^2 \sin^2 x}{x^2 - \sin^2 x}$$

Applying expansion

$$\lim_{x \rightarrow 0} \frac{x^2 \sin^2 x}{x^2 - \left(x - \frac{x^3}{3!} + \dots \right)^2}$$

$$\lim_{x \rightarrow 0} \frac{x^2 \sin^2 x}{x^2 - x^2 + \frac{2x^4}{3!} \dots} = \frac{3!}{2} = 3$$

Question21

For the function $f(x) = e^{\sin|x|} - |x|$, $x \in \mathbf{R}$, consider the following statements :

Statement I : f is differentiable for all $x \in \mathbf{R}$.

Statement II : f is increasing in $\left(-\pi, -\frac{\pi}{2}\right)$.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Both Statement I and Statement II are true

B.

Both Statement I and Statement II are false

C.

Statement I is true but Statement II is false

D.

Statement I is false but Statement II is true

Answer: D

Solution:

Here $f(x) = e^{\sin|x|} - |x|$. Since $|x|$ is not differentiable at $x = 0$, and also $\sin|x|$ has a “kink” at $x = 0$, the function f is not differentiable at $x = 0$. So, **Statement I is false**.

Now take $x \in (-\pi, -\frac{\pi}{2})$. Here $x < 0$, so $|x| = -x$ and hence $\sin|x| = \sin(-x) = -\sin x$. Therefore, on this interval we can write $f(x) = e^{-\sin x} + x$ and differentiate to check whether f is increasing.

$$f(x) = e^{-\sin x} + x$$

$$f'(x) = -e^{-\sin x} \cos x + 1$$

$$\text{for } x \in \left(-\pi, -\frac{\pi}{2}\right) \quad \cos x < 0$$

$$\therefore f'(x) > 0$$

$$\therefore f(x) \text{ is increasing}$$

Since $\cos x < 0$ on $(-\pi, -\frac{\pi}{2})$ and $e^{-\sin x} > 0$, we get $-e^{-\sin x} \cos x > 0$, so $f'(x) > 0$ on this interval. Hence, **Statement II is true**.

Question22

Let the function,

$$f(x) = \begin{cases} -3ax^2 - 2, & x < 1 \\ a^2 + bx, & x \geq 1 \end{cases}$$

be differentiable for all $x \in \mathbf{R}$, where $a > 1$, $b \in \mathbf{R}$. If the area of the region enclosed by $y = f(x)$ and the line $y = -20$ is $\alpha + \beta\sqrt{3}$, $\alpha, \beta \in \mathbf{Z}$, then the value of $\alpha + \beta$ is _____.

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Answer: 34

Solution:

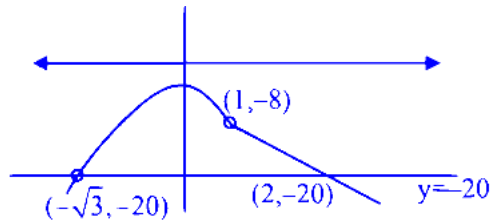
$f(x)$ is continuous and differentiable

$$\text{at } x = 1; \quad \text{LHL} = \text{RHL, LHD} = \text{RHD}$$

$$-3a - 2 = a^2 + b, -6a = b$$

$$a = 2, 1; b = -12$$

$$f(x) = \begin{cases} -6x^2 - 2, & x < 1 \\ 4 - 12x, & x \geq 1 \end{cases}$$



$$\text{Area} = \int_{-\sqrt{3}}^1 (-6x^2 - 2 + 20) dx + \int_1^2 (4 - 12x + 20) dx$$

$$= 16 + 12\sqrt{3} + 6 = 22 + 12\sqrt{3}$$

$$\therefore \alpha + \beta = 34$$

Question 23

$$\text{Let } f(x) = \begin{cases} 3x, & x < 0 \\ \min\{1 + x + [x], x + 2[x]\}, & 0 \leq x \leq 2 \\ 5, & x > 2 \end{cases}$$

where $[.]$ denotes greatest integer function. If α and β are the number of points, where f is not continuous and is not differentiable, respectively, then $\alpha + \beta$ equals _____.

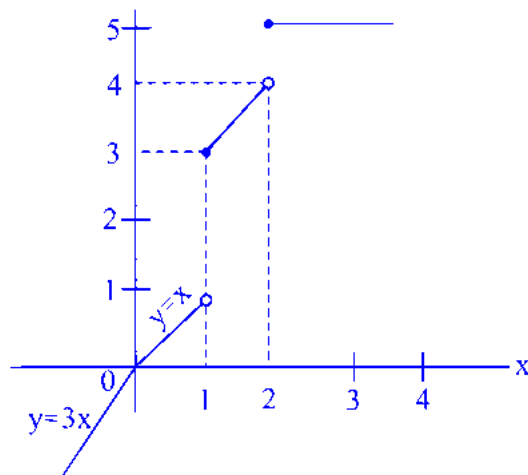
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Answer: 5

Solution:

$$f(x) = \begin{cases} 3x & ; \quad x < 0 \\ \min\{1 + x, x\} & ; \quad 0 \leq x < 1 \\ \min\{2 + x, x + 2\} & ; \quad 1 \leq x < 2 \\ 5 & ; \quad x > 2 \end{cases}$$

$$f(x) = \begin{cases} 3x & ; \quad x < 0 \\ x & ; \quad 0 \leq x < 1 \\ x + 2 & ; \quad 1 \leq x < 2 \\ 5 & ; \quad x > 2 \end{cases}$$



Not continuous at $x \in \{1, 2\} \Rightarrow \alpha = 2$

Not diff. at $x \in \{0, 1, 2\} \Rightarrow \beta = 3$

$$\alpha + \beta = 5$$

Question24

Let $f(x) = \lim_{n \rightarrow \infty} \sum_{r=0}^n \left(\frac{\tan(x/2^{r+1}) + \tan^3(x/2^{r+1})}{1 - \tan^2(x/2^{r+1})} \right)$ Then $\lim_{x \rightarrow 0} \frac{e^x - e^{f(x)}}{(x - f(x))}$ is equal to _____.

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Answer: 1

Solution:

$$f(x) = \lim_{n \rightarrow \infty} \sum_{r=0}^n \left(\tan \frac{x}{2^r} - \tan \frac{x}{2^{r+1}} \right) = \tan x$$
$$\lim_{x \rightarrow 0} \left(\frac{e^x - e^{\tan x}}{x - \tan x} \right) = \lim_{x \rightarrow 0} e^{\tan x} \frac{(e^{x - \tan x} - 1)}{(x - \tan x)}$$
$$= 1$$

Question25

Let $[t]$ be the greatest integer less than or equal to t . Then the least value of $p \in \mathbb{N}$ for which

$$\lim_{x \rightarrow 0^+} \left(x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{p}{x} \right] \right) - x^2 \left(\left[\frac{1}{x^2} \right] + \left[\frac{2^2}{x^2} \right] + \dots + \left[\frac{p^2}{x^2} \right] \right) \right) \geq 1 \text{ is equal to } \underline{\hspace{2cm}}.$$

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Answer: 24

Solution:

To find the least natural number p for which the following inequality holds:

$$\lim_{x \rightarrow 0^+} \left(x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{p}{x} \right] \right) - x^2 \left(\left[\frac{1}{x^2} \right] + \left[\frac{2^2}{x^2} \right] + \dots + \left[\frac{p^2}{x^2} \right] \right) \right) \geq 1$$

we simplify the expression inside the limit.

As $x \rightarrow 0^+$, $\left[\frac{n}{x} \right]$ approximates to $\frac{n}{x}$. Thus, the problem becomes finding:

$$(1 + 2 + \dots + p) - (1^2 + 2^2 + \dots + 9^2) \geq 1$$

The sum of the first p natural numbers is given by:

$$\frac{p(p+1)}{2}$$

And the sum of the squares of the first 9 natural numbers is:

$$1^2 + 2^2 + \dots + 9^2 = \frac{9 \cdot 10 \cdot 19}{6}$$

Thus, the inequality becomes:

$$\frac{p(p+1)}{2} - \frac{9 \cdot 10 \cdot 19}{6} \geq 1$$

Solving this, we rewrite:

$$p(p+1) \geq 572$$

The least natural number p satisfying this condition is 24.

Question26

If $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{x^2}} = p$, then $96 \log_e p$ is equal to _____

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Answer: 32

Solution:

To solve the given limit problem, we start by analyzing the expression:

$$\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{x^2}} = p$$

This limit exhibits the indeterminate form 1^∞ . To handle this form, we use the transformation:

$$p = e^{\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} - 1 \right) \frac{1}{x^2}}$$

Expanding $\tan x$ using its Taylor series near $x = 0$, we have:

$$\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots$$

Substituting this expansion into the limit, we get:

$$\frac{\tan x - x}{x^3} = \frac{\left(x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots - x \right)}{x^3} = \frac{\frac{x^3}{3} + \frac{2}{15}x^5 + \dots}{x^3}$$

This simplifies to:

$$\frac{x^3}{3x^3} = \frac{1}{3}$$

Thus, the limit becomes:

$$p = e^{\frac{1}{3}}$$

Therefore, the expression for $\log_e p$ is:

$$\log_e p = \frac{1}{3}$$

Finally, computing $96 \log_e p$:

$$96 \log_e p = 96 \cdot \frac{1}{3} = 32$$

Question 27

Let m and n be the number of points at which the function $f(x) = \max \{x, x^3, x^5, \dots, x^{21}\}$, $x \in \mathbb{R}$, is not differentiable and not continuous, respectively. Then $m + n$ is equal to _____.

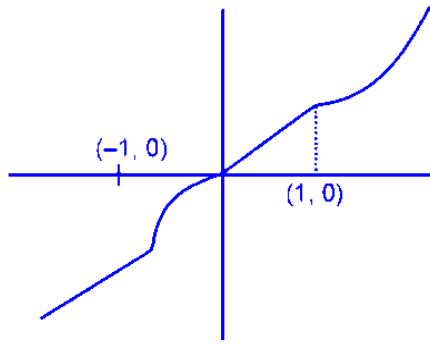
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Answer: 3

Solution:

$$\text{for } x \geq 1, x^{21} \geq x^{19} \geq \dots \geq x.$$

$$f(x) = \begin{cases} x & x < -1 \\ x^{21} & -1 \leq x \leq 0 \\ x & 0 < x < 1 \\ x^{21} & x \geq 1 \end{cases}$$



Clearly, $f(x)$ is continuous everywhere.

$$\Rightarrow \alpha = 0$$

$$f'(x) = \begin{cases} 1 & ; x < -1 \\ 21x^{20} & ; -1 \leq x \leq 0 \\ 1 & ; 0 < x < 1 \\ 21 \cdot x^{20} & ; x \geq 1 \end{cases}$$

$$\Rightarrow \beta = 3$$

$$\Rightarrow \alpha + \beta = 3$$

Question28

The number of points of discontinuity of the function $f(x) = \left\lfloor \frac{x^2}{2} \right\rfloor - [\sqrt{x}]$, $x \in [0, 4]$, where $[\cdot]$ denotes the greatest integer function, is _____.

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Answer: 8

Solution:

To determine the points of discontinuity of the function $f(x) = \left\lfloor \frac{x^2}{2} \right\rfloor - [\sqrt{x}]$, where $[\cdot]$ denotes the greatest integer function, we need to identify possible values of x where discontinuities might occur within the interval $[0, 4]$.

Discontinuity Analysis

For the term $\left\lfloor \frac{x^2}{2} \right\rfloor$:

The probable values of x that could cause discontinuities are the roots or specific values where the integer part changes between consecutive integers. The transitions happen when:

$$= 1, 2, 3, 4, 5, 6, 7, 8$$

$$\Rightarrow x = \sqrt{2}, 2, \sqrt{6}, 2\sqrt{2}, \sqrt{10}, 2\sqrt{3}, \sqrt{14}, 4$$

For the term $[\sqrt{x}]$:

The values of x where $[\sqrt{x}]$ changes are straightforward. They occur at:

$$x = 1, 2$$

Discontinuity Check

By evaluating $f(x)$ at all these potential points, we find the function is indeed discontinuous at:

$$x = 1, \sqrt{2}, 2, \sqrt{6}, 2\sqrt{2}, \sqrt{10}, 2\sqrt{3}, \sqrt{14}$$

Thus, the function $f(x)$ has 8 discontinuities on the interval $[0, 4]$.

Question29

For $t > -1$, let α_t and β_t be the roots of the equation

$$((t+2)^{1/7} - 1)x^2 + ((t+2)^{1/6} - 1)x + ((t+2)^{1/21} - 1) = 0. \text{ If } \lim_{t \rightarrow -1^+} \alpha_t = a \text{ and } \lim_{t \rightarrow -1^+} \beta_t = b,$$

then $72(a+b)^2$ is equal to _____.

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Answer: 98

Solution:

$$a + b = \lim_{t \rightarrow -1^+} (\alpha + \beta) = \lim_{t \rightarrow -1^+} -\frac{(t+2)^{\frac{1}{6}} - 1}{(t+2)^{\frac{1}{7}} - 1}$$

$$\text{let } t+2 = y$$

$$a + b = \lim_{y \rightarrow 1^+} \frac{y^{1/6} - 1}{y^{1/7} - 1} = \frac{7}{6}$$

$$72(a+b)^2 = 72 \frac{49}{36} = 98$$

Question30

If the function $f(x) = \frac{\tan(\tan x) - \sin(\sin x)}{\tan x - \sin x}$ is continuous at $x = 0$, then $f(0)$ is equal to _____.

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Answer: 2

Solution:

$$\lim_{x \rightarrow 0} \frac{\frac{\tan(\tan x) - \tan x}{\tan^3 x} + \frac{\tan x - \sin x}{x^3} + \frac{\sin x - \sin(\sin x)}{\sin^3 x} \cdot \frac{\sin^3 x}{x^3}}{\frac{\tan x - \sin x}{x^3}}$$
$$= \frac{\frac{1}{3} + \frac{1}{2} + \frac{1}{6}}{\frac{1}{2}} = 2$$

Question31

If $\sum_{r=1}^n T_r = \frac{(2n-1)(2n+1)(2n+3)(2n+5)}{64}$, then $\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{1}{T_r} \right)$ is equal to :

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Options:

A. $\frac{2}{3}$

B. $\frac{1}{3}$

C. 1

D. 0

Answer: A

Solution:

$$T_n = S_n - S_{n-1}$$

$$\Rightarrow T_n = \frac{1}{8}(2n-1)(2n+1)(2n+3)$$

$$\Rightarrow \frac{1}{T_n} = \frac{8}{(2n-1)(2n+1)(2n+3)}$$

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{T_r} = \lim_{n \rightarrow \infty} 8 \sum_{r=1}^n \frac{1}{(2r-1)(2r+1)(2r+3)}$$

$$= \lim_{n \rightarrow \infty} \frac{8}{4} \sum \left(\frac{1}{(2n-1)(2n+1)} - \frac{1}{(2n+1)(2n+3)} \right)$$

$$= \lim_{n \rightarrow \infty} 2 \left[\left(\frac{1}{1.3} - \frac{1}{3.5} \right) + \left(\frac{1}{3.5} - \frac{1}{5.7} \right) + \dots \right]$$

$$= \frac{2}{3}$$

Question 32

If $\lim_{x \rightarrow \infty} \left(\left(\frac{e}{1-e} \right) \left(\frac{1}{e} - \frac{x}{1+x} \right) \right)^x = \alpha$, then the value of $\frac{\log_e \alpha}{1 + \log_e \alpha}$ equals :

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Options:

A. e^{-2}

B. e^2

C. e

D. e^{-1}

Answer: C

Solution:

$$\alpha = \lim_{x \rightarrow \infty} \left[\left(\frac{e}{1-e} \right) \left(\frac{1}{e} - \frac{x}{1+x} \right) \right]^x.$$

Begin by rewriting the expression inside the limit. Notice that

$$\frac{x}{1+x} = 1 - \frac{1}{1+x},$$

so

$$\frac{1}{e} - \frac{x}{1+x} = \frac{1}{e} - \left(1 - \frac{1}{1+x} \right) = \frac{1}{e} - 1 + \frac{1}{1+x}.$$

Express the constant part as

$$1 - \frac{1}{e} = \frac{e-1}{e},$$

to obtain

$$\frac{1}{e} - \frac{x}{1+x} = -\frac{e-1}{e} + \frac{1}{1+x}.$$

Multiplying by the prefactor, we have

$$\left(\frac{e}{1-e} \right) \left(\frac{1}{e} - \frac{x}{1+x} \right) = \frac{e}{1-e} \left[-\frac{e-1}{e} + \frac{1}{1+x} \right].$$

Split the expression:

$$\frac{e}{1-e} \cdot \left(-\frac{e-1}{e} \right) + \frac{e}{1-e} \cdot \frac{1}{1+x} = -\frac{e-1}{1-e} + \frac{e}{(1-e)(1+x)}.$$

Since

$$1 - e = -(e - 1),$$

it follows that

$$-\frac{e-1}{1-e} = -\frac{e-1}{-(e-1)} = 1.$$

Hence, the expression simplifies to

$$1 + \frac{e}{(1-e)(1+x)}.$$

Again, replacing $1 - e$ by $-(e - 1)$,

$$1 + \frac{e}{-(e-1)(1+x)} = 1 - \frac{e}{(e-1)(1+x)}.$$

Thus, the limit becomes

$$\alpha = \lim_{x \rightarrow \infty} \left(1 - \frac{e}{(e-1)(1+x)} \right)^x.$$

For large x , note that

$$1 + x \sim x,$$

so we approximate

$$\alpha = \lim_{x \rightarrow \infty} \left(1 - \frac{e}{(e-1)x} \right)^x.$$

Recall the limit

$$\lim_{x \rightarrow \infty} \left(1 - \frac{a}{x} \right)^x = e^{-a},$$

with

$$a = \frac{e}{e-1}.$$

Therefore,

$$\alpha = e^{-\frac{e}{e-1}}.$$

Taking the natural logarithm,

$$\ln \alpha = -\frac{e}{e-1}.$$

Now, compute

$$\frac{\ln \alpha}{1 + \ln \alpha} = \frac{-\frac{e}{e-1}}{1 - \frac{e}{e-1}}.$$

Express the denominator with a common denominator:

$$1 - \frac{e}{e-1} = \frac{e-1}{e-1} - \frac{e}{e-1} = -\frac{1}{e-1}.$$

Thus,

$$\frac{-\frac{e}{e-1}}{-\frac{1}{e-1}} = \frac{e}{1} = e.$$

The final result is

$$\frac{\ln \alpha}{1 + \ln \alpha} = e.$$

Question33

If the function

$$f(x) = \begin{cases} \frac{2}{x} \{ \sin(k_1 + 1)x + \sin(k_2 - 1)x \}, & x < 0 \\ 4, & x = 0 \\ \frac{2}{x} \log_e \left(\frac{2+k_1x}{2+k_2x} \right), & x > 0 \end{cases}$$

is continuous at $x = 0$, then $k_1^2 + k_2^2$ is equal to :

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Options:

- A. 5
- B. 10
- C. 20
- D. 8

Answer: B

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0^-} \frac{2}{x} \{\sin(k_1 + 1)x + \sin(k_2 - 1)x\} &= 4 \\ \Rightarrow 2(k_1 + 1) + 2(k_2 - 1) &= 4 \\ \Rightarrow k_1 + k_2 &= 2 \\ \Rightarrow \lim_{x \rightarrow 0^+} \frac{2}{x} \ln \left(\frac{2 + k_1 x}{2 + k_2 x} \right) &= 4 \\ \Rightarrow \lim_{x \rightarrow 0^+} \frac{1}{x} \ln \left(1 + \frac{(k_1 - k_2)x}{2 + k_2 x} \right) &= 2 \\ \Rightarrow \frac{k_1 - k_2}{2} &= 2 \\ \Rightarrow k_1 - k_2 &= 4 \\ \therefore k_1 = 3, k_2 = -1 \\ k_1^2 + k_2^2 &= 9 + 1 = 10 \end{aligned}$$

Question34

$$\lim_{x \rightarrow \infty} \frac{(2x^2 - 3x + 5)(3x - 1)^{\frac{x}{2}}}{(3x^2 + 5x + 4)\sqrt{(3x + 2)^x}} \text{ is equal to :}$$

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Options:

- A. $\frac{2e}{3}$
- B. $\frac{2}{3\sqrt{e}}$
- C. $\frac{2e}{\sqrt{3}}$
- D. $\frac{2}{\sqrt{3e}}$

Answer: B

Solution:

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\left(2 - \frac{3}{x} + \frac{5}{x^2}\right) \left(1 - \frac{1}{3x}\right)^{x/2}}{\left(3 + \frac{5}{x} + \frac{4}{x^2}\right) \left(1 + \frac{2}{3x}\right)^{x/2}} \\ = \lim_{x \rightarrow \infty} \frac{2}{3} \cdot \frac{e^{\frac{x}{2} \left(1 - \frac{1}{3x} - 1\right)}}{e^{\frac{x}{2} \left(1 + \frac{2}{3x} - 1\right)}} \\ = \frac{2}{3} \cdot \frac{e^{-\frac{1}{6}}}{e^{1/3}} = \frac{2}{3} e^{-\frac{1}{2}} \end{aligned}$$

Question35

Let $f : \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ be a function such that $f(x) - 6f\left(\frac{1}{x}\right) = \frac{35}{3x} - \frac{5}{2}$. If the $\lim_{x \rightarrow 0} \left(\frac{1}{\alpha x} + f(x)\right) = \beta$; $\alpha, \beta \in \mathbb{R}$, then $\alpha + 2\beta$ is equal to

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Options:

A. 6

B. 5

C. 3

D. 4

Answer: D

Solution:

$$F(x) - 6f(1/x) = \frac{35}{3x} - \frac{5}{2} \quad \dots (1)$$

Replace $x \rightarrow \frac{1}{x}$

$$F(1/x) - 6(x) = \frac{35x}{3} - \frac{5}{2} \quad \dots (2)$$

Using (1) & (2)

$$f(x) = -2x - \frac{1}{3x} + \frac{1}{2}$$

$$B = \lim_{x \rightarrow 0} \left(\frac{1}{\alpha x} + f(x) \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{1}{\alpha x} - 2x - \frac{1}{3x} + \frac{1}{2} \right)$$

$$\alpha = 3, \quad B = 1/2$$

$$\text{So, } \alpha + 2B = 3 + 1 = 4$$

Question 36

$\lim_{x \rightarrow 0} \operatorname{cosec} x \left(\sqrt{2 \cos^2 x + 3 \cos x} - \sqrt{\cos^2 x + \sin x + 4} \right)$ is:

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Options:

A. $\frac{1}{\sqrt{15}}$

B. $\frac{1}{2\sqrt{5}}$

C. 0

D. $-\frac{1}{2\sqrt{5}}$

Answer: D

Solution:

$$\lim_{x \rightarrow 0} \operatorname{cosec} x \left(\sqrt{2 \cos^2 x + 3 \cos x} - \sqrt{\cos^2 x + \sin x + 4} \right)$$

$$\begin{aligned}
& \lim_{x \rightarrow 0} \frac{\operatorname{cosec}(\cos^2 x + 3 \cos x - \sin x - 4)}{\left(\sqrt{2 \cos^2 x + 3 \cos x} + \sqrt{\cos^2 x + \sin x + 4}\right)} \\
& \lim_{x \rightarrow 0} \frac{1}{\sin x} \frac{(\cos^2 x + 3 \cos x - 4) - \sin x}{\left(\sqrt{2 \cos^2 x + 3 \cos x} + \sqrt{\cos^2 x + \sin x + 4}\right)} \\
& \lim_{x \rightarrow 0} \frac{(\cos^x + 4)(\cos x - 1) - \sin x}{\sin x \left(\sqrt{2 \cos^2 x + 3 \cos x} + \sqrt{\cos^2 x + \sin x + 4}\right)} \\
& \lim_{x \rightarrow 0} \frac{-2 \sin^2 \frac{x}{2} (\cos x + 4) - 2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \sin \frac{x}{2} \cos \frac{x}{2} \left(\sqrt{2 \cos^2 x + 3 \cos x} + \sqrt{\cos^2 x + \sin x + 4}\right)} \\
& \lim_{x \rightarrow 0} \frac{-\left(\sin^x \frac{x}{2} (\cos x + 4) + \cos \frac{x}{2}\right)}{\cos \frac{x}{2} \left(\sqrt{2 \cos^2 x + 3 \cos x} + \sqrt{\cos^2 x + \sin x + 4}\right)} \\
& - \frac{1}{2\sqrt{5}}
\end{aligned}$$

Question37

Let $[x]$ denote the greatest integer function, and let m and n respectively be the numbers of the points, where the function $f(x) = [x] + |x - 2|$, $-2 < x < 3$, is not continuous and not differentiable. Then m + n is equal to :

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Options:

- A. 6
- B. 9
- C. 8
- D. 7

Answer: C

Solution:

$$f(x) = [x] + |x - 2| \quad -2 < x < 3$$

$$f(x) = \begin{cases} -x, & -2 < x < -1 \\ -x + 1, & -1 \leq x < 0 \\ -x + 2, & 0 \leq x < 1 \\ -x + 3, & 1 \leq x < 2 \\ x, & 2 \leq x < 3 \end{cases}$$

So $f(x)$ is not continuous at 4 points and not differentiable at 4 point

$$\text{So } m + n = 4 + 4 = 8$$

Question38

The value of $\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{k^3 + 6k^2 + 11k + 5}{(k+3)!} \right)$ is :

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Options:

A.

5/3

B.

2

C.

4/3

D.

7/3

Answer: A

Solution:

$$\begin{aligned} & \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^3 + 6k^2 + 11k + 5}{(k+3)!} \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^3 + 6k^2 + 11k + 6 - 1}{(k+3)!} \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{(k+1)(k+2)(k+3) - 1}{(k+3)!} \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{(k+1)(k+2)(k+3)}{(k+3)!} - \frac{1}{(k+3)!} \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{1}{k!} - \frac{1}{(k+3)!} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} \dots + \frac{1}{n!} - \frac{1}{4!} - \frac{1}{5!} - \frac{1}{6!} \dots - \frac{1}{(n+3)!} \right) \\ &= \frac{1}{1} + \frac{1}{2} + \frac{1}{6} = \frac{10}{6} = \frac{5}{3} \end{aligned}$$

Question 39

Let the function $f(x) = (x^2 - 1)|x^2 - ax + 2| + \cos|x|$ be not differentiable at the two points $x = \alpha = 2$ and $x = \beta$. Then the distance of the point (α, β) from the line $12x + 5y + 10 = 0$ is equal to :

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Options:

A.

5

B.

2

C. 4

D.

3

Answer: D

Solution:

$\cos|x|$ is always differentiable

$\therefore$ we have to check only for $|x^2 - ax + 2|$

$\therefore$ Not differentiable at

$$x^2 - ax + 2 = 0$$

One root is given, $\alpha = 2$

$$\therefore 4 - 2a + 2 = 0$$
$$a = 3$$

$\therefore$ other root $\beta = 1$

but for $x = 1$ $f(x)$ is differentiable

(Drop)

Question40

For $\alpha, \beta, \gamma \in \mathbf{R}$, if $\lim_{x \rightarrow 0} \frac{x^2 \sin \alpha x + (\gamma - 1)e^{x^2}}{\sin 2x - \beta x} = 3$, then $\beta + \gamma - \alpha$ is equal to :

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Options:

A. -1

B. 4

C. 6

D. 7

Answer: D

Solution:

At $x \rightarrow 0$

$$\sin 2x - \beta x \rightarrow 0$$

$$\Rightarrow \frac{0}{0} \text{ form}$$

$$\Rightarrow (\gamma - 1)e^0 + 0 \sin(\alpha x) \rightarrow 0$$

$$\Rightarrow (\gamma - 1) = 0$$

$$\Rightarrow \gamma = 1$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x^2 \sin(\alpha x)}{(\sin 2x - \beta x)} = 3$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x^2 \left[\alpha x - \frac{(\alpha x)^3}{3!} + \frac{(\alpha x)^5}{5!} - \dots \right]}{\left[(2x) - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} \right] - \beta x}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\alpha x^3 - \frac{\alpha^3 x^5}{3!} + \frac{\alpha^5 x^7}{5!} - \dots}{x(2 - \beta) - \frac{8x^3}{6} + \frac{32x^5}{315} - \dots} = 3$$

$$\Rightarrow 2 - \beta = 0 \text{ and } \frac{\alpha}{-\frac{8}{6}} = 3$$

$$\Rightarrow \beta = 2$$

$$\alpha = 3 \left(-\frac{8}{6} \right) = -4$$

$$\Rightarrow \gamma = 1, \beta = 2, \alpha = -4$$

$$\Rightarrow \beta + \gamma - \alpha = 7$$

Question41

If $\lim_{x \rightarrow 0} \frac{\cos(2x) + a \cos(4x) - b}{x^4}$ is finite, then $(a + b)$ is equal to :

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Options:

A. 0

B. $\frac{3}{4}$

C. -1

D. $\frac{1}{2}$

Answer: D

Solution:

To find the value of $a + b$ for which the following limit is finite:

$$\lim_{x \rightarrow 0} \frac{\cos(2x) + a \cos(4x) - b}{x^4}$$

we start by expanding the cosine functions using their Taylor series:

$$\cos(2x) = 1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \dots$$

$$\cos(4x) = 1 - \frac{(4x)^2}{2!} + \frac{(4x)^4}{4!} - \dots$$

Substitute these expansions into the limit expression:

$$\lim_{x \rightarrow 0} \frac{\left(1 - \frac{4x^2}{2!} + \frac{(2x)^4}{4!} - \dots\right) + a\left(1 - \frac{(4x)^2}{2!} + \frac{(4x)^4}{4!} - \dots\right) - b}{x^4}$$

To ensure that the limit is finite, the linear and quadratic terms must sum to zero. Let's equate the constant terms first:

For the limit to be finite:

$$1 + a - b = 0$$

Equating the coefficients of x^2 :

$$-2 - 8a = 0$$

Solving the second equation, we have:

$$8a = -2a = \frac{-1}{4}$$

Substitute a back into the first equation to find b :

$$1 + \frac{-1}{4} - b = 0 \Rightarrow b = 1 - \frac{1}{4} = \frac{3}{4}$$

Thus, the sum of a and b is:

$$a + b = \frac{-1}{4} + \frac{3}{4} = \frac{1}{2}$$

Therefore, the correct value of $a + b$ is $\frac{1}{2}$.

Question 42

Let $f(x) = \begin{cases} (1 + ax)^{1/x}, & x < 0 \\ 1 + b, & x = 0 \\ \frac{(x+4)^{1/2} - 2}{(x+c)^{1/3} - 2}, & x > 0 \end{cases}$ be continuous at $x = 0$. Then $e^a b c$ is equal to:

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Options:

A. 64

B. 48

C. 36

D. 72

Answer: B

Solution:

$$f(x) = \begin{cases} (1+ax)^{1/x} & , x < 0 \\ 1+b & , x = 0 \\ \frac{(x+4)^{1/2}-2}{(x+c)^{1/3}-2} & , x > 0 \end{cases}$$

Hence, $f(0) = 1 + 6$

$$\text{RHL} = \lim_{x \rightarrow 0^+} \frac{(x+4)^{1/2} - 2}{(x+c)^{1/3} - 2} \quad \left[\text{For } \frac{0}{0} \text{ form, } c = 8 \right]$$

$$= \lim_{x \rightarrow 0} \frac{2\left(1 + \frac{x}{4}\right)^{1/2} - 2}{\left(1 + \frac{x}{8}\right)^{1/3} - 2}$$

$$= \lim_{x \rightarrow 0} \frac{1 + \frac{x}{8} - 1}{1 + \frac{x}{8} \cdot \frac{1}{3} - 1} = \frac{\frac{1}{8}}{\frac{1}{8} \cdot \frac{1}{3}} = 3$$

$$\therefore 1 + b = 3$$

$$\Rightarrow b = 2$$

$$\text{LHL} = \lim_{x \rightarrow 0} (1+ax)^{1/x} = \lim_{x \rightarrow 0} e^{\frac{1+ax-1}{x}} = e^a = 3$$

$$\therefore e^a \cdot bc = 3 \cdot 2.8 = 48$$

Question43

If $\lim_{x \rightarrow 1^+} \frac{(x-1)(6+\lambda \cos(x-1))+\mu \sin(1-x)}{(x-1)^3} = -1$, where $\lambda, \mu \in \mathbb{R}$, then $\lambda + \mu$ is equal to

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Options:

A. 20

B. 19

C. 18

D. 17

Answer: C

Solution:

$$\lim_{x \rightarrow 1^+} \frac{(x-1)(6+\lambda \cos(x-1))+\mu \sin(1-x)}{(x-1)^3} = -1$$

Let $x-1 = t$

$$\lim_{t \rightarrow 0^+} \frac{6t + \lambda t \cos t - \mu \sin t}{t^3} = -1$$

$$= \lim_{t \rightarrow 0^+} \frac{6t + \lambda t \left(1 - \frac{t^2}{2!} + \frac{t^4}{4!} + \dots\right) - \mu \left(t - \frac{t^3}{3!} + \dots\right)}{t^3} = -1$$

$$= \lim_{t \rightarrow 0^+} \frac{t(6 + \lambda - \mu) + t^3 \left(-\frac{\lambda}{2} + \frac{\mu}{6}\right) + \dots}{t^3} = -1$$

$$\therefore \lambda - \mu + 6 = 0 \quad \dots (i)$$

$$\frac{\mu}{6} - \frac{\lambda}{2} = -1 \quad \dots (ii)$$

Solving (i) and (ii)

$$\lambda = 6, \mu = 12$$

$$\lambda + \mu = 18$$

Question44

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying $f(0) = 1$ and $f(2x) - f(x) = x$ for all $x \in \mathbb{R}$. If

$\lim_{n \rightarrow \infty} \left\{ f(x) - f\left(\frac{x}{2^n}\right) \right\} = G(x)$, then $\sum_{r=1}^{10} G(r^2)$ is equal to

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Options:

A. 215

B. 420

C. 385

D. 540

Answer: C

Solution:

$$f(0) = 1, f(2x) - f(x) = x$$

$$\text{Replace } x \rightarrow \frac{x}{2}$$

$$f(x) - f\left(\frac{x}{2}\right) = \frac{x}{2} \quad \dots (1)$$

$$\text{Again, Replace } x \rightarrow \frac{x}{2}$$

$$f\left(\frac{x}{2}\right) - f\left(\frac{x}{2^2}\right) = \frac{x}{2^2} \quad \dots (2)$$

$$\vdots \quad \vdots \quad \vdots$$

$$\vdots \quad \vdots \quad \vdots$$

$$\vdots \quad \vdots \quad \vdots$$

$$f\left(\frac{x}{2^{n-1}}\right) - f\left(\frac{x}{2^n}\right) = \frac{x}{2^n}$$

$$\text{Adding (1) + (2) + (3) + } \dots + (n)$$

$$\text{We get } f(x) - f\left(\frac{x}{2^n}\right) = \frac{x}{2} + \frac{x}{2^2} + \dots + \frac{x}{2^n}$$

$$\lim_{n \rightarrow \infty} \left(f(x) - f\left(\frac{x}{2^n}\right) \right) = \lim_{n \rightarrow \infty} \left(\frac{x}{2} + \frac{x}{2^n} + \dots + \frac{x}{2^n} \right)$$

$$f(x) - f(0) = \frac{x}{2}$$

$$\Rightarrow G(x) = x$$

$$\Rightarrow G(r^2) = r^2$$

$$\Rightarrow \sum_{r=1}^{10} G(r^2) = \sum_{r=1}^{10} G(r^2)$$

$$= \frac{(10)(11)(21)}{6} = (55)7$$

$$\Rightarrow 385$$

Question 45

Let f be a differentiable function on $\mathbb{R}$ such that $f(2) = 1, f'(2) = 4$. Let $\lim_{x \rightarrow 0} (f(2+x))^{3/x} = e^\alpha$. Then the number of times the curve $y = 4x^3 - 4x^2 - 4(\alpha - 7)x - \alpha$ meets x -axis is :

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Options:

A. 3

B. 1

C. 2

D. 0

Answer: C

Solution:

$$\begin{aligned}\lim_{x \rightarrow 0} (f(2+x))^{3/x} &= (1^\infty \text{ form}) \\ e^{\lim_{x \rightarrow 0} \frac{3}{x} (f(2+x) - 1)} &= e^{\lim_{x \rightarrow 0} 3f'(2+x)} \\ &= e^{3f'(2)} \\ &= e^{12} \\ \Rightarrow \alpha &= 12 \\ y &= 4x^3 - 4x^2 - 4(12-7)x - 12 \\ y &= 4x^3 - 4x^2 - 20x - 12 \\ y &= 4(x^3 - x^2 - 5x - 3) \\ &= 4(x+1)^2(x-3) \\ \text{It meets the } x\text{-axis at two points}\end{aligned}$$

Question46

$$\lim_{x \rightarrow 0^+} \frac{\tan\left(5(x)^{\frac{1}{3}}\right) \log_e(1+3x^2)}{(\tan^{-1} 3\sqrt{x})^2 \left(e^{5(x)^{\frac{4}{3}}} - 1\right)} \text{ is equal to}$$

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Options:

- A. $\frac{5}{3}$
- B. 1
- C. $\frac{1}{3}$
- D. $\frac{1}{15}$

Answer: C

Solution:

$$\begin{aligned}\lim_{x \rightarrow 0^+} \frac{\tan\left(5(x)^{\frac{1}{3}}\right) \ln(1+3x^3)}{(\tan^{-1} 3\sqrt{x})^2 \left(e^{5x^{\frac{4}{3}}} - 1\right)} \\ \lim_{x \rightarrow 0^+} \frac{\frac{\tan\left(5(x)^{\frac{1}{3}}\right)}{5(x)^{\frac{1}{3}}} \cdot \frac{\ln(1+3x^2)}{3x^2} \times 5(x)^{\frac{1}{3}} (3x^2)}{\frac{(\tan^2(3\sqrt{x}))^2}{(3\sqrt{x})^2} \cdot \frac{e^{5x^{\frac{4}{3}}} - 1}{5x^{\frac{4}{3}}} \times 9x \times 5x^{\frac{4}{3}}} \\ \lim_{x \rightarrow 0^+} \frac{\frac{\tan\left(5(x)^{\frac{1}{3}}\right)}{5(x)^{\frac{1}{3}}} \cdot \frac{\ln(1+3x^3)}{3x^2} \times 15x^{\frac{7}{3}}}{\frac{(\tan^2(3\sqrt{x}))^2}{(3\sqrt{x})^2} \cdot \frac{e^{5(x)^{\frac{4}{3}}} - 1}{5x^{\frac{4}{3}}} \times 45x^{\frac{7}{3}}} \\ = \frac{1}{3}\end{aligned}$$

Question47

Given below are two statements:

Statement I: $\lim_{x \rightarrow 0} \left(\frac{\tan^{-1} x + \log_e \sqrt{\frac{1+x}{1-x}} - 2x}{x^5} \right) = \frac{2}{5}$

Statement II: $\lim_{x \rightarrow 1} \left(x^{\frac{2}{1-x}} \right) = \frac{1}{e^2}$

In the light of the above statements, choose the correct answer from the options given below:

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Options:

A.

Statement I is false but Statement II is true

B.

Both Statement I and Statement II are false

C.

Both Statement I and Statement II are true

D.

Statement I is true but Statement II is false

Answer: C

Solution:

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\tan^{-1} x + \frac{1}{2} [\ln(1+x) - \ln(1-x)] - 2x}{x^5} \\ &= \lim_{x \rightarrow 0} \frac{\left(x - \frac{x^3}{3} + \frac{x^5}{5} \dots \right) + \frac{1}{2} \left[x - \frac{x^2}{2} + \frac{x^3}{3} \dots - \left(-x - \frac{x^2}{2} - \frac{x^3}{3} \dots \right) \right] - 2x}{x^5} \\ &= \lim_{x \rightarrow 0} \frac{2x + \frac{2x^5}{5} \dots - 2x}{x^5} = \frac{2}{5} \\ & \lim_{x \rightarrow 1} x^{\frac{2}{1-x}} = e^{\lim_{x \rightarrow 1} \left(\frac{2}{1-x} \right) (x-1)} = e^{-2} \\ & \Rightarrow \text{Both statements correct} \end{aligned}$$

Question48

Consider the function. $f(x) = \begin{cases} \frac{a(7x-12-x^2)}{b|x^2-7x+12|}, & x < 3 \\ 2^{\frac{\sin(x-3)}{x-[x]}}, & x > 3 \\ b, & x = 3 \end{cases}$

Where $[x]$ denotes the greatest integer less than or equal to x . If S denotes the set of all ordered pairs (a,b) such that $f(x)$ is continuous at $x = 3$, then the number of elements in S is :

[27-Jan-2024 Shift 1]

Options:

A.

2

B.

Infinitely many

C.

4

D.

1

Answer: D

Solution:

$$f(3^-) = \frac{a}{b} \frac{(7x - 12 - x^2)}{|x^2 - 7x + 12|} \text{ (for } f(x) \text{ to be cont.)}$$

$$\Rightarrow f(3^-) = \frac{-a}{b} \frac{(x-3)(x-4)}{(x-3)(x-4)}; x < 3 \Rightarrow \frac{-a}{b}$$

$$\text{Hence } f(3^-) = \frac{-a}{b}$$

$$\text{Then } f(3^+) = \lim_{x \rightarrow 3^+} \left(\frac{\sin(x-3)}{x-3} \right) = 1 \text{ and } f(3) = b.$$

$$\text{Hence } f(3) = f(3^+) = f(3^-)$$

$$\Rightarrow b = 2 = -\frac{a}{b}$$

$$b = 2, a = -4$$

Hence only 1 ordered pair $(-4, 2)$.

Question 49

Consider the function $f : (0, 2) \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{2} + \frac{2}{x}$ and the function $g(x)$ defined by

$$g(x) = \begin{cases} \min\{f(t)\} & 0 < t \leq x \text{ and } 0 < x \leq 1 \\ \frac{3}{2} + x & 1 < x < 2 \end{cases} \quad \text{.. Then}$$

[27-Jan-2024 Shift 2]

Options:

A.

g is continuous but not differentiable at $x = 1$

B.

g is not continuous for all $x \in (0, 2)$

C.

g is neither continuous nor differentiable at $x = 1$

D.

g is continuous and differentiable for all $x \in (0, 2)$

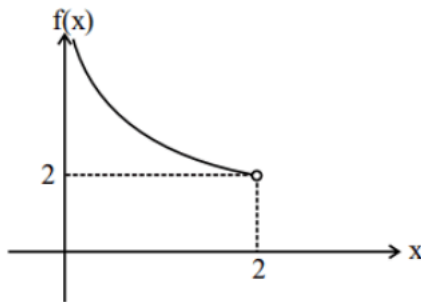
Answer: A

Solution:

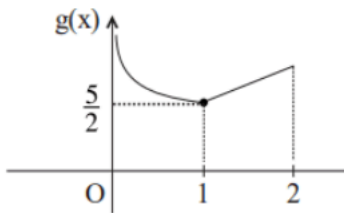
$$f: (0, 2) \rightarrow \mathbb{R}; f(x) = \frac{x}{2} + \frac{2}{x}$$

$$f'(x) = \frac{1}{2} - \frac{2}{x^2}$$

$\therefore f(x)$ is decreasing in domain.



$$g(x) = \begin{cases} \frac{x}{2} + \frac{2}{x} & 0 < x \leq 1 \\ \frac{3}{2} + x & 1 < x < 2 \end{cases}$$



Question 50

Let $f(x) = \sqrt{\lim_{r \rightarrow x} \left\{ \frac{2r^2[(f(r))^2 - f(x)f(r)]}{r^2 - x^2} - r^3 e^{\frac{f(r)}{r}} \right\}}$ be differentiable in $(-\infty, 0) \cup (0, \infty)$ and $f(1) = 1$. Then the value of ea , such that $f(a) = 0$, is equal to _____

[29-Jan-2024 Shift 2]

Answer: 2

Solution:

$$f(1) = 1, f(a) = 0$$

$$f^2(x) = \lim_{r \rightarrow x} \left(\frac{2r^2(f^2(r) - f(x)f(r))}{r^2 - x^2} - r^3 e^{\frac{f(r)}{r}} \right)$$

$$= \lim_{r \rightarrow x} \left(\frac{2r^2 f(r)}{r+x} \frac{(f(r) - f(x))}{r-x} - r^3 e^{\frac{f(r)}{r}} \right)$$

$$f^2(x) = \frac{2x^2 f(x)}{2x} f'(x) - x^3 e^{\frac{f(x)}{x}}$$

$$y^2 = xy \frac{dy}{dx} - x^3 e^{\frac{y}{x}}$$

$$\frac{y}{x} = \frac{dy}{dx} - \frac{x^2}{y} e^{\frac{y}{x}}$$

$$\text{Put } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v = v + x \frac{dv}{dx} - \frac{x}{v} e^v$$

$$\frac{dv}{dx} = \frac{e^v}{v} \Rightarrow e^{-v} v dv = dx$$

Integrating both side

$$e^v(x+c) + 1 + v = 0$$

$$f(1) = 1 \Rightarrow x = 1, y = 1$$

$$\Rightarrow c = -1 - \frac{2}{e}$$

$$e^v \left(-1 - \frac{2}{e} + x \right) + 1 + v = 0$$

$$e^{\frac{y}{x}} \left(-1 - \frac{2}{e} + x \right) + 1 + \frac{y}{x} = 0$$

$$x = a, y = 0 \Rightarrow a = \frac{2}{e}$$

$$ae = 2$$

Question 51

If the function $f(x) = \begin{cases} \frac{1}{|x|} & , |x| \geq 2 \\ ax^2 + 2b & , |x| < 2 \end{cases}$ is differentiable on $\mathbb{R}$, then $48(a+b)$ is equal to ____

[30-Jan-2024 Shift 1]

Answer: 15

Solution:

$$f(x) = \begin{cases} \frac{1}{x}; & x \geq 2 \\ ax^2 + 2b; & -2 < x < 2 \\ -\frac{1}{x}; & x \leq -2 \end{cases}$$

Continuous at $x = 2 \Rightarrow \frac{1}{2} = \frac{a}{4} + 2b$

Continuous at $x = -2 \Rightarrow \frac{1}{2} = \frac{a}{4} + 2b$

Since, it is differentiable at $x = 2$

$$-\frac{1}{x^2} = 2ax$$

Differentiable at $x = 2 \Rightarrow \frac{-1}{4} = 4a \Rightarrow a = \frac{-1}{16}, b = \frac{3}{8}$

Question 52

Let $f: \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ be a function satisfying $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$ for all $x, y, f(y) \neq 0$. If $f'(1) = 2024$, then

[30-Jan-2024 Shift 2]

Options:

A.

$$xf'(x) - 2024f(x) = 0$$

B.

$$xf'(x) + 2024f(x) = 0$$

C.

$$xf'(x) + f(x) = 2024$$

D.

$$xf'(x) - 2023f(x) = 0$$

Answer: A

Solution:

$$f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$$

$$f'(1) = 2024$$

$$f(1) = 1$$

Partially differentiating w. r. t. x

$$f'\left(\frac{x}{y}\right) \cdot \frac{1}{y} = \frac{1}{f(y)} f'(x)$$

$$y \rightarrow x$$

$$f'(1) \cdot \frac{1}{x} = \frac{f'(x)}{f(x)}$$

$$2024f(x) = xf'(x) \Rightarrow xf'(x) - 2024f(x) = 0$$

Question53

Let a and b be real constants such that the function f defined by $f(x) = \begin{cases} x^2 + 3x + a, & x \leq 1 \\ bx + 2, & x > 1. \end{cases}$ be differentiable on $\mathbb{R}$. Then, the value of $\int_{-2}^2 f(x) dx$ equals

[30-Jan-2024 Shift 2]

Options:

A.

15/6

B.

19/6

C.

21

D.

17

Answer: D

Solution:

f is continuous

$$\therefore 4 + a = b + 2$$

$$a = b - 2$$

$$f'(x) = \begin{cases} 2x + 3, & x < 1 \\ b, & x > 1 \end{cases}$$

f is differentiable

$$\therefore b = 5$$

$$\therefore a = 3$$

$$\begin{aligned} & \int_{-2}^1 (x^2 + 3x + 3) dx + \int_1^2 (5x + 2) dx \\ &= \left[\frac{x^3}{3} + \frac{3x^2}{2} + 3x \right]_{-2}^1 + \left[\frac{5x^2}{2} + 2x \right]_1^2 \\ &= \left(\frac{1}{3} + \frac{3}{2} + 3 \right) - \left(\frac{-8}{3} + 6 - 6 \right) + \left(10 + 4 - \frac{5}{2} - 2 \right) \\ &= 6 + \frac{3}{2} + 12 - \frac{5}{2} = 17 \end{aligned}$$

Question54

Consider the function $f : (0, \infty) \rightarrow \mathbb{R}$ defined by $f(x) = e^{-|\log_e x|}$. If m and n be respectively the number of points at which f is not continuous and f is not differentiable, then m+n is

[31-Jan-2024 Shift 2]

Options:

A.

0

B.

3

C.

1

D.

2

Answer: C

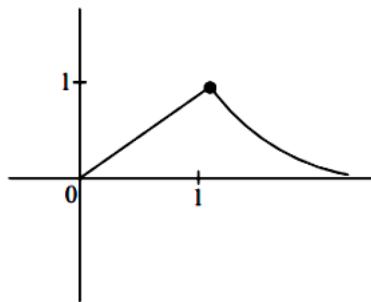
Solution:

$$f : (0, \infty) \rightarrow \mathbb{R}$$

$$f(x) = e^{-|\log_e x|}$$

$$f(x) = \frac{1}{e^{|\ln x|}} = \begin{cases} \frac{1}{e^{-\ln x}}; & 0 < x < 1 \\ \frac{1}{e^{\ln x}}; & x \geq 1 \end{cases}$$

$$\begin{cases} \frac{1}{\frac{1}{x}} = x; & 0 < x < 1 \\ \frac{1}{x}, & x \geq 1 \end{cases}$$



$m = 0$ (No point at which function is not continuous)

$n = 1$ (Not differentiable)

$$\therefore m + n = 1$$

Question 55

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} \frac{a - b \cos 2x}{x^2} & ; \quad x < 0 \\ x^2 + cx + 2 & ; \quad 0 \leq x \leq 1 \\ 2x + 1 & ; \quad x > 1 \end{cases}$$

If f is continuous everywhere in $\mathbb{R}$ and m is the number of points where f is NOT differentiable then $m + a + b + c$ equals :

[1-Feb-2024 Shift 1]

Options:

A.

1

B.

4

C.

3

D.

2

Answer: D

Solution:

At $x = 1$, $f(x)$ is continuous therefore,

$$f(1) = f(1) = f(1^+)$$

$$f(1) = 3 + c \dots\dots(1)$$

$$f(1^+) = \lim_{h \rightarrow 0} 2(1+h) + 1$$

$$f(1^+) = \lim_{h \rightarrow 0} 3 + 2h = 3 \dots\dots(2)$$

from (1) & (2)

$$c = 0$$

at $x = 0$, $f(x)$ is continuous therefore,

$$f(0^-) = f(0) = f(0^+) \dots\dots(3)$$

$$f(0) = f(0^+) = 2 \dots\dots(4)$$

$f(0^-)$ has to be equal to 2

$$\lim_{h \rightarrow 0} \frac{a - b \cos(2h)}{h^2}$$

$$\lim_{h \rightarrow 0} \frac{a - b \left\{ 1 - \frac{4h^2}{2!} + \frac{16h^4}{4!} + \dots \right\}}{h^2}$$

$$\lim_{h \rightarrow 0} \frac{a - b + b \left\{ 2h^2 - \frac{2}{3}h^4 \dots \right\}}{h^2}$$

for limit to exist $a - b = 0$ and limit is $2b \dots\dots(5)$

from (3), (4) & (5)

$$a = b = 1$$

checking differentiability at $x = 0$

$$\text{LHD : } \lim_{h \rightarrow 0} \frac{\frac{1 - \cos 2h}{h^2} - 2}{-h}$$

$$\lim_{h \rightarrow 0} \frac{1 - \left(1 - \frac{4h^2}{2!} + \frac{16h^4}{4!} \dots\right) - 2h^2}{-h^3} = 0$$

$$\text{RHD : } \lim_{h \rightarrow 0} \frac{(0+h)^2 + 2 - 2}{h} = 0$$

Function is differentiable at every point in its domain

$$\therefore m = 0$$

$$m + a + b + c = 0 + 1 + 1 + 0 = 2$$

Question56

Let $f(x) = 2|x^2 + 5|x| - 3|$, $x \in \mathbb{R}$. If m and n denote the number of points where f is not continuous and not differentiable respectively, then $m + n$ is equal to :

[1-Feb-2024 Shift 2]

Options:

A.

5

B.

2

C.

0

D.

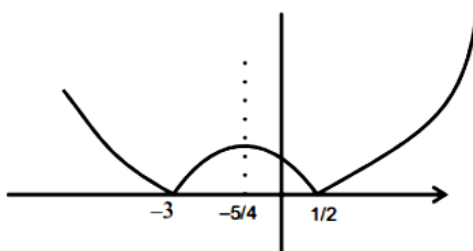
3

Answer: D

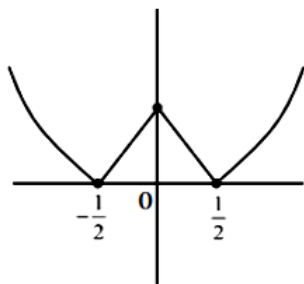
Solution:

$$f(x) = 2x^2 + 5|x| - 3|$$

Graph of $y = |2x^2 + 5x - 3|$



Graph of $f(x)$



Number of points of discontinuity = $0 = m$

Number of points of non-differentiability = $3 = n$

Question 57

Let $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$.; Then at $x = 0$

[24-Jan-2023 Shift 1]

Options:

- A. f is continuous but not differentiable
- B. f is continuous but f' is not continuous
- C. f and f' both are continuous
- D. f' is continuous but not differentiable

Answer: B

Solution:

Continuity of $f(x)$: $f(0^+) = h^2 \cdot \sin \frac{1}{h} = 0$

$$f(0^-) = (-h)^2 \cdot \sin \left(\frac{-1}{h} \right) = 0$$

$$f(0) = 0$$

$f(x)$ is continuous

$$f'(0^+) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \frac{h^2 \cdot \sin \left(\frac{1}{h} \right) - 0}{h} = 0$$

$$f'(0^-) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \frac{h^2 \cdot \sin \left(\frac{1}{-h} \right) - 0}{-h} = 0$$

$f(x)$ is differentiable.

$$f'(x) = 2x \cdot \sin\left(\frac{1}{x}\right) + x^2 \cdot \cos\left(\frac{1}{x}\right) \cdot \frac{-1}{x^2}$$

$$f'(x) = \begin{cases} 2x \cdot \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}.$$

(x) is not continuous (as $\cos\left(\frac{1}{x}\right)$ is highly oscillating at $x = 0$)

Question 58

If the function

$$f(x) = \begin{cases} (1 + |\cos x|) \frac{\lambda}{|\cos x|}, & 0 < x < \frac{\pi}{2} \\ \mu, & x = \frac{\pi}{2} \\ \frac{\cot 6x}{e^{\cot 4x}}, & \frac{\pi}{2} < x < \pi \end{cases}.$$

is continuous at $x = \frac{\pi}{2}$, then $9\lambda + 6\log_e \mu + \mu^6 - e^{6\lambda}$ is equal to
[25-Jan-2023 Shift 2]

Options:

- A. 11
- B. 8
- C. $2e^4 + 8$
- D. 10

Answer: D

Solution:

$$\Rightarrow \lim_{x \rightarrow \frac{\pi}{2}^+} \frac{\cot 6x}{e^{\cot 4x}} = \lim_{x \rightarrow \frac{\pi}{2}^+} e^{\frac{\sin 4x \cdot \cos 6x}{\sin 6x \cdot \cos 4x}} = e^{2/3}$$

$$\Rightarrow \lim_{x \rightarrow \frac{\pi}{2}^-} \left(1 + |\cos x| \right) \frac{\lambda}{|\cos x|} = e^\lambda.$$

$$\Rightarrow f(\pi/2) = \mu$$

For continuous function $\Rightarrow e^{2/3} = e^\lambda = \mu$

$$\lambda = \frac{2}{3}, \mu = e^{2/3}$$

$$\text{Now, } 9\lambda + 6\log_e \mu + \mu^6 - e^{6\lambda} = 10$$

Question 59

Let $a \in \mathbb{Z}$ and $[t]$ be the greatest integer $\leq t$. Then the number of points, where the function $f(x) = [a + 13 \sin x]$, $x \in (0, \pi)$ is not differentiable, is
[6-Apr-2023 shift 1]

Answer: 25

Solution:

$$\begin{aligned} f(x) &= [a + 13 \sin x] = a + [13 \sin x] \text{ in } (0, \pi) \\ x &\in (0, \pi) \\ \Rightarrow 0 &< 13 \sin x \leq 13 \\ \Rightarrow [13 \sin x] &= \{0, 1, 2, 3, \dots, 12, 13\} \\ \text{Total point of N.D.} &= 25. \end{aligned}$$

Question 60

Let $f : (-2, 2) \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x[x] & -2 < x < 0 \\ (x-1)[x] & 0 \leq x < 2 \end{cases}.$$

where $[x]$ denotes the greatest integer function. If m and n respectively are the number of points in $(-2, 2)$ at which $y = |f(x)|$ is not continuous and not differentiable, then $m + n$ is equal to _____.
[10-Apr-2023 shift 1]

Answer: 4

Solution:

$$f(x) = \begin{cases} -2x & -2 < x < -1 \\ -x & -1 \leq x < 0 \\ 0 & 0 \leq x < 1 \\ x-1 & 1 \leq x < 2 \end{cases}.$$

Clearly $f(x)$ is discontinuous at $x = -1$ also non differentiable.

$$\therefore m = 1$$

Now for differentiability

$$f'(x) = \begin{cases} -2 & -2 < x < -1 \\ -1 & -1 < x < 0 \\ 0 & 0 < x < 1 \\ -1 & 1 < x < 2 \end{cases}.$$

Clearly $f(x)$ is non-differentiable at $x = -1, 0, 1$

Also, $|f(x)|$ remains same.

$$\therefore n = 3$$

$$\therefore m + n = 4$$

Question 61

Let $f(x) = [x^2 - x] + |-x + [x]|$, where $x \in \mathbb{R}$ and $[t]$ denotes the greatest integer less than or equal to t . Then, f is :

[11-Apr-2023 shift 1]

Options:

- A. not continuous at $x = 0$ and $x = 1$
- B. continuous at $x = 0$ and $x = 1$
- C. continuous at $x = 1$, but not continuous at $x = 0$
- D. continuous at $x = 0$, but not continuous at $x = 1$

Answer: C

Solution:

Here $f(x) = [x(x-1)] + \{x\}$
 $f(0^+) = -1 + 0 = -1$ $f(1^+) = 0 + 0 = 0$
 $f(0) = 0$ $f(1) = 0$
 $f(1^-) = -1 + 1 = 0$
 $\therefore f(x)$ is continuous at $x = 1$, discontinuous at $x = 0$

Question62

Let f and g be two functions defined by $f(x) = \begin{cases} x+1 & x < 0 \\ |x-1|, & x \geq 0 \end{cases}$ and $g(x) = \begin{cases} x+1 & x < 0 \\ 1 & x \geq 0 \end{cases}$. Then $(g \circ f)(x)$ is

[11-Apr-2023 shift 2]

Options:

- A. continuous everywhere but not differentiable at $x = 1$
- B. continuous everywhere but not differentiable exactly at one point
- C. differentiable everywhere
- D. not continuous at $x = -1$

Answer: B

Solution:

$$f(x) = \begin{cases} x+1 & x < 0 \\ 1-x & 0 \leq x < 1 \\ x-1 & 1 \leq x \end{cases}$$

$$g(x) = \begin{cases} x+1 & x < 0 \\ 1 & x \geq 0 \end{cases}$$

$$g(f(x)) = \begin{cases} x+2 & x < -1 \\ 1 & x \geq -1 \end{cases}$$

$\therefore g(f(x))$ is continuous everywhere
 $g(f(x))$ is not differentiable at $x = -1$
Differentiable everywhere else

Question63

Let $[x]$ be the greatest integer $\leq x$. Then the number of points in the interval $(-2, 1)$, where the function $f(x) = |[x]| + \sqrt{x - [x]}$ is discontinuous, is _____.

[12-Apr-2023 shift 1]

Answer: 2

Solution:

Need to check at doubtful points

discont at $x = 1$ only

$$\text{at } x = -1 \Rightarrow f(-1^+) = 1 + 0 = 1$$

$$\Rightarrow f(-1^-) = 2 + 1 = 3$$

$$\text{at } x = 0 \Rightarrow f(0^+) = 0 + 0 = 0$$

$$\Rightarrow f(0^-) = 1 + 1 = 2$$

$$\text{at } x = 1 \Rightarrow f(1^+) = 1 + 0 = 1$$

$$\Rightarrow f(1^-) = 0 + 1 = 1$$

discont. at two points

Question 64

Let $[x]$ denote the greatest integer function and $f(x) = \max\{1 + x + [x], 2 + x, x + 2[x]\}$, $0 \leq x \leq 2$. Let m be the number of points in $[0, 2]$, where f is not continuous and n be the number of points in $(0, 2)$, where f is not differentiable. Then $(m + n)^2 + 2$ is equal to
[15-Apr-2023 shift 1]

Options:

A. 6

B. 3

C. 2

D. 11

Answer: B

Solution:

$$\text{Let } g(x) = 1 + x + [x] = \begin{cases} 1 + x; & x \in [0, 1) \\ 2 + x; & x \in [1, 2) \\ 5; & x = 2 \end{cases}$$

$$h(x) = x + 2[x] = \begin{cases} x; & x \in [0, 1) \\ x + 2; & x \in [1, 2) \\ 6; & x = 2 \end{cases}$$

$$r(x) = 2 + x$$

$$f(x) = \begin{cases} 2 + x; & x \in [0, 2) \\ 6; & x = 2 \end{cases}$$

$f(x)$ is discontinuous only at $x = 2 \Rightarrow m = 1$

$f(x)$ is differentiable in $(0, 2) \Rightarrow n = 0$

$$(m + n)^2 + 2 = 3$$

Question 65

$$\text{Let } f(x) = \begin{cases} \frac{\sin(x - [x])}{x - [x]} & x \in (-2, -1) \\ \max\{2x, 3[x]\} & |x| < 1 \\ 1 & \text{otherwise} \end{cases} \quad \text{where } [t] \text{ denotes greatest integer } \leq t. \text{ If } m \text{ is the number of points}$$

where f is not continuous and n is the number of points where f is not differentiable, then the ordered

pair (m, n) is :
[24-Jun-2022-Shift-2]

Options:

- A. (3, 3)
- B. (2, 4)
- C. (2, 3)
- D. (3, 4)

Answer: C

Solution:

$$f(x) = \begin{cases} \frac{\sin(x - [x])}{x[x]} & x \in (-2, -1) \\ \max\{2x, 3[[x]]\} & |x| < 1 \\ 1 & \text{otherwise} \end{cases}$$

$$f(x) = \begin{cases} \frac{\sin(x+2)}{x+2} & x \in (-2, -1) \\ 0 & x \in (-1, 0] \\ 2x & x \in (0, 1) \\ 1 & \text{otherwise} \end{cases}$$

It clearly shows that $f(x)$ is discontinuous

At $x = -1, 1$ also non differentiable

$$\text{and at } x = 0, L.H.D = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = 0$$

$$R.H.D = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = 2$$

$\therefore f(x)$ is not differentiable at $x = 0$

$$\therefore m = 2, n = 3$$

Question 66

Let $f(x) = [2x^2 + 1]$ and $g(x) = \begin{cases} 2x-3 & x < 0 \\ 2x+3 & x \geq 0 \end{cases}$, where $[t]$ is the greatest integer $\leq t$. Then, in the open interval $(-1, 1)$, the number of points where $f \circ g$ is discontinuous is equal to _____
[25-Jun-2022-Shift-2]

Answer: 62

Solution:

$$f(g(x)) = [2g^2(x)] + 1$$

$$= \begin{cases} [2(2x-3)^2] + 1; & x < 0 \\ [2(2x+3)^2] + 1; & x \geq 0 \end{cases}$$

$\therefore f \circ g$ is discontinuous whenever $2(2x-3)^2$ or $2(2x+3)^2$ belongs to integer except $x = 0$.

$\therefore 62$ points of discontinuity.

Question67

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be two real valued functions defined as $f(x) = \begin{cases} -|x+3| & x < 0 \\ e^x & x \geq 0 \end{cases}$

$$\text{and } g(x) = \begin{cases} x^2 + k_1 x & x < 0 \\ 4x + k_2 & x \geq 0 \end{cases}$$

where k_1 and k_2 are real constants. If $(g \circ f)$ is differentiable at $x = 0$, then $(g \circ f)(-4) + (g \circ f)(4)$ is equal to
:
[26-Jun-2022-Shift-1]

Options:

- A. $4(e^4 + 1)$
- B. $2(2e^4 + 1)$
- C. $4e^4$
- D. $2(2e^4 - 1)$

Answer: D

Solution:

$\because$ $g \circ f$ is differentiable at $x = 0$

So R.H.D = L.H.D

$$\frac{d}{dx}(4e^x + k_2) = \frac{d}{dx}((-|x+3|)^2 - k_1|x+3|)$$

$$\Rightarrow 4 = 6 - k_1 \Rightarrow k_1 = 2$$

$$\text{Also } f'(f(0^+)) = g'(f(0^-))$$

$$\Rightarrow 4 - k_2 = 9 - 3k_1 \Rightarrow k_2 = -1$$

$$\text{Now } g(f(-4)) + g(f(4))$$

$$= g(-1) - g(e^4) = (1 - k_1) + (4e^4 + k_2)$$

$$= 4e^4 - 2$$

$$= 2(2e^4 - 1)$$

Question68

Let $f(x) = \min\{1, 1 + x \sin x\}$, $0 \leq x \leq 2\pi$. If m is the number of points, where f is not differentiable and n is the number of points, where f is not continuous, then the ordered pair (m, n) is equal to
[26-Jun-2022-Shift-2]

Options:

- A. $(2, 0)$
- B. $(1, 0)$
- C. $(1, 1)$
- D. $(2, 1)$

Answer: B

Solution:

$$f(x) = \min\{1, 1 + x \sin x\}, 0 \leq x \leq \pi$$

$$f(x) = \begin{cases} 1 & 0 \leq x < \pi \\ 1 + x \sin x & \pi \leq x \leq 2\pi \end{cases}$$

$$\text{Now at } x = \pi, \lim_{x \rightarrow \pi^-} f(x) = 1 = \lim_{x \rightarrow \pi^+} f(x)$$

$\therefore f(x)$ is continuous in $[0, 2\pi]$

$$\text{Now, at } x = \pi, \lim_{h \rightarrow 0} \frac{f(\pi - h) - f(\pi)}{-h} = 0$$

$$\lim_{h \rightarrow 0} \frac{f(\pi + h) - f(\pi)}{h} = 1 - \frac{(\pi + h) \sin h - 1}{h} = -\pi$$

$\therefore f(x)$ is not differentiable at $x = \pi$

$$\therefore (m, n) = (1, 0)$$

Question 69

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} [e^x], & x < 0 \\ ae^x + [x - 1], & 0 \leq x < 1 \\ b + [\sin(\pi x)], & 1 \leq x < 2 \\ [e^{-x}] - c, & x \geq 2. \end{cases}$$

where $a, b, c \in \mathbb{R}$ and $[t]$ denotes greatest integer less than or equal to t . Then, which of the following statements is true?

[28-Jun-2022-Shift-1]

Options:

- A. There exists $a, b, c \in \mathbb{R}$ such that f is continuous on $\mathbb{R}$.
- B. If f is discontinuous at exactly one point, then $a + b + c = 1$
- C. If f is discontinuous at exactly one point, then $a + b + c \neq 1$
- D. f is discontinuous at at least two points, for any values of a, b and c

Answer: C

Solution:

$$f(x) = \begin{cases} 0 & x < 0 \\ ae^x - 1 & 0 \leq x < 1 \\ b & x = 1 \\ b - 1 & 1 < x < 2 \\ -c & x \geq 2 \end{cases}$$

To be continuous at $x = 0$

$$a - 1 = 0$$

to be continuous at $x = 1$

$$ae - 1 = b = b - 1 \Rightarrow \text{not possible}$$

to be continuous at $x = 2$

$$b - 1 = -c \Rightarrow b + c = 1$$

If $a = 1$ and $b + c = 1$ then $f(x)$ is discontinuous at exactly one point.

Question 70

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be functions defined by

$$f(x) = \begin{cases} [x], & x < 0 \\ |1-x|, & x \geq 0. \end{cases} \text{ and}$$

$$g(x) = \begin{cases} e^x - x, & x < 0 \\ (x-1)^2 - 1, & x \geq 0. \end{cases}$$

where $[x]$ denote the greatest integer less than or equal to x . Then, the function $f \circ g$ is discontinuous at exactly :

[28-Jun-2022-Shift-2]

Options:

- A. one point
- B. two points
- C. three points
- D. four points

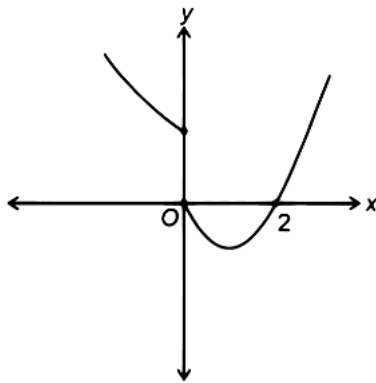
Answer: B

Solution:

$$f(x) = \begin{cases} [x], & x < 0 \\ |1-x|, & x \geq 0 \end{cases}$$

$$\text{and } g(x) = \begin{cases} e^x - x, & x < 0 \\ (x-1)^2 - 1, & x \geq 0 \end{cases}$$

$$f \circ g(x) = \begin{cases} [g(x)], & g(x) < 0 \\ |1-g(x)|, & g(x) \geq 0 \end{cases}.$$



(graph of $y = g(x)$)

$$= \begin{cases} |1+x-e^x|, & x < 0 \\ 1, & x = 0 \\ [(x-1)^2 - 1], & 0 < x < 2 \\ |2-(x-1)^2|, & x \geq 2 \end{cases}$$

So, $x = 0, 2$ are the two points where $f \circ g$ is discontinuous.

Question71

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by :

$$f(x) = \begin{cases} \max\{t^3 - 3t\}; & x \leq 2 \\ t \leq x & \\ x^2 + 2x - 6; & 2 < x < 3 \\ [x - 3] + 9; & 3 \leq x \leq 5 \\ 2x + 1; & x > 5. \end{cases}$$

where $[t]$ is the greatest integer less than or equal to t . Let m be the number of points where f is not differentiable and $I = \int_{-2}^2 f(x) dx$. Then the ordered pair (m, I) is equal to :

[29-Jun-2022-Shift-1]

Options:

A. $\left(3, \frac{27}{4}\right)$

B. $\left(3, \frac{23}{4}\right)$

C. $\left(4, \frac{27}{4}\right)$

D. $\left(4, \frac{23}{4}\right)$

Answer: C

Solution:

$$f(x) = \begin{cases} x^3 - 3x & x \leq -1 \\ 2 & -1 < x < 2 \\ x^2 + 2x - 6 & 2 < x < 3 \\ 9 & 3 \leq x < 4 \\ 10 & 4 \leq x < 5 \\ 11 & x = 5 \\ 2x + 1 & x > 5 \end{cases}$$

Clearly $f(x)$ is not differentiable at

$$x = 2, 3, 4, 5 \Rightarrow m = 4$$

$$I = \int_{-2}^{-1} (x^3 - 3x) dx + \int_{-1}^2 2 \cdot dx = \frac{27}{4}$$

Question 72

The number of points where the function

$$f(x) = \begin{cases} |2x^2 - 3x - 7| & \text{if } x \leq -1 \\ [4x^2 - 1] & \text{if } -1 < x < 1 \\ |x + 1| + |x - 2| & \text{if } x \geq 1. \end{cases}$$

$[t]$ denotes the greatest integer $\leq t$, is discontinuous is

[24-Jun-2022-Shift-1]

Answer: 7

Solution:

$$\therefore f(-1) = 2 \text{ and } f(1) = 3$$

$$\text{For } x \in (-1, 1), (4x^2 - 1) \in [-1, 3)$$

hence $f(x)$ will be discontinuous at $x = 1$ and also

$$\text{whenever } 4x^2 - 1 = 0, 1 \text{ or } 2$$

$$\Rightarrow x = \pm \frac{1}{2}, \pm \frac{1}{\sqrt{2}} \text{ and } \pm \frac{\sqrt{3}}{2}$$

So there are total 7 points of discontinuity.

Question 73

$$\text{Let } f(x) = \begin{cases} |4x^2 - 8x + 5|, & \text{if } 8x^2 - 6x + 1 \geq 0 \\ [4x^2 - 8x + 5], & \text{if } 8x^2 - 6x + 1 < 0 \end{cases},$$

where $[a]$ denotes the greatest integer less than or equal to a . Then the number of points in $\mathbb{R}$ where f is not differentiable is

[25-Jul-2022-Shift-1]

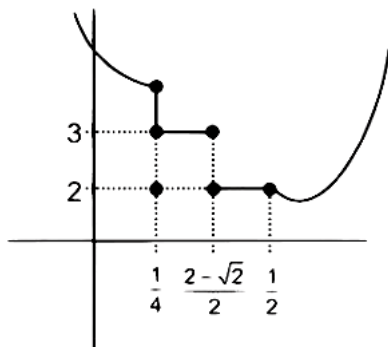
Answer: 3

Solution:

$$f(x) = \begin{cases} |4x^2 - 8x + 5| & \text{if } 8x^2 - 6x + 1 \geq 0 \\ [4x^2 - 8x + 5], & \text{if } 8x^2 - 6x + 1 < 0. \end{cases}$$

$$= \begin{cases} 4x^2 - 8x + 5 & \text{if } x \in \left[-\infty, \frac{1}{4}\right] \cup \left[\frac{1}{2}, \infty\right) \\ [4x^2 - 8x + 5] & \text{if } x \in \left(\frac{1}{4}, \frac{1}{2}\right). \end{cases}$$

$$f(x) = \begin{cases} 4x^2 - 8x + 5 & \text{if } x \in \left(-\infty, \frac{1}{4}\right] \cup \left[\frac{1}{2}, \infty\right) \\ 3 & x \in \left(\frac{1}{4}, \frac{2-\sqrt{2}}{2}\right) \\ 2 & x \in \left[\frac{2-\sqrt{2}}{2}, \frac{1}{2}\right). \end{cases}$$



$$\therefore \text{Non-diff at } x = \frac{1}{4}, \frac{2-\sqrt{2}}{2}, \frac{1}{2}$$

Question 74

If $f(x) = \begin{cases} x+a, & x \leq 0 \\ |x-4|, & x > 0. \end{cases}$ and $g(x) = \begin{cases} x+1, & x < 0 \\ (x-4)^2+b, & x \geq 0. \end{cases}$ are continuous on $\mathbb{R}$, then $(g \circ f)(2) + (f \circ g)(-2)$ is

equal to :

[26-Jul-2022-Shift-1]

Options:

A. -10

B. 10

C. 8

D. -8

Answer: D

Solution:

$\because f(x)$ and $g(x)$ are continuous on $\mathbb{R}$

$\therefore a = 4$ and $b = 1 - 16 = -15$

then $(g \circ f)(2) + (f \circ g)(-2)$

$= g(2) + f(-1)$

$= -11 + 3 = -8$

Question 75

If for $p \neq q \neq 0$, the function $f(x) = \frac{\sqrt[7]{p(729+x)} - 3}{\sqrt[3]{729+qx} - 9}$ is continuous at $x = 0$, then :

[27-Jul-2022-Shift-2]

Options:

A. $7pqf(0) - 1 = 0$

B. $63qf(0) - p^2 = 0$

C. $21qf(0) - p^2 = 0$

D. $7pqf(0) - 9 = 0$

Answer: B

Solution:

$$f(x) = \frac{\sqrt[7]{p(729+x)} - 3}{\sqrt[3]{729+qx} - 9}$$

for continuity at $x = 0$, $\lim_{x \rightarrow 0} f(x) = f(0)$

$$\text{Now, } \therefore \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\sqrt[7]{p(729+x)} - 3}{\sqrt[3]{729+qx} - 9}$$

$\Rightarrow p = 3$ (To make indeterminate form)

$$\text{So, } \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{(3^7 + 3x)^{\frac{1}{7}} - 3}{(729 + qx)^{\frac{1}{3}} - 9}$$

$$= \lim_{x \rightarrow 0} \frac{3 \left[\left(1 + \frac{x}{3^6} \right)^{\frac{1}{7}} - 1 \right]}{9 \left[\left(1 + \frac{q}{729}x \right)^{\frac{1}{3}} - 1 \right]} = \frac{1}{3} \cdot \frac{\frac{1}{7} \cdot \frac{1}{3^6}}{\frac{1}{3} \cdot \frac{q}{729}}$$

$$\therefore f(0) = \frac{1}{7q}$$

$\therefore$ Option (B) is correct.

Question76

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$f(x) = \lim_{n \rightarrow \infty} \frac{\cos(2\pi x) - x^{2n} \sin(x-1)}{1 + x^{2n+1} - x^{2n}}$ is continuous for all x in :

[28-Jul-2022-Shift-2]

Options:

A. $\mathbb{R} - \{-1\}$

B. $\mathbb{R} - \{-1, 1\}$

C. $\mathbb{R} - \{1\}$

D. $\mathbb{R} - \{0\}$

Answer: B

Solution:

$$f(x) = \lim_{n \rightarrow \infty} \frac{\cos(2\pi x) - x^{2n} \sin(x-1)}{1 + x^{2n+1} - x^{2n}}$$

For $|x| < 1$, $f(x) = \cos 2\pi x$, continuous function

$$|x| > 1, f(x) = \lim_{n \rightarrow \infty} \frac{\frac{1}{x^{2n}} \cos 2\pi x - \sin(x-1)}{\frac{1}{x^{2n}} + x - 1}$$

$$= \frac{-\sin(x-1)}{x-1}, \text{ continuous}$$

$$\text{For } |x| = 1, f(x) = \begin{cases} 1 & \text{if } x = 1 \\ -(1 + \sin 2) & \text{if } x = -1. \end{cases}$$

Now,

$$\lim_{x \rightarrow 1^+} f(x) = -1, \lim_{x \rightarrow 1^-} f(x) = 1, \text{ so discontinuous at } x = 1$$

$$\lim_{x \rightarrow 1^+} f(x) = 1, \lim_{x \rightarrow -1^-} f(x) = -\frac{\sin 2}{2}, \text{ so discontinuous at } x = -1$$

$\therefore f(x)$ is continuous for all $x \in \mathbb{R} - \{-1, 1\}$

Question77

The number of points, where the function

$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = |x-1| \cos |x-2| \sin |x-1| + (x-3)x^2 - 5x + 4|$, is NOT differentiable, is :
[29-Jul-2022-Shift-1]

Options:

A. 1

B. 2

C. 3

D. 4

Answer: B

Solution:

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = |x-1| \cos |x-2| \sin |x-1| + (x-3)x^2 - 5x + 4|$$

$$= |x-1| \cos |x-2| \sin |x-1| + (x-3)|x-1||x-4|$$

$$= |x-1| [\cos |x-2| \sin |x-1| + (x-3)|x-4|]$$

Sharp edges at $x = 1$ and $x = 4$

$\therefore$ Non-differentiable at $x = 1$ and $x = 4$

Question78

Let the function $f(x) = \begin{cases} \frac{\log_e(1+5x) - \log_e(1+\alpha x)}{x} & ; \text{ if } x \neq 0 \\ 10 & ; \text{ if } x = 0. \end{cases}$ be continuous at $x = 0$.

Then α is equal to
[29-Jul-2022-Shift-2]

Options:

- A. 10
- B. -10
- C. 5
- D. -5

Answer: D

Solution:

$f(x)$ is continuous at $x = 0$

$$\therefore f(0) = \lim_{x \rightarrow 0} f(x)$$

$$\Rightarrow 10 = \lim_{x \rightarrow 0} \frac{\log_e(1+5x) - \log_e(1+\alpha x)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\log(1+5x)}{5x} \times 5 - \frac{\log_e(1+\alpha x)}{\alpha x} \times \alpha$$

$$= 1 \times 5 - \alpha$$

$$\Rightarrow \alpha = 5 - 10 = -5$$

Question79

If $[t]$ denotes the greatest integer $\leq t$, then the number of points, at which the function

$f(x) = 4 \left| 2x + 3 \right| + 9 \left[x + \frac{1}{2} \right] - 12[x + 20]$ is not differentiable in the open interval $(-20, 20)$, is _____.

[29-Jul-2022-Shift-2]

Answer: 79

Solution:

$$f(x) = 4 \left| 2x + 3 \right| + 9 \left[x + \frac{1}{2} \right] - 12[x + 20]$$

$$= 4 \left| 2x + 3 \right| + 9 \left[x + \frac{1}{2} \right] - 12[x] - 240$$

$$f(x) \text{ is non differentiable at } x = -\frac{3}{2}$$

and $f(x)$ is discontinuous at $\{-19, -18, \dots, 18, 19\}$

as well as $\left\{ -\frac{39}{2}, -\frac{37}{2}, \dots, -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \dots, \frac{39}{2} \right\}$,

at same point they are also non differentiable

$\therefore$ Total number of points of non differentiability

$$= 39 + 40$$

$$= 79$$

Question80

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = \sin \pi x$. If f is continuous on $\mathbb{R}$, then $a + b$ equals
[2021, 26 Feb. Shift-11]

Options:

- A. -3
- B. -1
- C. 3
- D. 1

Answer: B

Solution:

Given, $f(x)$ is continuous on $\mathbb{R}$.

If $f(x)$ is continuous, then f is continuous at $x = 1$

$$\Rightarrow \lim_{x \rightarrow 1^-} f(x) = f(1) = \lim_{x \rightarrow 1^+} f(x)$$

$$\Rightarrow |a + 1 + b| = \sin \pi = 0$$

$$\Rightarrow a + b = -1 \dots (i)$$

Also, f is continuous at $x = -1$

$$\Rightarrow \lim_{x \rightarrow -1^-} f(x) = f(-1) = \lim_{x \rightarrow -1^+} f(x)$$

$$\Rightarrow 2 \sin \left(\frac{-\pi}{2}(-1) \right) = |a - 1 + b|$$

$$\Rightarrow 2 = |a + b - 1| \dots (ii)$$

Eq. (ii) is satisfied.

$$\therefore a + b = -1$$

Question81

Let f be any function defined on $\mathbb{R}$ and let it satisfy the condition $|f(x) - f(y)| \leq |x - y|^2, \forall (x, y) \in \mathbb{R}$. If $f(0) = 1$, then
[2021, 26 Feb. Shift-1]

Options:

- A. $f(x)$ can take any value in $\mathbb{R}$
- B. $f(x) < 0, \forall x \in \mathbb{R}$
- C. $f(x) = 0, \forall x \in \mathbb{R}$
- D. $f(x) > 0, \forall x \in \mathbb{R}$

Answer: D

Solution:

Given, $|f(x) - f(y)| \leq |x - y|^2$

$$\Rightarrow \frac{|f(x) - f(y)|}{|x - y|} \leq |x - y|$$

Now, taking the limit,

$$\lim_{x \rightarrow y} \left| \frac{f(x) - f(y)}{x - y} \right| \leq \lim_{x \rightarrow y} |x - y|$$

$$\Rightarrow |f'(y)| \leq 0 \text{ [using the definition of } f'(y) \text{]}$$

$$\Rightarrow f'(y) = 0 \text{ [since, modulus value can never be less than 0]}$$

On integrating it, we get

$f(y) = c$ (constant)

Given, $f(0) = 1$ gives $c = 1$

$$\therefore f(y) = 1 \forall y \in \mathbb{R}$$

From given options, $f(x) > 0 \forall x \in \mathbb{R}$ is satisfied only.

Question82

Let $f(x)$ be a differentiable function at $x = a$ with $f'(a) = 2$ and $f(a) = 4$. Then, $\lim_{x \rightarrow a} \frac{xf(a) - af(x)}{x - a}$ equals
[2021, 26 Feb. Shift-II]

Options:

- A. $2a + 4$
- B. $4 - 2a$
- C. $2a - 4$
- D. $a + 4$

Answer: B

Solution:

$$\begin{aligned}
 \lim_{x \rightarrow a} \frac{xf(a) - af(x)}{x - a} &= \lim_{x \rightarrow a} \frac{xf(a) - af(x) + af(a) - af(a)}{x - a} \\
 &= \lim_{x \rightarrow a} \frac{(x - a)f(a) - a[f(x) - f(a)]}{x - a} \\
 &= \lim_{x \rightarrow a} \frac{(x - a)f(a)}{x - a} - a \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\
 &= f(a) - af'(a) \\
 &= 4 - a(2) \quad [\text{Given, } f(a) = 4, f'(a) = 2] \\
 &= 4 - 2a
 \end{aligned}$$

Question83

A function f is defined on $[-3, 3]$ as $f(x) = \begin{cases} \min\{|x|, 2 - x^2\} & -2 \leq x \leq 2 \\ [x] & , 2 < |x| \leq 3. \end{cases}$

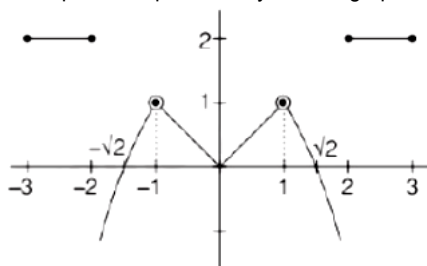
where, $[x]$ denotes the greatest integer $\leq x$. The number of points, where f is not differentiable in $(-3, 3)$ is

[2021, 25 Feb. Shift-II]

Answer: 5

Solution:

For this particular problem, try to draw graph in the region $(-3, 3)$, it will be as follows,



Thus, points of discontinuity will be at $-2, 2$ because the curve breaks at these points and at $-1, 0, 1$ because curve has sharp points.
 $\therefore$ Point of discontinuity are $-2, -1, 0, 1, 2$ i.e. 5 points.

Question84

The number of points at which the function $f(x) = |2x + 1| - 3x + 2 + x^2 + x - 2| x \in \mathbb{R}$ is not differentiable, is
[2021, 25 Feb. Shift-1]

Answer: 2

Solution:

Given,

$$f(x) = |2x + 1| - 3x + 2 + x^2 + x - 2|$$

$$= |2x + 1| - 3x + 2 + |x + 2| |xx - 1|$$

Here, critical points are $x = \frac{-1}{2}, -2, 1$

$$\therefore f(x) = \begin{cases} x^2 + 2x + 3 & x < -2 \\ -x^2 - 6x - 5 & -2 < x < \frac{-1}{2} \\ -x^2 - 2x - 3 & \frac{-1}{2} < x < 1. \end{cases}$$

$$\text{Now, } f'(x) = \begin{cases} 2x + 2 & x < -2 \\ -2x - 6 & -2 < x < \frac{-1}{2} \\ -2x - 2 & \frac{-1}{2} < x < 1 \\ 2x & x > 1. \end{cases}$$

Now, $f'(x)$ at 1, -2 and $-1/2$.

For $x = 1$,

$$f'(x) = 2x = 2 \times 1 = 2$$

and $-2x - 2 = -(2 \times 1) - 2 = -4$ both are not equal.

$\therefore$ Non-differentiable at $x = 1$

Similarly, for $x = -2$,

$$f'(x) = 2x + 2 = 2 \times (-2) + 2 = -2$$

and $-2x - 6 = -2 \times (-2) - 6 = -2$ both are equal.

$\therefore$ Differentiable at $x = -2$

and for $x = -1/2$, $f'(x) = -2x - 6$

$$= -2 \times \left(\frac{-1}{2} \right) - 6 = -5 \text{ and}$$

$$-2x - 2 = -2 \times \left(\frac{-1}{2} \right) - 2 = -1 \text{ both are not equal.}$$

$\therefore$ Non-differentiable at $x = -1/2$

$\therefore$ The number of points at which $f(x)$ is non-differentiable is 2 .

Question85

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function defined by $f(x) = [x - 1] \cos \left(\frac{2x - 1}{2} \right) \pi$, where $[.]$ denotes the greatest integer function, then f is
[2021, 24 Feb. Shift-1]

Options:

- A. discontinuous only at $x = 1$
- B. discontinuous at all integral values of x except at $x = 1$
- C. continuous only at $x = 1$
- D. continuous for every real x

Answer: D

Solution:

Given, $f(x) = [x - 1] \cos\left(\frac{2x-1}{2}\right) \pi$ where $[\cdot]$ is greatest integer function and $f : \mathbb{R} \rightarrow \mathbb{R}$

$\therefore$ It is a greatest integer function then we need to check its continuity at $x \in \mathbb{I}$ except these it is continuous.
Let $x = n$ where $n \in \mathbb{I}$

$$\begin{aligned}\text{Then, LHL} &= \lim_{x \rightarrow n^-} [x - 1] \cos\left(\frac{2x-1}{2}\right) \pi \\ &= (n-2) \cos\left(\frac{2n-1}{2}\right) \pi = 0\end{aligned}$$

$$\begin{aligned}\text{RHL} &= \lim_{x \rightarrow n^+} [x - 1] \cos\left(\frac{2x-1}{2}\right) \pi \\ &= (n-2) \cos\left(\frac{2n-1}{2}\right) \pi = 0\end{aligned}$$

and $f(n) = 0$.

Here, $\lim_{x \rightarrow n^-} f(x) = \lim_{x \rightarrow n^+} f(x) = f(n)$

$\therefore$ It is continuous at every integers.

Therefore, the given function is continuous for all real x .

Question86

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function defined by $f(x) = [x - 1] \cos\left(\frac{2x-1}{2}\right) \pi$, where $[\cdot]$ denotes the greatest integer function, then f is:
24 Feb 2021 Shift 1

Options:

- A. discontinuous at all integral values of x except at $x = 1$
- B. continuous only at $x = 1$
- C. continuous for every real x
- D. discontinuous only at $x = 1$

Answer: C

Solution:

For $x = n, n \in \mathbb{Z}$

$$\text{LHL} = \lim_{x \rightarrow n^-} f(x) = \lim_{x \rightarrow n^-} [x - 1] \cos\left(\frac{2x-1}{2}\right) \pi = 0$$

$$\text{RHL} = \lim_{x \rightarrow n^+} f(x) = \lim_{x \rightarrow n^+} [x - 1] \cos\left(\frac{2x-1}{2}\right) \pi = 0$$

$f(n) = 0$

$\Rightarrow \text{LHL} = \text{RHL} = f(n)$

$\Rightarrow f(x)$ is continuous for every real x .

Question87

If $f(x) = \begin{cases} \frac{1}{|x|} & |x| \geq 1; ax^2 + b, & |x| < 1. \end{cases}$ is differentiable at every point of the domain, then the values of a and b are respectively
[2021, 18 March shift-1]

Options:

A. $\frac{1}{2}, \frac{1}{2}$

B. $\frac{1}{2}, -\frac{3}{2}$

C. $\frac{5}{2}, -\frac{3}{2}$

D. $-\frac{1}{2}, \frac{3}{2}$

Answer: D

Solution:

$$\text{Given, } f(x) = \begin{cases} \frac{1}{|x|} & |x| \geq 1; ax^2 + b, & |x| < 1. \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} \frac{1}{|x|} & x \leq -1 \text{ or } x \geq 1 \\ ax^2 + b & -1 < x < 1. \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} \frac{-1}{x} & x \leq -1 \\ ax^2 + b & -1 < x < 1 \\ \frac{1}{x} & x \geq 1. \end{cases}$$

$$\text{Given, } f(x) \text{ is differentiable at every point of domain. } \therefore f'(x) = \begin{cases} \frac{1}{x^2} & x < -1 \\ 2ax & -1 < x < 1 \\ \frac{-1}{x^2} & x > 1. \end{cases}$$

$$\because f(x) \text{ is differentiable at } x = 1 \\ \therefore (\text{LHD at } x = 1) = (\text{RHD at } x = 1)$$

$$\Rightarrow f'(1^-) = f'(1^+)$$

$$\Rightarrow 2a = -1 \Rightarrow a = -\frac{1}{2}$$

As, we know that, a function is differentiable at $x = a$, if it is continuous at $x = a$.

Hence, $f(x)$ is also continuous at $x = 1$.

$$\text{i.e., } (\text{LHL at } x = 1) = (\text{RHL at } x = 1) = f(1)$$

$$\Rightarrow a + b = 1$$

$$\Rightarrow \left(-\frac{1}{2}\right) + b = 1$$

$$\Rightarrow b = \frac{3}{2}$$

$$\text{Hence, } a = -\frac{1}{2}, b = \frac{3}{2}$$

Note You can also (or apply) continuity and differentiability at $x = -1$.

Question 88

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as

$$f(x) = \begin{cases} \frac{\sin(a+1)x + \sin 2x}{2x} & \text{if } x < 0 \\ b & \text{if } x = 0 \\ \frac{\sqrt{x+bx^3} - \sqrt{x}}{bx^{5/2}} & \text{if } x > 0. \end{cases}$$

If f is continuous at $x = 0$, then the value of $a + b$ is equal to
[2021, 18 March Shift-II]

Options:

A. $-\frac{5}{2}$

B. -2

C. -3

D. $-\frac{3}{2}$

Answer: D

Solution:

$$\text{Given, } f(x) = \begin{cases} \frac{\sin(a+1)x + \sin 2x}{2x} & x < 0 \\ b & x = 0 \\ \frac{\sqrt{x+bx^3} - \sqrt{x}}{bx^{5/2}} & x > 0. \end{cases}$$

$\therefore f(x)$ is continuous at $x = 0$.

$$\therefore \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0) \dots (i)$$

$$\therefore f(0) = b \dots (ii)$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \left(\frac{\sin(a+1)x + \sin 2x}{2x} \right)$$

$$\Rightarrow \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \left(\frac{\sin(a+1)x}{2x} + \frac{\sin 2x}{2x} \right)$$

$$= \lim_{x \rightarrow 0^-} \left(\frac{\sin(a+1)x}{(a+1)x} \times \left(\frac{a+1}{2} \right) + \frac{\sin 2x}{2x} \right)$$

$$= \frac{a+1}{2} + 1 \dots (iii)$$

$$\text{Again, } \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \left(\frac{\sqrt{x+bx^3} - \sqrt{x}}{bx^{5/2}} \right)$$

$$= \lim_{x \rightarrow 0^+} \frac{(\sqrt{x+bx^3} - \sqrt{x})(\sqrt{x+bx^3} + \sqrt{x})}{bx^{5/2}(\sqrt{x+bx^3} + \sqrt{x})}$$

$$= \lim_{x \rightarrow 0^+} \frac{(x+bx^3-x)}{bx^{5/2}(\sqrt{x+bx^3} + \sqrt{x})}$$

$$= \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{\sqrt{x}(\sqrt{1+bx^2} + 1)}$$

$$\Rightarrow \lim_{x \rightarrow 0^+} f(x) = \frac{1}{2} \dots (iv)$$

From Eq. (i), (ii), (iii) and (iv)

$$\frac{1}{2} = b = \frac{a+1}{2} + 1$$

$$\Rightarrow b = \frac{1}{2}, a = -2$$

$$\therefore a+b = \frac{-3}{2}$$

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy the equation $f(x + y) = f(x) \cdot f(y)$ for all $x, y \in \mathbb{R}$ and $f(x) \neq 0$ for any $x \in \mathbb{R}$. If the function f is differentiable at $x = 0$ and $f'(0) = 3$, then $\lim_{h \rightarrow 0} \frac{1}{h}(f(h) - 1)$ is equal to.....

[2021, 18 March Shift-II]

Answer: 3

Solution:

Method 1

Given, $f(x + y) = f(x) \cdot f(y) \forall x, y \in \mathbb{R}$

$$\therefore f(x) = a^x$$

$$\Rightarrow f'(x) = a^x \cdot \log(a)$$

$$\text{Now, } f'(0) = \log(a)$$

$$\Rightarrow 3 = \log(a)$$

$$\Rightarrow a = e^3$$

$$\therefore f(x) = (e^3)^x = e^{3x}$$

$$f(h) = e^{3h}$$

$$\text{Now, } \lim_{h \rightarrow 0} \left(\frac{f(h) - 1}{h} \right) = \lim_{h \rightarrow 0} \left(\frac{e^{3h} - 1}{h} \right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{e^{3h} - 1}{3h} \times 3 \right)$$

$$= 3 \times 1 = 3$$

Method(2)

$$\text{Let } L = \lim_{h \rightarrow 0} \frac{1}{h}(f(h) - 1) \left(\frac{0}{0} \text{ form} \right)$$

$$f(x + y) = f(x) + f(y)$$

$$\text{Put } x = y = 0$$

$$\therefore f(0) = f(0) + f(0)$$

$$\Rightarrow [f(0)]^2 = f(0)$$

$$\Rightarrow [f(0)]^2 - f(0) = 0$$

$$\Rightarrow f(0)[f(0) - 1] = 0$$

$$\Rightarrow f(0) = 0, f(0) = 1$$

Rejected because $f(x) \neq 0, \forall x \in \mathbb{R}$

$$\therefore f(0) = 1$$

Using L-Hospital Rule,

$$L = \lim_{h \rightarrow 0} \frac{f'(h) - 0}{1}$$

$$= f'(0) = 3$$

Question90

If the function $f(x) = \frac{\cos(\sin x) - \cos x}{x^4}$ is continuous at each point in its domain and $f(0) = \frac{1}{k}$, then k is

[2021, 17 March Shift-1]

Answer: 6

Solution:

$$f(x) = \frac{\cos(\sin x) - \cos x}{x^4}$$

As, $f(x)$ is continuous everywhere, so

$$f(0) = \lim_{x \rightarrow 0} f(x)$$

$$= \lim_{x \rightarrow 0} \frac{\cos(\sin x) - \cos x}{x^4}$$

On expanding the numerator and only identifying the coefficient of x^4 will give us the required limit.

$$\begin{aligned}
\cos(\sin x) &= \left(1 - \frac{\sin^2 x}{2} + \frac{\sin^4 x}{24}\right) \\
&= 1 - \frac{1}{2} \left(x - \frac{x^3}{6}\right)^2 + \frac{1}{24} (x)^4 \\
&= 1 - \frac{1}{2} \left(x^2 - \frac{x^4}{3}\right) + \frac{x^4}{24} \\
&= 1 - \frac{x^2}{2} + \frac{x^4}{6} + \frac{x^4}{24} \\
\cos x &= 1 - \frac{x^2}{2} + \frac{x^4}{24} \\
\therefore \frac{\cos(\sin x) - \cos x}{x^4} &= \frac{\left(1 - \frac{x^2}{2} + \frac{x^4}{6} + \frac{x^4}{24}\right) - \left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right)}{x^4} \\
&= \frac{1}{6} \\
\therefore f(0) &= \frac{1}{6} = \frac{1}{k} \\
\text{Hence, } k &= 6.
\end{aligned}$$

Question91

Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by Then, f is $f(x) = \begin{cases} \left[2 - \sin\left(\frac{1}{x}\right)\right] |x|, & x \neq 0 \\ 0 & x = 0 \end{cases}$.

[2021, 17 March Shift-II]

Options:

- A. monotonic on $(-\infty, 0) \cup (0, \infty)$
- B. not monotonic on $(-\infty, 0)$ and $(0, \infty)$
- C. monotonic on $(0, \infty)$ only
- D. monotonic on $(-\infty, 0)$ only

Answer: B

Solution:

Method (1)

$$\text{Given, } f(x) = \begin{cases} \left[2 - \sin\left(\frac{1}{x}\right)\right] |x|, & x \neq 0 \\ 0 & x = 0 \end{cases}.$$

Here, $f'(x)$ is an oscillating function which is non-monotonic in $(-\infty, 0) \cup (0, \infty)$.
Method (II)

$$\therefore f(x) = \begin{cases} -\left(2 - \sin \frac{1}{x}\right)x & x < 0 \\ 0 & x = 0 \\ \left(2 - \sin \frac{1}{x}\right)x & x > 0. \end{cases}$$

From above we observe that, $f(x)$ is continuous and $f\left(\frac{1}{\pi}\right) = f\left(\frac{2}{\pi}\right) = \frac{2}{\pi}$ So, $f(x)$ is non-monotonic in $(0, \infty)$.

Further, $\lim_{x \rightarrow -\infty} f(x) \rightarrow \infty$ and $\lim_{x \rightarrow \infty} f(x) \rightarrow \infty$
and $f(0) = 0$

Hence, $f(x)$ is non-monotonic on $(-\infty, 0) \cup (0, \infty)$.

Question92

Let the functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} x+2 & x < 0 \\ x^2 & x \geq 0. \end{cases}$$

$$\text{and } g(x) = \begin{cases} x^3 & x < 1 \\ 3x-2 & x \geq 1. \end{cases}$$

Then, the number of points in $\mathbb{R}$, where $(f \circ g)(x)$ is not differentiable is equal to
[2021, 16 March Shift-1]

Options:

- A. 3
- B. 1
- C. 0
- D. 2

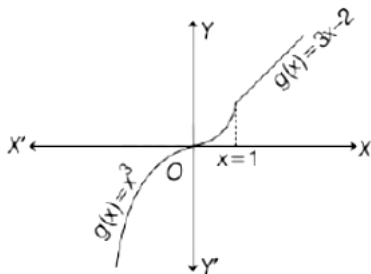
Answer: B

Solution:

$$f(x) = \begin{cases} x+2 & x < 0 \\ x^2 & x \geq 0. \end{cases}$$

$$g(x) = \begin{cases} x^3 & x < 1 \\ 3x-2 & x \geq 1. \end{cases}$$

$$f[g(x)] = \begin{cases} g(x)+2 & g(x) < 0 \\ g^2(x) & g(x) \geq 0. \end{cases}$$



When $g(x) < 0 \Rightarrow g(x) = x^3, x < 0$

When $g(x) \geq 0 \Rightarrow g(x) = \begin{cases} x^3 & 0 \leq x < 1 \\ 3x-2 & x \geq 1. \end{cases}$

$$f[g(x)] = \begin{cases} x^3+2 & x < 0 \\ x^6 & 0 \leq x < 1 \\ (3x-2)^2 & x \geq 1. \end{cases}$$

As, polynomial function is continuous everywhere in its domain. So, $f[g(x)]$ will be continuous everywhere at $x < 0$, $0 < x < 1$ and $x > 1$. We will check the behaviour of $f \circ g(x)$ only at boundary points which is $x = 0$ and $x = 1$.

$$\lim_{x \rightarrow 0^-} x^6 = 0^- \rightarrow 0^-(x^3+2) = 2$$

Clearly, $L^+ H L \neq R H L$ at $x = 0$

So, $f \circ g(x)$ is discontinuous at $x = 0$.

$$\lim_{x \rightarrow 1^+} (3x-2)^{2x \rightarrow 1^-} (\lim^6 = 1$$

Also $f(1) = 1$ fog (x) is continuous at $x = 1$

Derivative test at $x = 1$,

$$LH D = \lim_{h \rightarrow 0} \frac{f(1)-f(1-h)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1-(1-h)^6}{h}$$

$$= \lim_{h \rightarrow 0} 6(1-h)^5 = 6$$

$$RH D = \lim_{h \rightarrow 0} \frac{f(1+h)-f(1)}{h}$$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{[3(1+h)-2]^2 - 1}{h} \\
&= \lim_{h \rightarrow 0} 2[3(1+h)-2] \cdot 3 = 6 \\
&\therefore \text{fog}(x) \text{ is continuous and differentiable at } x = 1.
\end{aligned}$$

Question93

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} x+a & x < 0 \\ |x-1| & x \geq 0. \end{cases} \text{ and}$$

$$g(x) = \begin{cases} x+1 & x < 0 \\ (x-1)^2 + b & x \geq 0. \end{cases}$$

where a, b are non-negative real numbers. If $(g \circ f)(x)$ is continuous for all $x \in \mathbb{R}$, then $a + b$ is equal to.....

[2021, 16 March Shift-II]

Answer: 1

Solution:

$$g(x) = \begin{cases} x+1 & x < 0 \\ (x-1)^2 + b & x \geq 0. \end{cases}$$

$$g[f(x)] = \begin{cases} f(x)+1 & f(x) < 0 \\ [f(x)-1]^2 + b, & f(x) \geq 0. \end{cases}$$

$$f(x) < 0$$

$$\text{Case I } x+a < 0 \text{ and } x < 0 \Rightarrow x < -a$$

$$\text{Case II } |x-1| < 0 \text{ and } x \geq 0 \Rightarrow \text{Not possible } f(x) \geq 0$$

$$\text{Case I } x+a \geq 0 \text{ and } x < 0 \Rightarrow x \in [-a, 0)$$

$$\text{Case II } |x-1| \geq 0 \text{ and } x \geq 0 \Rightarrow x \geq 0$$

$$g[f(x)] = \begin{cases} x+a+1 & x < -a \\ (x+a-1)^2 + b & -a \leq x < 0 \\ (|x-1|-1)^2 + b & x \geq 0. \end{cases}$$

This is continuous function.

Since, $g[f(x)]$ is continuous for all $x \in \mathbb{R}$

So, $g(f(x))$ will be continuous at $x = -a$ and $x = 0$

Now, at $x = -a$

$$\text{LHL} = \text{RHL} = \text{value of function}$$

$$\Rightarrow 1 = 1 + b = 1 + b \Rightarrow b = 0$$

At $x = 0$

$$\text{LHL} = \text{RHL} = \text{value of function}$$

$$\Rightarrow (a-1)^2 + b = b$$

$$\Rightarrow (a-1)^2 = 0$$

$$\Rightarrow a = 1$$

$$\text{Hence, } a + b = 1$$

Question94

$$\text{Let } \alpha \in \mathbb{R} \text{ be such that the function } f(x) = \begin{cases} \frac{\cos^{-1}(1 - \{x\}^2) \sin^{-1}(1 - \{x\})}{\{x\} - \{x\}^3} & x \neq 0 \\ \alpha & x = 0. \end{cases} \text{ is continuous at } x = 0, \text{ where}$$

$\{x\} = x - [x]$, $[x]$ is the greatest integer less than or equal to x .

**Then,
[2021, 16 March Shift-II]**

Options:

A. $\alpha = \frac{\pi}{\sqrt{2}}$

B. $\alpha = 0$

C. no such α exists

D. $\alpha = \frac{\pi}{4}$

Answer: C

Solution:

$$\text{Given, } f(x) = \begin{cases} \frac{\cos^{-1}(1 - \{x\}^2) \sin^{-1}(1 - \{x\})}{\{x\} - \{x\}^3} & x \neq 0 \\ \alpha & x = 0. \end{cases}$$

$$\{x\} = x - [x]$$

So, when $x \rightarrow 0^+$

$$\Rightarrow \{x\} = x - 0 = x$$

And, when $x \rightarrow 0^-$

$$\Rightarrow \{x\} = x + 1$$

$$\text{LHL} = \lim_{x \rightarrow 0^-} f(x)$$

$$= \lim_{x \rightarrow 0^-} \frac{\cos^{-1}[1 - (1+x)^2] \sin^{-1}[1 - (1+x)]}{(1+x) - (1+x)^3}$$

$$= \lim_{x \rightarrow 0^-} \frac{\cos^{-1}(-x^2 - 2x) \sin^{-1}(-x)}{(1+x)(1+1+x)(1-1-x)}$$

$$= \lim_{x \rightarrow 0^-} \frac{\cos^{-1}(-x^2 - 2x)}{(1+x)(x+2)}$$

$$= \frac{\cos^{-1}(0)}{1 \cdot 2} = \frac{\pi}{4}$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} f(x)$$

$$= \lim_{x \rightarrow 0^+} \frac{\cos^{-1}(1 - x^2) \sin^{-1}(1 - x)}{x(1-x)(1+x)}$$

$$= \frac{\pi}{2} \lim_{x \rightarrow 0^+} \frac{\cos^{-1}(1 - x^2)}{x}$$

Applying L-Hospital Rule,

$$= \frac{\pi}{2} \lim_{x \rightarrow 0^+} \frac{(-1)(-2x)}{\sqrt{1 - (1 - x^2)^2}}$$

$$= \frac{\pi}{2} \cdot 2 \lim_{x \rightarrow 0^+} \frac{x}{\sqrt{2x^2 - x^4}}$$

$$= \pi \lim_{x \rightarrow 0^+} \frac{1}{\sqrt{2 - x^2}}$$

$$= \frac{\pi}{\sqrt{2}}$$

$$\text{LHL} = \frac{\pi}{4} \text{ and } \text{RHL} = \frac{\pi}{\sqrt{2}}$$

Hence, $\text{LHL} \neq \text{RHL}$

So, the function will be discontinuous for every value of $\alpha \in \mathbb{R}$.

$\therefore$ No such α exist.

Question95

Let $f : S \rightarrow S$, where $S = (0, \infty)$ be a twice differentiable function, such that $f(x+1) = xf(x)$. If $g : S \rightarrow \mathbb{R}$ be defined as $g(x) = \log_e f(x)$, then the value of $g''(5) - g''(1)$ is equal to

[2021, 16 March Shift-II]

Options:

A. $\frac{205}{144}$

B. $\frac{197}{144}$

C. $\frac{187}{144}$

D. 1

Answer: A

Solution:

We have, $f : S \rightarrow S, S = (0, \infty)$

$$f(x+1) = x \cdot f(x)$$

$$g : S \rightarrow R$$

$$g(x) = \log_e f(x)$$

To find $|g''(5) - g''(1)|$

$$\Rightarrow g(x+1) = \log_e f(x+1)$$

$$\Rightarrow g(x+1) = \log[x \cdot f(x)]$$

$$\Rightarrow g(x+1) = \log x + \log f(x)$$

$$\Rightarrow g(x+1) = \log x + g(x)$$

$$\Rightarrow g(x+1) - g(x) = \log x$$

$$\Rightarrow g'(x+1) - g'(x) = 1/x$$

$$\Rightarrow g''(x+1) - g''(x) = \frac{-1}{x^2}$$

$$x = 1, g''(2) - g''(1) = -1 \dots (i)$$

$$x = 2, g''(3) - g''(2) = -1/4 \dots (ii)$$

$$x = 3, g''(4) - g''(3) = -1/9 \dots (iii)$$

$$x = 4, g''(5) - g''(4) = -1/16 \dots (iv)$$

Adding Eqs. (i), (ii), (iii) and (iv),

$$g''(5) - g''(1) = -1 - \frac{1}{4} - \frac{1}{9} - \frac{1}{16}$$

$$= -\left(\frac{144 + 36 + 16 + 9}{144}\right)$$

$$= \frac{-205}{144}$$

$$\text{So, } |g''(5) - g''(1)| = \frac{205}{144}$$

Question96

Let $f : [0, 3] \rightarrow R$ be defined by $f(x) = \min\{x - [x], 1 + [x] - x\}$ where $[x]$ is the greatest integer less than or equal to x . Let P denote the set containing all x in $(0, 3)$, where f is discontinuous and Q denote the set containing all x in $(0, 3)$, where f is not differentiable.

Then the sum of number of elements in P and Q is equal to

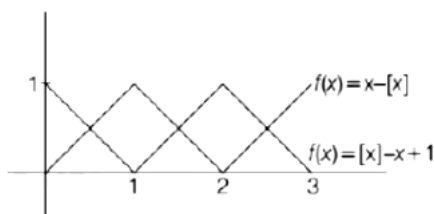
[2021, 27 July Shift-1]

Answer: 5

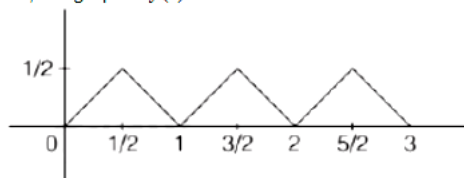
Solution:

$$f(x) = \min\{x - [x], 1 + [x] - x\}$$

$$f(x) = \min(\{x\}, 1 - \{x\})$$



So, the graph of $f(x)$ will be



f is continuous everywhere for $0 \leq x \leq 3$. But f is non-differentiable at $x = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}$ and $x = 1, 2$

So, if set A denotes the points of discontinuity, then $n(A) = 0$.

And if set B denotes the points of non-differentiable, then

$n(B) = 5$

$\therefore n(A) + n(B) = 0 + 5 = 5$

Question 97

Let $f : \left(-\frac{\pi}{4}, \frac{\pi}{4}\right) \rightarrow \mathbb{R}$ be defined as $f(x) = \begin{cases} (1 + |\sin x|)^{\frac{3a}{|\sin x|}} & -\frac{\pi}{4} < x < 0 \\ b_1 & x = 0 \\ \frac{\cot 4x}{e^{\cot 2x}}, & 0 < x < \frac{\pi}{4} \end{cases}$

If f is continuous at $x = 0$, then the value of $6a + b^2$ is equal to
[2021, 27 July Shift I]

Options:

A. $1 - e$

B. $e - 1$

C. $1 + e$

D. e

Answer: C

Solution:

$f : \left(-\frac{\pi}{4}, \frac{\pi}{4}\right) \rightarrow \mathbb{R}$

$f(x) = \begin{cases} (1 + |\sin x|)^{\frac{3a}{|\sin x|}} & -\frac{\pi}{4} < x < 0 \\ b & x = 0 \\ \frac{\cot 4x}{e^{\cot 2x}} & 0 < x < \frac{\pi}{4} \end{cases}$

Given $f(x)$ is continuous at $x = 0$

LHL at $x = 0$

Put $x = 0 - h$

$$\text{we get } \lim_{h \rightarrow 0} (1 - \sin h)^{\frac{3a}{\sin h}}$$

$$\lim_{h \rightarrow 0} (1 - \sinh^{-1})^{\frac{3a}{-\sinh}} = e^{3a}$$

$$\lim_{x \rightarrow 0^+} (1 + |\sin x|)^{\frac{3a}{|\sin x|}}$$

$$= \lim_{x \rightarrow 0^+} (1 + |\sin x|)^{\frac{3a}{|\sin x|}} = e^{3a}$$

RHL at $x = 0$

$$\lim_{x \rightarrow 0^+} e^{\frac{\cot 4x}{\cot 2x}}$$

Put $x = 0 + h$

$$\text{we get } \lim_{h \rightarrow 0} \frac{\cot 4h}{e^{\cot 2h}}$$

$$\lim_{h \rightarrow 0} \frac{\cos 4h}{e^{\cos 2h}} \times \frac{\sin 2h}{\sin 4h}$$

$$\frac{\cos 4h}{\cos 2h} \times \frac{\frac{\sin 2h}{2h} \times 2h}{\frac{\sin 4h}{4h} \times 4h} = e^{1/2}$$

As, $f(x)$ is continuous at $x = 0$.

So, LHL = $f(0)$ = RHL

$$e^{3a} = b = e^{\frac{1}{2}}$$

$$\therefore a = \frac{1}{6}, b = \sqrt{e}$$

$$\therefore 6a + b^2 = 6\left(\frac{1}{6}\right) + (\sqrt{e})^2$$

$$= 1 + e$$

Question 98

Let $f : [0, \infty) \rightarrow [0, 3]$ be a function defined by $f(x) = \begin{cases} \max\{\sin t : 0 \leq t \leq x\} & 0 \leq x < \pi \\ 2 + \cos x & x \geq \pi. \end{cases}$

Then which of the following is true?
[2021, 27 July Shift-11]

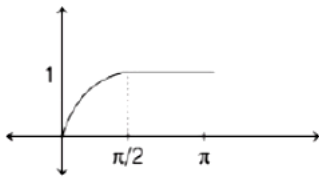
Options:

- A. f is continuous everywhere but not differentiable exactly at one point in $(0, \infty)$
- B. f is differentiable everywhere in $(0, \infty)$
- C. f is not continuous exactly at two points in $(0, \infty)$
- D. f is continuous everywhere but not differentiable exactly at two points in $(0, \infty)$

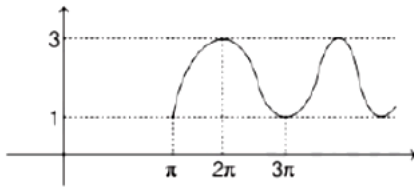
Answer: B

Solution:

Graph of $\max(\sin t : 0 \leq t \leq x)$ in $x \in [0, \pi]$

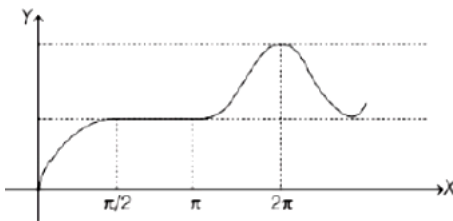


and graph of $2 + \cos x$ for $x \in [\pi, \infty]$



So, graph of

$$f(x) = \begin{cases} \max[\sin t : 0 \leq t \leq x], & 0 \leq x \leq \pi \\ 2 + \cos x, & x > \pi \end{cases}$$



So, $f(x)$ is differentiable everywhere in $(0, \infty)$.

Question99

Let $f : (a, b) \rightarrow \mathbb{R}$ be twice differentiable function such that $f(x) = \int_a^x g(t) dt$ for a differentiable function $g(x)$. If $f(x) = 0$ has exactly five distinct roots in (a, b) , then $g(x)g'(x) = 0$ has at least [2021, 27 July Shift-II]

Options:

- A. twelve roots in (a, b)
- B. five roots in (a, b)
- C. seven roots in (a, b)
- D. three roots in (a, b)

Answer: C

Solution:

We have, $f(x) = \int_a^x g(t) dt$

So, $f'(x) = g(x)$ and $f''(x) = g'(x)$

$$\left\{ \begin{array}{l} \because f(x) = \int_{g(x)}^{h(x)} F(t) \cdot dt \Rightarrow f'(x) \\ = F[h(x)] \cdot h'(x) - F[g(x)] \cdot g'(x) \end{array} \right\}$$

Now, $g'(x)g(x) = 0$

$\Rightarrow f''(x)f'(x) = 0$

If $f(x)$ has five roots, then $f'(x)$ has atleast 4 roots and $f''(x)$ has atleast 3 roots.

So, $f''(x) \cdot f'(x) = 0$ has atleast 7 roots. Hence, the minimum number of roots of the equation $g'(x)g(x) = 0$ is 7.

Question100

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} \frac{\lambda x^2 - 5x + 6}{\mu(5x - x^2 - 6)} & x < 2 \\ e^{\frac{\tan(x-2)}{x - [x]}} & x > 2 \\ \mu & x = 2. \end{cases}$$

where, $[x]$ is the greatest integer less than or equal to x . If f is continuous at $x = 2$, then $\lambda + \mu$ is equal to
[2021, 25 July Shift-1]

Options:

- A. $e(-e + 1)$
- B. $e(e - 2)$
- C. 1
- D. $2e - 1$

Answer: A

Solution:

$$\text{We have } f(x) = \begin{cases} \frac{\lambda |x^2 - 5x + 6|}{\mu(5x - x^2 - 6)} & x < 2 \\ e^{\frac{\tan(x-2)}{x - [x]}} & x > 2 \\ \mu & x = 2. \end{cases}$$

$f(x)$ is continuous at $x = 2$.

$$\therefore \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2)$$

$$\therefore \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{\lambda}{\mu} \frac{|(x-3)(x-2)|}{-(x-3)(x-2)}$$

$$= \lim_{x \rightarrow 2^-} \frac{\lambda}{\mu} \frac{(x-3)(x-2)}{-(x-3)(x-2)} = -\frac{\lambda}{\mu}$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} e^{\frac{\tan(x-2)}{x-2}} = e$$

As, $f(x)$ is continuous.

So, LHL = $f(2)$ = RHL

$$-\frac{\lambda}{\mu} = \mu = e$$

$$\lambda = -e^2$$

$$\mu = e$$

$$\lambda + \mu = e(-e + 1)$$

Question 101

Let $f : [0, \infty) \rightarrow [0, \infty)$ be defined as $f(x) = \int_0^x [y] dy$ where, $[x]$ is the greatest integer less than or equal to x . Which of the following is true?
[2021, 25 July Shift-1]

Options:

- A. f is continuous at every point in $[0, \infty)$ and differentiable except at the integer points
- B. f is both continuous and differentiable except at the integer points in $[0, \infty)$
- C. f is continuous everywhere except at the integer points in $[0, \infty)$
- D. f is differentiable at every point in $[0, \infty)$

Answer: A

Solution:

$$f : [0, \infty) \rightarrow [0, \infty)$$

$$f(x) = \int_0^x [y] dy$$

$$\text{Let } x = 1 + f, 0 < f < 1$$

$$f(x) = \int_0^1 [y] dy + \int_1^2 [y] dy + \int_2^3 [y] dy + \dots + \int_{1+f}^{1+f} [y] dy$$

$$+ \int_{1+f}^{1+f} [y] dy$$

$$f(x) = 0 + 1 + 2 + \dots + (1-1) + 1 \cdot f$$

$$= \frac{(1-1)(1-1+1)}{2} + 1 \cdot f$$

$$= \frac{1(1-1)}{2} + 1 \cdot f$$

$$f(x) = \frac{[x]([x]-1)}{2} + [x][x]$$

$$f(x) = \frac{[x]([x]-1)}{2} + [x](x - [x])$$

$$f(1) = \frac{1(1-1)}{2}$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{h \rightarrow 0} \frac{1(1-1)}{2} + 1(1+h-1)$$

$$= \frac{1(1-1)}{2}$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{h \rightarrow 0} \frac{1(1-1)(1-2)}{2} + (1-1)(1+h-1+1)$$

$$= \frac{(1-1)(1-2)}{2} + (1-1)$$

$$= \frac{(1-1)1}{2}$$

$\therefore f(x)$ is continuous and differentiable except at integer points.

Question 102

$$\text{If } f(x) = \begin{cases} \int_0^x (5 + |1-t|) dt & x > 2 \\ 5x + 1 & x \leq 2 \end{cases}, \text{ then}$$

[2021, 25 July Shift-11]

Options:

A. $f(x)$ is not continuous at $x = 2$

B. $f(x)$ is everywhere differentiable

C. $f(x)$ is continuous but not differentiable at $x = 2$

D. $f(x)$ is not differentiable at $x = 1$

Answer: C

Solution:

$$f(x) = \begin{cases} \int_0^x (5 + |1-t|) dt, & x > 2; \\ 5x + 1, & x \leq 2. \end{cases}$$

$$\int_0^x 5 + |1-t| dt$$

$$= \int_0^1 5 + (1-t) dt + \int_1^x 5 + (t-1) dt$$

$$= \int_0^1 (6-t) dt + \int_1^x (4+t) dt$$

$$= \left[6t - \frac{t^2}{2} \right]_0^1 + \left[4t + \frac{t^2}{2} \right]_1^x = 1 + 4x + \frac{x^2}{2}$$

$$\Rightarrow f(x) = \begin{cases} 1 + 4x + \frac{x^2}{2} & x > 2 \\ 5x + 1 & x \leq 2. \end{cases}$$

At $x = 2$

$$\text{LHL} = \lim_{x \rightarrow 2^-} (5x + 1) = 11$$

$$\text{RHL} = \lim_{x \rightarrow 2^+} \left(1 + 4x + \frac{x^2}{2} \right) = 1 + 8 + 2 = 11$$

$$\therefore f(2) = 11$$

So, $f(x)$ is continuous at $x = 2$.

$$f'(x) = \begin{cases} 4 + x & x > 2 \\ 5 & x \leq 2 \end{cases}$$

$$\text{Now, LHD at } x = 2 \text{ is } \frac{d}{dx}(5x + 1)_{x=2} = 5$$

$$\text{RHD at } x = 2 \text{ is } 4 + 2 = 6$$

Here, $\text{LHD} \neq \text{RHD}$

So, function is not differentiable at $x = 2$.

Question 103

Consider the function $f(x) = \begin{cases} \frac{P(x)}{\sin(x-2)} & x \neq 2 \\ 7 & x = 2. \end{cases}$

where, $P(x)$ is a polynomial such that $P'''(x)$ is always a constant and $P(3) = 9$. If $f(x)$ is continuous at $x = 2$, then $P(5)$ is equal to

[2021, 25 July Shift-11]

Answer: 39

Solution:

$$f(x) = \begin{cases} P(x)/\sin(x-2) & x \neq 2 \\ 7 & x = 2. \end{cases}$$

Given, that $P'''(x)$ is always a constant.

$\Rightarrow P(x)$ is a 2 degree polynomial.

$f(x)$ is continuous at $x = 2$

$$\lim_{x \rightarrow 2^+} P(x)/\sin(x-2) = 7$$

$$\Rightarrow \lim_{x \rightarrow 2^+} (x-2)(ax+b)/\sin(x-2) = 7$$

$$\Rightarrow 2a + b = 7 \dots (i)$$

$$\text{Now, } P(x) = (x-2)(ax+b)$$

$$P(3) = 9 \text{ (given)}$$

$$\Rightarrow 3a + b = 9$$

Subtracting Eq. (ii) from Eq. (i),

$$a = 2$$

$$\text{From Eq. (i), } b = 3$$

$$\text{Hence, } P(x) = (x-2)(2x+3)$$

$$\text{So, } P(5) = (5-2)(2 \times 5 + 3) = 3 \times 13 = 39$$

Question 104

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = \begin{cases} \frac{x^3}{(1 - \cos 2x)^2} \log_e \left(\frac{1 + 2xe^{-2x}}{(1 - xe^{-x})^2} \right) & x \neq 0 \\ \alpha & x = 0. \end{cases}$

If f is continuous at $x = 0$, then α is equal to

[2021, 22 July Shift-II]

Options:

A. 1

B. 3

C. 0

D. 2

Answer: A

Solution:

$$f(x) = \begin{cases} \frac{x^3}{(1 - \cos 2x)^2} \log_e \left[\frac{1 + 2xe^{-2x}}{(1 - xe^{-x})^2} \right] & x \neq 0 \\ \alpha & x = 0. \end{cases}$$

For continuity, $\lim_{x \rightarrow 0} \frac{x^3}{4\sin^4 x} [\log_e(1 + 2xe^{-2x}) - \log_e(1 - xe^{-x})^2] = \alpha$ (by expansion) ... (i)

$$\therefore \log(1 + 2xe^{-2x}) = 2xe^{-2x} - \frac{(2xe^{-2x})^2}{2} + \dots \text{ and } \log(1 - xe^{-x}) = -xe^{-x} - \frac{(xe^{-x})^2}{2} - \dots$$

On putting the values in Eq. (i), we get

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{1}{4} \cdot \frac{x}{x} \right) \left(\frac{x^3}{\sin^4 x} \right) [2xe^{-2x} - 2(-xe^{-x})] \\ = \lim_{x \rightarrow 0} \left(\frac{1}{4x} \right) \left(\frac{x}{\sin x} \right)^4 (2xe^{-2x} + 2xe^{-x}) \\ = \lim_{x \rightarrow 0} \left(\frac{1}{4x} \right) \cdot \left(\frac{x}{\sin x} \right)^4 \cdot 2x \cdot (e^{-2x} + e^{-x}) \\ = \left(\frac{1}{4} \right) \cdot (1) \cdot (2) \cdot (2) \Rightarrow \alpha = 1 \end{aligned}$$

Question 105

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined $f(x) = \begin{cases} 3\left(1 - \frac{|x|}{2}\right) & \text{if } |x| \leq 2; \\ 0, & \text{if } |x| > 2. \end{cases}$

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be given by $g(x) = f(x+2) - f(x-2)$

If n and m denote the number of points in $\mathbb{R}$,

where g is not continuous and not differentiable respectively, then $n + m$ is equal to

[2021, 22 July Shift-II]

Answer: 4

Solution:

$$f(x) = \begin{cases} 3\left(\frac{1 - |x|}{2}\right) & \text{if } |x| \leq 2; \\ 0, & \text{if } |x| > 2. \end{cases}$$
$$g(x) = f(x+2) - f(x-2)$$

$$f(x) = \begin{cases} 0 & x < -2 \\ \frac{3}{2}(1+x) & -2 \leq x < 0 \\ \frac{3}{2}(1-x) & 0 \leq x < 2 \\ 0 & x > 2. \end{cases}$$

$$f(x+2) = \begin{cases} 0 & x < -4 \\ \frac{3}{2}(3+x) & -4 \leq x < -2 \\ \frac{3}{2}(-1-x) & -2 \leq x < 0 \\ 0 & x > 4. \end{cases}$$

$$f(x-2) = \begin{cases} 0 & x < 0 \\ \frac{3}{2}(x-1) & 0 \leq x < 2 \\ \frac{3}{2}(-1-x) & 2 \leq x < 4 \\ 0 & x > 4. \end{cases}$$

$$g(x) = f(x+2) + f(x-2)$$

$$= \begin{cases} \frac{3x}{2} + 6 & -4 \leq x \leq 2 \\ -\frac{3x}{2} & -2 < x < 2 \\ \frac{3x}{2} - 6 & 2 \leq x \leq 4 \\ 0 & x > 4. \end{cases}$$

So, $n = 0$ and $m = 4$

$$\therefore m + n = 4$$

Question 106

Let a function $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = \begin{cases} \sin x - e^x & \text{if } x \leq 0 \\ a + [-x] & \text{if } 0 < x < 1 \\ 2x - b & \text{if } x \geq 1. \end{cases}$

where, $[x]$ is the greatest integer less than or equal to x . If f is continuous on $\mathbb{R}$, then $(a + b)$ is [2021, 20 July Shift-1]

Options:

A. 4

B. 3

C. 2

D. 5

Answer: B

Solution:

$$f(x) = \begin{cases} \sin x - e^x & x < 0 \\ a + [-x] & 0 < x < 1 \\ 2x - b & x \geq 1. \end{cases}$$

$f(x)$ is continuous.

$$\text{So, } \lim_{h \rightarrow 0} f(0^-) = 0 - e^0 = -1$$

$$\lim_{h \rightarrow 0} f(0^+) = a - 1$$

$$\Rightarrow a - 1 = -1 \Rightarrow a = 0$$

$$\lim_{h \rightarrow 0} f(1^-) = \lim_{h \rightarrow 0} a + [-1 - h] = a - 1$$

$$\lim_{h \rightarrow 0} f(1^+) = 2(1 + h) - b = 2 - b$$

$$\therefore 2 - b = a - 1 \Rightarrow b = 2 + 1 = 3$$

$$\therefore a + b = 3$$

Question 107

Let a function $g : [0, 4] \rightarrow \mathbb{R}$ be defined as $g(x) = \begin{cases} \max_{0 \leq t \leq x} (t^3 - 6t^2 + 9t - 3) & 0 \leq x \leq 3 \\ 4 - x & 3 < x \leq 4. \end{cases}$

then the number of points in the interval $(0, 4)$ where $g(x)$ is not differentiable, is
[2021, 20 July Shift-II]

Answer: 1

Solution:

$$\text{Let } f(x) = x^3 - 6x^2 + 9x - 3$$

$$f'(x) = 3x^2 - 12x + 9$$

$$f'(x) = 0 \text{ gives}$$

$$3x^2 - 12x + 9 = 0$$

$$\Rightarrow 3(x-1)(x-3) = 0$$

$$\therefore x = 1 \text{ or } x = 3$$

$$\text{Now, } f(1) = 1 \text{ and } f(3) = -3$$

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq 1 \\ 1 & 1 \leq x \leq 3 \\ 4 - x & 3 < x \leq 4. \end{cases}$$

$g(x)$ is continuous.

$$g'(x) = \begin{cases} 3(x-1)(x-3) & 0 \leq x \leq 1 \\ 0 & 1 \leq x < 3 \\ -1 & 3 < x \leq 4. \end{cases}$$

$g(x)$ is non-differentiable at $x = 3$.

So, the number of points in $(0, 4)$ where $g(x)$ is not differentiable is 1.

Question108

The function $f(x) = x^2 - 2x - 3 \mid \cdot e^{9x^2 - 12x + 4}$ is not differentiable at exactly
[2021, 31 Aug. Shift-1]

Options:

- A. four points
- B. three points
- C. two points
- D. one point

Answer: C

Solution:

$$f(x) = \begin{cases} (x-3)(x+1)e^{(3x-2)^2} & x > 3 \\ -(x-3)(x+1)e^{(3x-2)^2} & -1 \leq x \leq 3 \\ (x-3)(x+1)e^{(3x-2)^2} & x < -1. \end{cases}$$

At $x = -1$, let LHD be α , then its clear that RHD be $-\alpha$.

Similarly, at $x = 3$, if LHD is β , then RHD at $x = 3$ will be $-\beta$.

So, $f(x)$ is not differentiable at $x = -1, x = 3$

At, all other points $f(x)$ will be differentiable.

Question109

If the function $f(x) = \begin{cases} \frac{1}{x} \log_e \left(\frac{1 + \frac{x}{a}}{1 - \frac{x}{b}} \right) & x < 0 \\ k & x = 0 \\ \frac{\cos^2 x - \sin^2 x - 1}{\sqrt{x^2 + 1} - 1} & x > 0. \end{cases}$ is continuous at $x = 0$, then $\frac{1}{a} + \frac{1}{b} + \frac{4}{k}$ is equal to

[2021,31 Aug. Shift-1]

Options:

A. -5-2

B. 5

C. -4

D. 4

Answer: A

Solution:

$f(x)$ is continuous at $x = 0$

LHL at $x = 0 = f(0) = \text{RHL at } x = 0$

$$\lim_{x \rightarrow 0^-} \frac{\ln \left(\frac{1 + \frac{x}{a}}{1 - \frac{x}{b}} \right)}{x} = \lim_{x \rightarrow 0^-} \left(\frac{1}{a} \right) \ln \left(1 + \frac{x}{a} \right) \left(-\frac{1}{b} \right) x = \left(\frac{1}{a} \right) x$$

$$- \lim_{x \rightarrow 0^-} \frac{\left(-\frac{1}{b} \right) \ln \left(1 - \frac{x}{b} \right)}{\left(-\frac{1}{b} \right) x} = \left(\frac{1}{a} + \frac{1}{b} \right)$$

$$\lim_{x \rightarrow 0^+} \frac{\cos^2 x - \sin^2 x - 1}{\sqrt{x^2 + 1} - 1}$$

$$\lim_{x \rightarrow 0^+} \frac{-2\sin^2 x}{\sqrt{x^2 + 1} - 1} = \lim_{x \rightarrow 0^+} - \left(\frac{2\sin^2 x}{x^2} \right)$$

$$\left(\sqrt{x^2 + 1} + 1 \right) = -4$$

$$\Rightarrow \frac{1}{a} + \frac{1}{b} = -4 = k$$

$$\left(\frac{1}{a} + \frac{1}{b} \right) + \left(\frac{4}{k} \right) = -4 - 1 = -5$$

Question 110

Let $a, b \in \mathbb{R}$, $b \neq 0$. Define a function

$f(x) = \begin{cases} \sin \frac{\pi}{2}(x-1), & \text{for } x \leq 0 \\ \frac{\tan 2x - \sin 2x}{bx^3}, & \text{for } x > 0 \end{cases}$ end cases If f is continuous at $x = 0$,

then $10 - ab$ is equal to

[2021, 26 Aug. Shift-1]

Answer: 14

Solution:

For continuity

LHL at 0 = f(0) = RHL at 0

$$\text{LHL} = \lim_{x \rightarrow 0^-} a \sin \frac{\pi}{2}(x-1)$$

$$= -a \sin \frac{\pi}{2} = -a \dots (i)$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} \frac{\tan 2x - \sin 2x}{bx^3}$$

$$= \lim_{x \rightarrow 0^+} \frac{\sin 2x(1 - \cos 2x)}{bx^3 \cdot \cos 2x}$$

$$= \lim_{x \rightarrow 0^+} 2 \left(\frac{\sin 2x}{2x} \right) \cdot \frac{(2\sin^2 x)}{x^2} \cdot \frac{1}{b \cos 2x} = \frac{4}{b} \dots (ii)$$

From Eqs. (i) and (ii), we get

$$-a = \frac{4}{b}$$

$$\Rightarrow ab = -4$$

$$\Rightarrow 10 - ab = 14$$

Question 111

Let $[t]$ denote the greatest integer less than or equal to t . Let $f(x) = x - [x]$, $g(x) = 1 - x + [x]$, and $h(x) = \min\{f(x), g(x)\}$, $x \in [-2, 2]$. Then h is
[2021, 26 Aug. Shift-II]

Options:

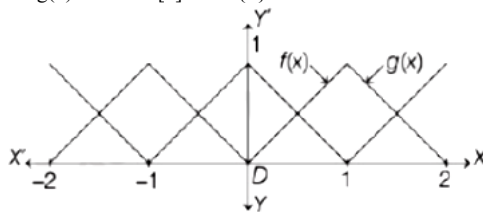
- A. continuous in $[-2, 2]$ but not differentiable at more than four points in $(-2, 2)$
- B. not continuous at exactly three points in $[-2, 2]$
- C. continuous in $[-2, 2]$ but not differentiable at exactly three points in $(-2, 2)$
- D. not continuous at exactly four points in $[-2, 2]$

Answer: A

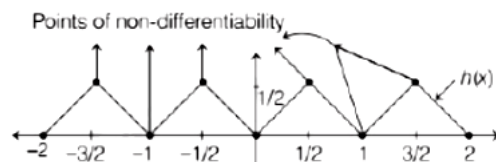
Solution:

We have, $f(x) = x - [x] = \{x\}$

and $g(x) = 1 - x + [x] = 1 - \{x\}$



Again, $h(x) = \min[f(x), g(x)]$, so graph of $h(x)$ will be



From graph, it is clear that $h(x)$ is continuous in $[-2, 2]$ but not differentiable at $x = -\frac{3}{2}, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}$ in $(-2, 2)$

Question 112

Let $x^k + y^k = a^k$, ($a, k > 0$) and $\frac{dy}{dx} + \left(\frac{y}{x}\right)^{\frac{1}{3}} = 0$, then k is:

[Jan. 7, 2020 (I)]

Options:

A. $\frac{3}{2}$

B. $\frac{4}{3}$

C. $\frac{2}{3}$

D. $\frac{1}{3}$

Answer: C

Solution:

$$k \cdot x^{k-1} + k \cdot y^{k-1} \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\left(\frac{x}{y}\right)^{k-1}$$

$$\Rightarrow \frac{dy}{dx} + \left(\frac{x}{y}\right)^{k-1} = 0$$

$$\Rightarrow k-1 = -\frac{1}{3}$$

$$\Rightarrow k = 1 - \frac{1}{3} = \frac{2}{3}$$

Question113

The value of c in the Lagrange's mean value theorem for the function $f(x) = x^3 - 4x^2 + 8x + 11$, when $x \in [0, 1]$ is:

[Jan. 7, 2020 (II)]

Options:

A. $\frac{4-\sqrt{5}}{3}$

B. $\frac{4-\sqrt{7}}{3}$

C. $\frac{2}{3}$

D. $\frac{\sqrt{7}-2}{3}$

Answer: B

Solution:

Since, $f(x)$ is a polynomial function.

$\therefore$ It is continuous and differentiable in $[0, 1]$

Here, $f(0) = 11$, $f(1) = 1 - 4 + 8 + 11 = 16$

$$f'(x) = 3x^2 - 8x + 8$$

$$\therefore f'(c) = \frac{f(1) - f(0)}{1 - 0} = \frac{16 - 11}{1}$$

$$= 3c^2 - 8c + 8$$

$$\Rightarrow 3c^2 - 8c + 3 = 0$$

$$\Rightarrow c = \frac{8 \pm 2\sqrt{7}}{6} = \frac{4 \pm \sqrt{7}}{3}$$

$$\therefore c = \frac{4 - \sqrt{7}}{3} \in (0, 1)$$

Question114

Let $[t]$ denote the greatest integer $\leq t$ and $\lim_{x \rightarrow 0} x \left[\frac{4}{x} \right] = A$. Then the function, $f(x) = [x^2] \sin(\pi x)$ is discontinuous, when x is equal to :
[Jan. 9, 2020 (II)]

Options:

A. $\sqrt{A+1}$

B. $\sqrt{A+5}$

C. $\sqrt{A+21}$

D. $\sqrt{A}$

Answer: A

Solution:

$$\lim_{x \rightarrow 0} x \left[\frac{4}{x} \right] = A \Rightarrow \lim_{x \rightarrow 0} x \left[\frac{4}{x} - \left\{ \frac{4}{x} \right\} \right] = A$$

$$\Rightarrow \lim_{x \rightarrow 0} 4 - x \left\{ \frac{4}{x} \right\} = A \Rightarrow 4 - 0 = A$$

As, $f(x) = [x^2] \sin(\pi x)$ will be discontinuous at non-integers And, when $x = \sqrt{A+1} \Rightarrow x = \sqrt{5}$, which is not an integer.

Hence, $f(x)$ is discontinuous when x is equal to $\sqrt{A+1}$

Question 115

If the function f defined on $\left(-\frac{1}{3}, \frac{1}{3}\right)$ by

$$f(x) = \begin{cases} \frac{1}{x} \log_e \left(\frac{1+3x}{1-2x} \right), & \text{when } x \neq 0 \\ k, & \text{when } x = 0 \end{cases}$$

is continuous, then k is equal to _____.

[NA Jan. 7, 2020 (II)]

Answer: 5

Solution:

$$\begin{aligned} \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \left(\frac{1}{x} \ln \left(\frac{1+3x}{1-2x} \right) \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{\ln(1+3x)}{x} - \frac{\ln(1-2x)}{x} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{3 \ln(1+3x)}{3x} - \frac{2 \ln(1-2x)}{-2x} \right) \\ &= 3 + 2 = 5 \end{aligned}$$

$\therefore f(x)$ will be continuous

$$\therefore k = f(0) = \lim_{x \rightarrow 0} f(x) = 5$$

Question 116

If c is a point at which Rolle's theorem holds for the function, $f(x) = \log_e \left(\frac{x^2 + a}{7x} \right)$ in the interval $[3, 4]$, where $a \in \mathbb{R}$, then $f''(c)$ is equal to:
[Jan. 8, 2020 (I)]

Options:

A. $-\frac{1}{12}$

B. $\frac{1}{12}$

C. $-\frac{1}{24}$

D. $\frac{\sqrt{3}}{7}$

Answer: B

Solution:

Since, Rolle's theorem is applicable

$$\therefore f(a) = f(b)$$

$$f(3) = f(4) \Rightarrow a = 12$$

$$f'(x) = \frac{x^2 - 12}{x(x^2 + 12)}$$

As $f'(c) = 0$ (by Rolle's theorem)

$$x = \pm\sqrt{12}, \therefore c = \sqrt{12}, \therefore f''(c) = \frac{1}{12}$$

Question 117

$$\text{If } f(x) = \begin{cases} \frac{\sin(a+2)x + \sin x}{x} & : x < 0 \\ b & : x = 0 \\ \frac{(x+3x^2)^{1/3} - x^{1/3}}{x^{4/3}} & : x > 0 \end{cases}$$

is continuous at $x = 0$, then $a + 2b$ is equal to:
[Jan. 9, 2020 (I)]

Options:

A. 1

B. -1

C. 0

D. -2

Answer: C

Solution:

$$\text{LHL} = \lim_{x \rightarrow 0} \frac{\sin(a+2)x + \sin x}{x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin(a+2)x}{(a+2)x} \right) (a+2) + \lim_{x \rightarrow 0} \frac{\sin x}{x} = a+3$$

$$f(0) = b$$

$$\text{RHL} = \lim_{h \rightarrow 0} \left(\frac{(1+3h)^{\frac{1}{3}} - 1}{h} \right) = 1$$

$\therefore$ Function $f(x)$ is continuous
 $\therefore \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$
 $\therefore a + 3 = 1 \Rightarrow a = -2$
 and $b = 1$
 Hence, $a + 2b = 0$

Question118

Let f and g be differentiable functions on $\mathbb{R}$ such that $f \circ g$ is the identity function. If for some $a, b \in \mathbb{R}$, $g'(a) = 5$ and $g(a) = b$, then $f'(b)$ is equal to:
 [Jan. 9, 2020 (II)]

Options:

- A. $\frac{1}{5}$
- B. 1
- C. 5
- D. $\frac{2}{5}$

Answer: A

Solution:

It is given that functions f and g are differentiable and $f \circ g$ is identity function.
 $\therefore (f \circ g)(x) = x \Rightarrow f(g(x)) = x$
 Differentiating both sides, we get
 $f'(g(x)) \cdot g'(x) = 1$
 Now, put $x = a$, then
 $f'(g(a)) \cdot g'(a) = 1$
 $f'(b) \cdot 5 = 1$
 $f'(b) = \frac{1}{5}$

Question119

Let S be the set of all functions $f : [0, 1] \rightarrow \mathbb{R}$, which are continuous on $[0, 1]$ and differentiable on $(0, 1)$.
 Then for every f in S , there exists a $c \in (0, 1)$, depending on f , such that:
 [Jan. 8, 2020 (II)]

Options:

- A. $|f(c) - f(1)| < (1 - c) |f'(c)|$
- B. $\frac{f(1) - f(c)}{1 - c} = f'(c)$
- C. $|f(c) + f(1)| < (1 + c) |f'(c)|$
- D. Bonus

Answer: D

Solution:

For a constant function $f(x)$, option (1), (3) and (4) doesn't hold and by LMVT theorem, option (2) is incorrect.

Question120

Let the function, $f : [-7, 0] \rightarrow \mathbb{R}$ be continuous on $[-7, 0]$ and differentiable on $(-7, 0)$. If $f(-7) = -3$ and $f'(x) \leq 2$, for all $x \in (-7, 0)$, then for all such functions f , $f'(-1) + f(0)$ lies in the interval:
[Jan. 7, 2020 (I)]

Options:

A. $(-\infty, 20]$

B. $[-3, 11]$

C. $(-\infty, 11]$

D. $[-6, 20]$

Answer: A

Solution:

From, LMVT for $x \in [-7, -1]$

$$\frac{f(-1) - f(-7)}{(-1 + 7)} \leq 2 \Rightarrow \frac{f(-1) + 3}{6} \leq 2 \Rightarrow f(-1) \leq 9$$

From, LMVT for $x \in [-7, 0]$

$$\frac{f(0) - f(-7)}{(0 + 7)} \leq 2$$

$$\frac{f(0) + 3}{7} \leq 2 \Rightarrow f(0) \leq 11$$

$$\therefore f(0) + f(-1) \leq 20$$

Question 121

Let S be the set of points where the function, $f(x) = |2 - |x - 3||$, $x \in \mathbb{R}$, is not differentiable. Then $\sum_{x \in S} f(f(x))$ is equal to _____.

[NA Jan. 7, 2020 (I)]

Answer: 3

Solution:

$\therefore f(x)$ is non differentiable at $x = 1, 3, 5$

[$\therefore x - 3$ is not differentiable at $x = 3$]

$$\begin{aligned} \sum f(f(x)) &= f(f(1)) + f(f(3)) + f(f(5)) \\ &= 1 + 1 + 1 = 3 \end{aligned}$$

Question 122

If $x = 2 \sin \theta - \sin 2\theta$ and $y = 2 \cos \theta - \cos 2\theta$, $\theta \in [0, 2\pi]$, then $\frac{d^2y}{dx^2}$ at $\theta = \pi$ is :

[Jan. 9, 2020 (II)]

Options:

A. $\frac{3}{4}$

B. $\frac{3}{8}$

C. $\frac{3}{2}$

D. $-\frac{3}{4}$

Answer: B

Solution:

It is given that

$$x = 2 \sin \theta - \sin 2\theta \dots (i)$$

$$y = 2 \cos \theta - \cos 2\theta \dots (ii)$$

Differentiating (i) w.r.t. θ , we get

$$\frac{dx}{d\theta} = 2 \cos \theta - 2 \cos 2\theta$$

Differentiating (ii) w.r.t. θ ; we get

$$\frac{dy}{d\theta} = -2 \sin \theta + 2 \sin 2\theta$$

From (ii) \div (i), we get

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{\sin 2\theta - \sin \theta}{\cos \theta - \cos 2\theta} \\ &= \frac{2 \sin \frac{\theta}{2} \cdot \cos \frac{3\theta}{2}}{2 \sin \frac{\theta}{2} \cdot \sin \frac{3\theta}{2}} = \cot \frac{3\theta}{2} \dots (iii) \end{aligned}$$

Again, differentiating eqn. (iii), we get

$$\frac{d^2y}{dx^2} = \frac{-3}{2} \operatorname{cosec}^2 \frac{3\theta}{2} \cdot \frac{d\theta}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{\frac{-3}{2} \operatorname{cosec}^2 \frac{3\theta}{2}}{2(\cos \theta - \cos 2\theta)}$$

$$\frac{d^2y}{dx^2}(\theta = \pi) = -\frac{3}{4(-1-1)} = \frac{3}{8}$$

Question 123

If $y(\alpha) = \sqrt{2 \left(\frac{\tan \alpha + \cot \alpha}{1 + \tan^2 \alpha} \right) + \frac{1}{\sin^2 \alpha}}$, $\alpha \in \left(\frac{3\pi}{4}, \pi \right)$, then $\frac{dy}{d\alpha}$ at $\alpha = \frac{5\pi}{6}$ is:

[Jan. 7, 2020 (I)]

Options:

A. 4

B. $\frac{4}{3}$

C. -4

D. $-\frac{1}{4}$

Answer: A

Solution:

$$y(\alpha) = \sqrt{\frac{2 \sin \alpha + \cos \alpha}{\sec^2 \alpha}} = \sqrt{\frac{2 \cos^2 \alpha}{\sin \alpha \cos \alpha} + \frac{1}{\sin^2 \alpha}}$$

$$= \sqrt{2 \cot \alpha + \operatorname{cosec}^2 \alpha} = \sqrt{2 \cot \alpha + 1 + \cot^2 \alpha}$$

$$= |1 + \cot \alpha| = -1 - \cot \alpha \left[\because \alpha \in \left(\frac{3\pi}{4}, \pi \right) \right]$$

$$\frac{dy}{d\alpha} = \operatorname{cosec}^2 \alpha \Rightarrow \left(\frac{dy}{d\alpha} \right)_{\alpha = \frac{5\pi}{6}} = 4$$

Question 124

Let $y = y(x)$ be a function of x satisfying $y\sqrt{1-x^2} = k - x\sqrt{1-y^2}$ where k is a constant and $y\left(\frac{1}{2}\right) = -\frac{1}{4}$.

Then $\frac{dy}{dx}$ at $x = \frac{1}{2}$, is equal to:

[Jan. 7, 2020 (II)]

Options:

A. $-\frac{\sqrt{5}}{4}$

B. $-\frac{\sqrt{5}}{2}$

C. $\frac{2}{\sqrt{5}}$

D. $\frac{\sqrt{5}}{2}$

Answer: B

Solution:

Given, $x = \frac{1}{2}, y = -\frac{1}{4} \Rightarrow xy = -\frac{1}{8}$

$$y \cdot \frac{1 \cdot (-2x)}{2\sqrt{1-x^2}} + y' \sqrt{1-x^2}$$

$$= - \left\{ 1 \cdot \sqrt{1-y^2} + \frac{x \cdot (-2y)}{2\sqrt{1-y^2}} y' \right\}$$

$$\Rightarrow -\frac{xy}{\sqrt{1-x^2}} + y' \sqrt{1-x^2} = -\sqrt{1-y^2} + \frac{xy \cdot y'}{\sqrt{1-y^2}}$$

$$\Rightarrow y' \left(\sqrt{1-x^2} - \frac{xy}{\sqrt{1-y^2}} \right) = \frac{xy}{\sqrt{1-x^2}} - \sqrt{1-y^2}$$

$$\Rightarrow y' \left(\frac{\sqrt{3}}{2} + \frac{1}{8 \cdot \frac{\sqrt{15}}{4}} \right) = \frac{-1}{8 \cdot \sqrt{\frac{3}{2}}} - \frac{\sqrt{15}}{4}$$

$$\Rightarrow y' \left(\frac{\sqrt{45}+1}{2\sqrt{15}} \right) = -\frac{(1+\sqrt{45})}{4\sqrt{3}}$$

$$\therefore y' = -\frac{\sqrt{5}}{2}$$

Question 125

For all twice differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$, with $f(0) = f(1) = f'(0) = 0$

[Sep. 06, 2020 (II)]

Options:

A. $f''(x) \neq 0$ at every point $x \in (0,1)$

B. $f''(x) = 0$, for some $x \in (0,1)$

C. $f''(0) = 0$

D. $f''(x) = 0$, at every point $x \in (0,1)$

Answer: B

Solution:

Let $f : \mathbb{R} \rightarrow \mathbb{R}$, with $f(0) = f(1) = 0$ and $f'(0) = 0$

$\therefore f(x)$ is differentiable and continuous and

$f(0) = f(1) = 0$

Then by Rolle's theorem, $f'(c) = 0, c \in (0, 1)$

Now again

$\therefore f'(c) = 0, f'(0) = 0$

Then, again by Rolle's theorem,
 $f''(x) = 0$ for some $x \in (0, 1)$

Question126

If $y^2 + \log_e(\cos^2 x) = y$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, then :

[Sep. 03, 2020 (I)]

Options:

A. $y''(0) = 0$

B. $|y'(0)| + |y''(0)| = 1$

C. $|y''(0)| = 2$

D. $|y'(0)| + |y''(0)| = 3$

Answer: C

Solution:

$$y^2 + 2\log_e(\cos x) = y \dots\dots(i)$$

$$\Rightarrow 2yy' - 2 \tan x = y' \dots\dots(ii)$$

From (i), $y(0) = 0$ or 1

$$\therefore y'(0) = 0$$

Again differentiating (ii) we get,

$$2(y')^2 + 2yy'' - 2\sec^2 x = y''$$

Put $x = 0$, $y(0) = 0, 1$ and $y'(0) = 0$

we get, $|y''(0)| = 2$.

Question127

Let $f(x) = x \cdot \left[\frac{x}{2} \right]$, for $-10 < x < 10$, where $[t]$ denotes the greatest integer function. Then the number of points of discontinuity of f is equal to _____.

[NA Sep. 05, 2020 (I)]

Answer: 0

Solution:

We know $[x]$ discontinuous for $x \in \mathbb{Z}$

$f(x) = x \cdot \left[\frac{x}{2} \right]$ may be discontinuous where $\frac{x}{2}$ is an integer.

So, points of discontinuity are,

$x = \pm 2, \pm 4, \pm 6, \pm 8$ and 0

but at $x = 0$

$$\lim_{x \rightarrow 0^+} f(x) = 0 = f(0) = \lim_{x \rightarrow 0^-} f(x)$$

So, $f(x)$ will be discontinuous at $x = \pm 2, \pm 4, \pm 6$ and ± 8 .

Question128

If a function $f(x)$ defined by

$$f(x) = \begin{cases} ae^x + be^{-x}, & -1 \leq x < 1 \\ cx^2, & 1 \leq x \leq 3 \\ ax^2 + 2cx, & 3 < x \leq 4 \end{cases}$$

be continuous for some $a, b, c \in \mathbb{R}$ and $f'(0) + f'(2) = e$, then the value of a is :
[Sep. 02, 2020 (I)]

Options:

A. $\frac{1}{e^2 - 3e + 13}$

B. $\frac{e}{e^2 - 3e - 13}$

C. $\frac{e}{e^2 + 3e + 13}$

D. $\frac{e}{e^2 - 3e + 13}$

Answer: D

Solution:

Since, function $f(x)$ is continuous at $x = 1, 3$

$$\therefore f(1) = f(1^+)$$

$$\Rightarrow ae + be^{-1} = c \dots (i)$$

$$f(3) = f(3^+)$$

$$\Rightarrow 9c = 9a + 6c \Rightarrow c = 3a \dots (ii) \text{ From (i) and (ii),}$$

$$b = ae(3 - e) \dots (iii)$$

$$f'(x) = \begin{cases} ae^x - be^{-x} & -1 < x < 1 \\ 2cx & 1 < x < 3 \\ 2ax + 2c & 3 < x < 4 \end{cases}$$

$$f'(0) = a - b, f'(2) = 4c$$

$$\text{Given, } f'(0) + f'(2) = e$$

$$a - b + 4c = e$$

From eqs. (i), (ii), (iii) and (iv),

$$a - 3ae + ae^2 + 12a = e$$

$$\Rightarrow 13a - 3ae + ae^2 = e$$

$$\Rightarrow a = \frac{e}{e^2 - 3e + 13}$$

Question 129

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \max\{x, x^2\}$. Let S denote the set of all points in $\mathbb{R}$, where f is not differentiable. Then:
[Sep. 06, 2020 (II)]

Options:

A. $\{0, 1\}$

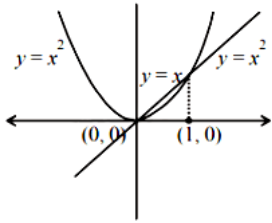
B. $\{0\}$

C. $\emptyset$ (an empty set)

D. $\{1\}$

Answer: A

Solution:



$$f(x) = \max\{x, x^2\}$$

$$\Rightarrow f(x) = \begin{cases} x^2, & x < 0 \\ x, & 0 \leq x < 1 \\ x^2, & x \geq 1 \end{cases}$$

$\therefore f(x)$ is not differentiable at $x = 0, 1$

Question 130

If the function $f(x) = \begin{cases} k_1(x-\pi)^2 - 1, & x \leq \pi \\ k_2 \cos x, & x > \pi \end{cases}$ is twice differentiable, then the ordered pair (k_1, k_2) is equal to:

[Sep. 05, 2020 (I)]

Options:

A. $\left(\frac{1}{2}, 1\right)$

B. $(1, 0)$

C. $\left(\frac{1}{2}, -1\right)$

D. $(1, 1)$

Answer: A

Solution:

$f(x)$ is differentiable then, $f(x)$ is also continuous.

$$\therefore \lim_{x \rightarrow \pi^+} f(x) = \lim_{x \rightarrow \pi^-} f(x) = f(\pi)$$

$$\Rightarrow -1 = -K_2 \Rightarrow K_2 = 1$$

$$\therefore f'(x) = \begin{cases} 2K_1(x-\pi) & : x \leq \pi \\ -K_2 \sin x & x > \pi \end{cases}$$

$$\text{Then, } \lim_{x \rightarrow \pi^+} f'(x) = \lim_{x \rightarrow \pi^-} f'(x) = 0$$

$$f''(x) = \begin{cases} 2K_1 & ; x \leq \pi \\ -K_2 \cos x & ; x > \pi \end{cases}$$

$$\text{Then, } \lim_{x \rightarrow \pi^+} f''(x) = \lim_{x \rightarrow \pi^-} f''(x)$$

$$\Rightarrow 2K_1 = K_2 \Rightarrow K_1 = \frac{1}{2}$$

$$\text{So, } (K_1, K_2) = \left(\frac{1}{2}, 1\right)$$

Question 131

Let f be a twice differentiable function on $(1, 6)$. If $f(2) = 8, f'(2) = 5, f'(x) \geq 1$ and $f''(x) \geq 4$, for all $x \in (1, 6)$, then

[Sep. 04, 2020 (I)]

Options:

- A. $f(5) + f'(5) \leq 26$
- B. $f(5) + f'(5) \geq 28$
- C. $f'(5) + f''(5) \leq 20$
- D. $f(5) \leq 10$

Answer: B

Solution:

Let f be twice differentiable function

$$\because f'(x) \geq 1$$

$$\Rightarrow \frac{f(5) - f(2)}{3} \geq 1$$

$$\Rightarrow f(5) \geq 3 + f(2)$$

$$\Rightarrow f(5) \geq 3 + 8 \Rightarrow f(5) \geq 11$$

and also $f''(x) \geq 4$

$$\Rightarrow \frac{f'(5) - f'(2)}{5 - 2} \geq 4 \Rightarrow f'(5) \geq 12 + f'(2)$$

$$\Rightarrow f'(5) \geq 17$$

$$\text{Hence, } f(5) + f'(5) \geq 28$$

Question 132

Suppose a differentiable function $f(x)$ satisfies the identity $f(x + y) = f(x) + f(y) + xy^2 + x^2y$, for all real x and y . If $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$, then $f'(3)$ is equal to _____.

[NA Sep. 04, 2020 (I)]

Answer: 10

Solution:

$$f(x + y) = f(x) + f(y) + xy^2 + x^2y$$

Differentiate w.r.t. x :

$$f'(x + y) = f'(x) + 0 + y^2 + 2xy$$

Put $y = -x$

$$f'(0) = f'(x) + x^2 - 2x^2 \dots\dots(i)$$

$$\because \lim_{x \rightarrow 0} \frac{f(x)}{x} = 1 \Rightarrow f(0) = 0$$

$$\therefore f'(0) = 1 \dots\dots(ii)$$

From equations (i) and (ii),

$$f'(x) = (x^2 + 1) \Rightarrow f'(3) = 10$$

Question 133

The function

$$f(x) = \begin{cases} \frac{\pi}{4} + \tan^{-1}x, & |x| \leq 1 \\ \frac{1}{2}(|x| - 1), & |x| > 1 \end{cases}$$

is :

[Sep. 04, 2020 (II)]

Options:

- A. continuous on $\mathbb{R} - \{1\}$ and differentiable on $\mathbb{R} - \{-1, 1\}$

B. both continuous and differentiable on $\mathbb{R} - \{1\}$.

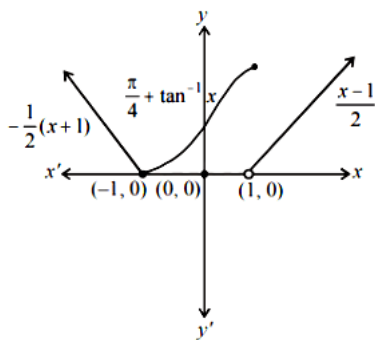
C. continuous on $\mathbb{R} - \{-1\}$ and differentiable on $\mathbb{R} - \{-1, 1\}$

D. both continuous and differentiable on $\mathbb{R} - \{-1\}$.

Answer: A

Solution:

$$f(x) = \begin{cases} \frac{-x-1}{2}, & x < -1 \\ \frac{\pi}{4} + \tan^{-1}x, & -1 \leq x \leq 1 \\ \frac{1}{2}(x-1), & x > 1 \end{cases}$$



It is clear from above graph that,

$f(x)$ is discontinuous at $x = 1$.

i.e. continuous on $\mathbb{R} - \{1\}$

$f(x)$ is non-differentiable at $x = -1, 1$

i.e. differentiable on $\mathbb{R} - \{-1, 1\}$.

Question 134

The derivative of $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ with respect to $\tan^{-1}\left(\frac{2x\sqrt{1-x^2}}{1-2x^2}\right)$ at $x = \frac{1}{2}$ is :

[Sep. 05, 2020 (II)]

Options:

A. $\frac{2\sqrt{3}}{5}$

B. $\frac{\sqrt{3}}{12}$

C. $\frac{2\sqrt{3}}{3}$

D. $\frac{\sqrt{3}}{10}$

Answer: D

Solution:

Let $u = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$

Put $x = \tan \theta \Rightarrow \theta = \tan^{-1}x$

$\therefore u = \tan^{-1}\left(\frac{\sec \theta - 1}{\tan \theta}\right) = \tan^{-1}\left(\tan \frac{\theta}{2}\right)$

$= \frac{\theta}{2} = \frac{1}{2}\tan^{-1}x$

$\therefore \frac{du}{dx} = \frac{1}{2} \times \frac{1}{(1+x^2)}$

$$\text{Let } v = \tan^{-1} \left(\frac{2x\sqrt{1-x^2}}{1-2x^2} \right)$$

$$\text{Put } x = \sin \phi \Rightarrow \phi = \sin^{-1} x$$

$$v = \tan^{-1} \left(\frac{2 \sin \phi \cos \phi}{\cos 2\phi} \right) = \tan^{-1}(\tan 2\phi)$$

$$= 2\phi = 2\sin^{-1} x$$

$$\frac{dv}{dx} = 2 \frac{1}{\sqrt{1-x^2}}$$

$$\frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{1-x^2}{4(1+x^2)}$$

$$\therefore \left(\frac{du}{dv} \right)_{\left(x = \frac{1}{2} \right)} = \frac{\sqrt{3}}{10}$$

Question135

If $(a + \sqrt{2}b \cos x)(a - \sqrt{2}b \cos y) = a^2 - b^2$, where $a > b > 0$ then $\frac{dx}{dy}$ at $\left(\frac{\pi}{4}, \frac{\pi}{4} \right)$ is

[Sep. 04, 2020 (I)]

Options:

A. $\frac{a-2b}{a+2b}$

B. $\frac{a-b}{a+b}$

C. $\frac{a+b}{a-b}$

D. $\frac{2a+b}{2a-b}$

Answer: C

Solution:

$$(a + \sqrt{2}b \cos x)(a - \sqrt{2}b \cos y) = a^2 - b^2$$

Differentiating both sides,

$$(-\sqrt{2}b \sin x)(a - \sqrt{2}b \cos y) + (a + \sqrt{2}b \cos x)$$

$$(\sqrt{2}b \sin y) \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{(\sqrt{2}b \sin x)(a - \sqrt{2}b \cos y)}{(a + \sqrt{2}b \cos x)(\sqrt{2}b \sin y)}$$

$$\therefore \left[\frac{dy}{dx} \right] \left(\frac{\pi}{4}, \frac{\pi}{4} \right) = \frac{a-b}{a+b} \Rightarrow \frac{dx}{dy} = \frac{a+b}{a-b}$$

Question136

If $y = \sum_{k=1}^6 k \cos^{-1} \left\{ \frac{3}{5} \cos kx - \frac{4}{5} \sin kx \right\}$, then $\frac{dy}{dx}$ at $x = 0$ is _____.

[NA Sep. 02, 2020 (II)]

Answer: 91

Solution:

$$y = \sum_{k=1}^6 k \cos^{-1} \left\{ \frac{3}{5} \cos kx - \frac{4}{5} \sin kx \right\}$$

$$\text{Let } \cos a = \frac{3}{5} \text{ and } \sin a = \frac{4}{5}$$

$$\begin{aligned}
 \therefore y &= \sum_{k=1}^6 k \cos^{-1} \{ \cos a \cos kx - \sin a \sin kx \} \\
 &= \sum_{k=1}^6 k \cos^{-1} (\cos(kx + a)) \\
 &= \sum_{k=1}^6 k(kx + a) = \sum_{k=1}^6 (k^2 x + ak) \\
 \therefore \frac{dy}{dx} &= \sum_{k=1}^6 k^2 = \frac{6(7)(13)}{6} = 91
 \end{aligned}$$

Question137

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as

$$f(x) = \begin{cases} 5, & \text{if } x \leq 1 \\ a + bx, & \text{if } 1 < x < 3 \\ b + 5x, & \text{if } 3 \leq x < 5 \\ 30, & \text{if } x \geq 5 \end{cases}$$

Then, f is :

[Jan 09, 2019 (I)]

Options:

- A. continuous if $a = 5$ and $b = 5$
- B. continuous if $a = -5$ and $b = 10$
- C. continuous if $a = 0$ and $b = 5$
- D. not continuous for any values of a and b

Answer: D

Solution:

Let $f(x)$ is continuous at $x = 1$, then

$$f(1^-) = f(1) = f(1^+)$$

$$\Rightarrow 5 = a + b \dots (1)$$

Let $f(x)$ is continuous at $x = 3$, then

$$f(3^-) = f(3) = f(3^+)$$

$$\Rightarrow a + 3b = b + 15 \dots (2)$$

Let $f(x)$ is continuous at $x = 5$, then

$$f(5^-) = f(5) = f(5^+)$$

$$\Rightarrow b + 25 = 30$$

$$\Rightarrow b = 30 - 25 = 5$$

From (1), $a = 0$

But $a = 0$, $b = 5$ do not satisfy equation (2)

Hence, $f(x)$ is not continuous for any values of a and b

Question138

Let f be a differentiable function such that $f(1) = 2$ and $f'(x) = f(x)$ for all $x \in \mathbb{R}$. If $h(x) = f(f(x))$, then $h'(1)$ is equal to :

[Jan. 12, 2019 (II)]

Options:

- A. $2e^2$
- B. $4e$
- C. $2e$
- D. $4e^2$

Answer: B

Solution:

Since, $f'(x) = f(x)$

Then, $\frac{f'(x)}{f(x)} = 1$

$$\Rightarrow \frac{f'(x)}{f(x)} = d x \Rightarrow \frac{f'(x)}{f(x)} d x = \int d x$$

$$\Rightarrow \ln |f(x)| = x + c$$

$$f(x) = \pm e^{x+c} \dots (1)$$

Since, the given condition

$$f(1) = 2$$

$$\text{From eq } ^n(1) f(x) = e^{x+c} = e^c e^x$$

$$\text{Then, } f(1) = e^c \cdot e^1$$

$$\Rightarrow 2 = e^c \cdot e$$

$$\Rightarrow \frac{2}{e} = e^c$$

Then, from eq ⁿ(1)

$$f(x) = \frac{2}{e} e^x$$

$$\Rightarrow f'(x) = \frac{2}{e} e^x$$

Now $h(x) = f(f(x))$

$$\Rightarrow h'(x) = f'(f(x)) \cdot f'(x)$$

$$h'(1) = f'(2) \cdot f'(1) = \frac{2}{e} e^2 \cdot \frac{2}{e} \cdot e = 4e$$

Question 139

Let $f(x) = \begin{cases} -1, & -2 \leq x < 0 \\ x^2 - 1, & 0 \leq x \leq 2 \end{cases}$ and $g(x) = |f(x)| + f(|x|)$. Then, in the interval $(-2, 2)$, g is :

[Jan. 11, 2019 (I)]

Options:

- A. differentiable at all points
- B. not continuous
- C. not differentiable at two points
- D. not differentiable at one point

Answer: D

Solution:

$$f(x) = \begin{cases} -1, & -2 \leq x < 0 \\ x^2 - 1, & 0 \leq x \leq 2 \end{cases}$$

$$\text{Then, } f(|x|) = \begin{cases} -1, & -2 \leq |x| < 0 \\ |x|^2 - 1, & 0 \leq |x| \leq 2 \end{cases}$$

$$\Rightarrow f(|x|) = x^2 - 1, -2 \leq x \leq 2$$

$$g(x) = \begin{cases} -1 + x^2 - 1, & -2 \leq x < 0 \\ (x^2 - 1) + |x^2 - 1|, & 0 \leq x \leq 2 \end{cases}$$

$$= \begin{cases} x^2, & -2 \leq x < 0 \\ 0, & 0 \leq x < 1 \\ 2(x^2 - 1), & 1 \leq x \leq 2 \end{cases}$$

$$g'(0^-) = 0, g'(0^+) = 0, g'(1^-) = 0, g'(1^+) = 4$$

$\Rightarrow g(x)$ is non-differentiable at $x = 1$

$\Rightarrow g(x)$ is not differentiable at one point.

Question140

If $x \log_e(\log_e x) - x^2 + y^2 = 4 (y > 0)$, then $\frac{dy}{dx}$ at $x = e$ is equal to :

[Jan. 11, 2019 (I)]

Options:

A. $\frac{(1+2e)}{2\sqrt{4+e^2}}$

B. $\frac{(2e-1)}{2\sqrt{4+e^2}}$

C. $\frac{(1+2e)}{\sqrt{4+e^2}}$

D. $\frac{e}{\sqrt{4+e^2}}$

Answer: B

Solution:

Consider the equation,

$$x \log_e(\log_e x) - x^2 + y^2 = 4$$

Differentiate both sides w.r.t. x ,

$$\log_e(\log_e x) + x \cdot \frac{1}{x \cdot \log_e x} - 2x + 2y \frac{dy}{dx} = 0$$

$$\log_e(\log_e x) + \frac{1}{\log_e x} - 2x + 2y \frac{dy}{dx} = 0$$

When $x = e$, $y = \sqrt{4+e^2}$. Put these values in (1),

$$0 + 1 - 2e + 2\sqrt{4+e^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{2e-1}{2\sqrt{4+e^2}}$$

Question141

Let K be the set of all real values of x where the function $f(x) = \sin |x| - |x| + 2(x - \pi) \cos |x|$ is not differentiable.

Then the set K is equal to

[Jan. 11, 2019 (II)]

Options:

A. ϕ (an empty set)

B. $\{\pi\}$

C. $\{0\}$

D. $\{0, \pi\}$

Answer: A

Solution:

$$f(x) = \sin |x| - |x| + 2(x - \pi) \cos |x|$$

There are two cases,

Case (1), $x > 0$

$$f(x) = \sin x - x + 2(x - \pi) \cos x$$

$$f'(x) = \cos x - 1 + 2(1 - 0) \cos x - 2 \sin(x - \pi)$$

$$f'(x) = 3 \cos x - 2(x - \pi) \sin x - 1$$

Then, function $f(x)$ is differentiable for all $x > 0$

Case (2) $x < 0$

$$f(x) = -\sin x + x + 2(x - \pi) \cos x$$

$$f'(x) = -\cos x + 1 - 2(x - \pi) \sin x + 2 \cos x$$

$$f'(x) = \cos x + 1 - 2(x - \pi) \sin x$$

Then, function $f(x)$ is differentiable for all $x < 0$

Now check for $x = 0$

$$f'(0^+) \text{ R.H.D.} = 3 - 1 = 2$$

$$f'(0^-) \text{ L.H.D.} = 1 + 1 = 2$$

$$\text{L.H.D.} = \text{R.H.D.}$$

Then, function $f(x)$ is differentiable for $x = 0$. So it is differentiable everywhere

Hence, $k = \phi$

Question 142

$$\text{Let } f(x) = \begin{cases} \max\{|x|, x^2\}, & |x| \leq 2 \\ 8 - 2|x|, & 2 < |x| \leq 4 \end{cases}$$

Let S be the set of points in the interval $(-4, 4)$ at which f is not differentiable. Then S :
[Jan 10, 2019 (I)]

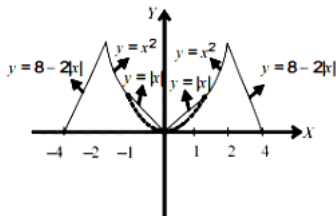
Options:

- A. is an empty set
- B. equals $\{-2, -1, 0, 1, 2\}$
- C. equals $\{-2, -1, 1, 2\}$
- D. equals $\{-2, 2\}$

Answer: B

Solution:

$$\text{Given } f(x) = \begin{cases} \max\{|x|, x^2\}, & |x| \leq 2 \\ 8 - 2|x|, & 2 < |x| \leq 4 \end{cases}$$



$\therefore f(x)$ is not differentiable at $-2, -1, 0, 1$ and 2 .

$$\therefore S = \{-2, -1, 0, 1, 2\}$$

Question 143

Let $f : (-1, 1) \rightarrow \mathbb{R}$ be a function defined by $f(x) = \max\{-|x|, -\sqrt{1-x^2}\}$. If K be the set of all points at which f is not differentiable, then K has exactly:
[Jan. 10, 2019 (II)]

Options:

- A. five elements
- B. one element
- C. three elements
- D. two elements

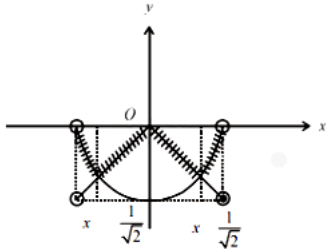
Answer: C

Solution:

Consider the function

$$f(x) = \max \{ -|x|, -\sqrt{1-x^2} \}$$

Now, the graph of the function



From the graph, it is clear that $f(x)$ is not differentiable at x

$$= 0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$$

$$\text{Then, } K = \left\{ -\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right\}$$

Hence, K has exactly three elements.

Question 144

Let S be the set of all points in $(-\pi, \pi)$ at which the function $f(x) = \min\{\sin x, \cos x\}$ is not differentiable. Then S is a subset of which of the following?

[Jan. 12, 2019 (I)]

Options:

A. $\left\{ -\frac{\pi}{4}, 0, \frac{\pi}{4} \right\}$

B. $\left\{ -\frac{3\pi}{4}, -\frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{4} \right\}$

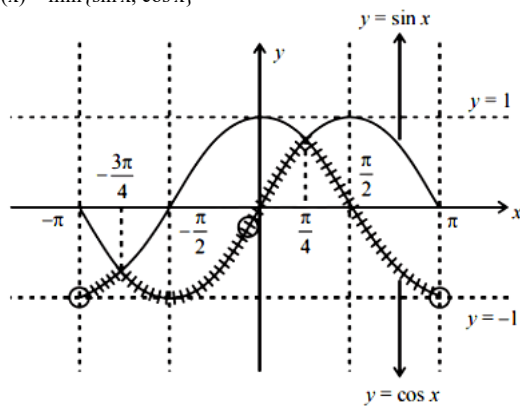
C. $\left\{ -\frac{\pi}{2}, -\frac{\pi}{4}, \frac{\pi}{4}, \frac{\pi}{2} \right\}$

D. $\left\{ -\frac{3\pi}{4}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{4} \right\}$

Answer: B

Solution:

$$f(x) = \min\{\sin x, \cos x\}$$



$$\therefore f(x) \text{ is not differentiable at } x = -\frac{3\pi}{4}, \frac{\pi}{4}$$

$$\therefore S = \left\{ -\frac{3\pi}{4}, \frac{\pi}{4} \right\}$$

$$\Rightarrow S \subseteq \left\{ -\frac{3\pi}{4}, -\frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{4} \right\}$$

Question145

For $x > 1$, if $(2x)^{2y} = 4e^{2x-2y}$, then $(1 + \log_e 2x) \frac{2dy}{dx}$ is equal to:

[Jan. 12, 2019 (I)]

Options:

A. $\frac{x \log_e 2x - \log_e 2}{x}$

B. $\log_e 2x$

C. $\frac{x \log_e 2x + \log_e 2}{x}$

D. $x \log_e 2x$

Answer: A

Solution:

Consider the equation,

$$(2x)^{2y} = 4e^{2x-2y}$$

Taking log on both sides

$$2y \ln(2x) = \ln 4 + (2x - 2y) \dots\dots(1)$$

Differentiating both sides w.r.t. x ,

$$2y \frac{1}{2x} + 2 \ln(2x) \frac{dy}{dx} = 0 + 2 - 2 \frac{dy}{dx}$$

$$2 \frac{dy}{dx} \left(1 + \ln(2x) \right) = 2 - \frac{2y}{x} = \frac{2x - 2y}{x} \dots\dots(2)$$

From (1) and (2),

$$\frac{dy}{dx} (1 + \ln 2x) = 1 - \frac{1}{x} \left(\frac{\ln 2 + x}{1 + \ln 2x} \right)$$

$$(1 + \ln 2x)^2 \frac{dy}{dx} = 1 + \ln(2x) - \left(\frac{x + \ln 2}{x} \right)$$

$$= \frac{x \ln(2x) - \ln 2}{x}$$

Question146

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that

$f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$, $x \in \mathbb{R}$. Then $f(2)$ equals:

[Jan 10, 2019 (I)]

Options:

A. -4

B. 30

C. -2

D. 8

Answer: C

Solution:

$$\text{Let } f(x) = x^3 + ax^2 + bx + c$$

$$f'(x) = 3x^2 + 2ax + b \Rightarrow f'(1) = 3 + 2a + b$$

$$f''(x) = 6x + 2a \Rightarrow f''(2) = 12 + 2a$$

$$f'''(x) = 6 \Rightarrow f'''(3) = 6$$

$$\therefore f(x) = x^3 + f'(1)x^2 + f''(2)x + f'''(3)$$

$$\therefore f'(1) = a \Rightarrow 3 + 2a + b = a \Rightarrow a + b = -3 \dots\dots(1)$$

$$\text{also } f''(2) = b \Rightarrow 12 + 2a = b \Rightarrow 2a - b = -12 \dots\dots(2)$$

$$\text{and } f'''(3) = c \Rightarrow c = 6$$

Add (1) and (2)

$$\begin{aligned}
 3a &= -15 \Rightarrow a = -5 \Rightarrow h = 2 \\
 \Rightarrow f(x) &= x^3 - 5x^2 + 2x + 6 \\
 \Rightarrow f(2) &= 8 - 20 + 4 + 6 = -2
 \end{aligned}$$

Question 147

If $x = 3 \tan t$ and $y = 3 \sec t$, then the value of $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$, is:

[Jan. 09, 2019 (II)]

Options:

A. $\frac{1}{3\sqrt{2}}$

B. $\frac{1}{6\sqrt{2}}$

C. $\frac{3}{2\sqrt{2}}$

D. $\frac{1}{6}$

Answer: B

Solution:

$$\begin{aligned}
 \because x &= 3 \tan t \Rightarrow \frac{dx}{dt} = 3 \sec^2 t \\
 \text{and } y &= 3 \sec t \Rightarrow \frac{dy}{dt} = 3 \sec t \cdot \tan t \\
 \therefore \frac{dy}{dx} &= \frac{dy/dt}{dx/dt} \therefore \frac{dy}{dx} = \frac{\tan t}{\sec t} = \sin t \\
 \therefore \frac{d^2y}{dx^2} &= \frac{d}{dt}(\sin t) \cdot \frac{dt}{dx} \\
 &= \cos t \cdot \frac{1}{3 \sec^2 t} \\
 \therefore \frac{d^2y}{dx^2} \left(\text{at } t = \frac{\pi}{4} \right) &= \frac{1}{3} \cdot \left(\frac{1}{\sqrt{2}} \right)^3 \\
 &= \frac{1}{6\sqrt{2}}
 \end{aligned}$$

Question 148

If the function f defined on $\left(\frac{\pi}{6}, \frac{\pi}{3} \right)$ by

$$f(x) = \begin{cases} \frac{\sqrt{2} \cos x - 1}{\cot x - 1}, & x \neq \frac{\pi}{4} \\ k, & x = \frac{\pi}{4} \end{cases}$$

is continuous, then k is equal to:
[April 09, 2019 (I)]

Options:

A. 2

B. $\frac{1}{2}$

C. 1

D. $\frac{1}{\sqrt{2}}$

Answer: B

Solution:

Since, $f(x)$ is continuous, then

$$\lim_{x \rightarrow \frac{\pi}{4}} f(x) = f\left(\frac{\pi}{4}\right)$$

$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} \cos x - 1}{\cot x - 1} = k$$

Now by L- hospital's rule

$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} \sin x}{\operatorname{cosec}^2 x} = k \Rightarrow \frac{\sqrt{2} \left(\frac{1}{\sqrt{2}}\right)}{(\sqrt{2})^2} = k \Rightarrow k = \frac{1}{2}$$

Question149

If $f(x) = [x] - \left\lfloor \frac{x}{4} \right\rfloor$, $x \in \mathbb{R}$, where $[x]$ denotes the greatest integer function, then:

[April 09, 2019 (II)]

Options:

A. f is continuous at $x = 4$.

B. $\lim_{x \rightarrow 4^+} f(x)$ exists but $\lim_{x \rightarrow 4} f(x)$ does not exist.

C. Both $\lim_{x \rightarrow 4^-} f(x)$ and $\lim_{x \rightarrow 4^+} f(x)$ exist but are not equal.

D. $\lim_{x \rightarrow 4^-} f(x)$ exists but $\lim_{x \rightarrow 4^+} f(x)$ does not exist.

Answer: A

Solution:

$$\text{L.H.L. } \lim_{x \rightarrow 4^-} \left([x] - \left\lfloor \frac{x}{4} \right\rfloor \right) = 3 - 0 = 3$$

$$\text{R.H.L. } \lim_{x \rightarrow 4^+} [x] - \left\lfloor \frac{x}{4} \right\rfloor = 4 - 1 = 3$$

$$f(4) = [4] - \left\lfloor \frac{4}{4} \right\rfloor = 4 - 1 = 3$$

$$\therefore \text{L.H.L.} = f(4) = \text{R.H.L.}$$

$$\therefore f(x) \text{ is continuous at } x = 4$$

Question150

If the function

$$f(x) = \begin{cases} a|\pi - x| + 1, & x \leq 5 \\ b|x - \pi| + 3, & x > 5 \end{cases}$$

is continuous at $x = 5$, then the value of $a - b$ is:

[April 09, 2019 (II)]

Options:

A. $\frac{2}{\pi + 5}$

B. $\frac{-2}{\pi + 5}$

C. $\frac{2}{\pi-5}$

D. $\frac{2}{5-\pi}$

Answer: D

Solution:

$$\text{R.H.L. } \lim_{x \rightarrow 5^+} b|(x-\pi)| + 3 = (5-\pi)b + 3$$

$$f(5) = \text{L.H.L. } \lim_{x \rightarrow 5^-} a|(\pi-x)| + 1 = a(5-\pi) + 1$$

$\therefore$ function is continuous at $x = 5$

$\therefore \text{LHL} = \text{RHL}$

$$(5-\pi)b + 3 = (5-\pi)a + 1$$

$$\Rightarrow 2 = (a-b)(5-\pi) \Rightarrow a-b = \frac{2}{5-\pi}$$

Question 151

Let $f : [-1, 3] \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} |x| + [x], & -1 \leq x < 1 \\ x + |x|, & 1 \leq x < 2 \\ x + [x], & 2 \leq x \leq 3 \end{cases}$$

where $[t]$ denotes the greatest integer less than or equal to t . Then, f is discontinuous at :
[April 08, 2019 (II)]

Options:

A. only one point

B. only two points

C. only three points

D. four or more points

Answer: C

Solution:

Given function is,

$$f(x) = \begin{cases} |x| + [x], & -1 \leq x < 1 \\ x + |x|, & 1 \leq x < 2 \\ x + [x], & 2 \leq x \leq 3 \end{cases}$$

$$= \begin{cases} -x - 1, & -1 \leq x < 0 \\ x, & 0 \leq x < 1 \\ 2x, & 1 \leq x < 2 \\ x + 2, & 2 \leq x < 3 \\ 6, & x = 3 \end{cases}$$

$$\Rightarrow f(-1) = 0, f(-1^+) = 0$$

$$f(0^-) = -1, f(0) = 0, f(0^+) = 0$$

$$f(1^-) = 1, f(1) = 2, f(1^+) = 2$$

$$f(2^-) = 4, f(2) = 4, f(2^+) = 4;$$

$$f(3^-) = 5, f(3) = 6$$

$f(x)$ is discontinuous at $x = \{0, 1, 3\}$

Hence, $f(x)$ is discontinuous at only three points.

Question 152

If

$$f(x) = \begin{cases} \frac{\sin(p+1)x + \sin x}{x} & , x < 0 \\ q & , x = 0 \\ \frac{\sqrt{x+x^2} - \sqrt{x}}{x^{3/2}} & , x > 0 \end{cases} \text{ is continuous at } x = 0,$$

then the ordered pair (p, q) is equal to:

[April 10, 2019 (I)]

Options:

A. $\left(-\frac{3}{2}, -\frac{1}{2}\right)$

B. $\left(-\frac{1}{2}, \frac{3}{2}\right)$

C. $\left(-\frac{3}{2}, \frac{1}{2}\right)$

D. $\left(\frac{5}{2}, \frac{1}{2}\right)$

Answer: C

Solution:

$$f(x) = \begin{cases} \frac{\sin(p+1)x + \sin x}{x} & , x < 0 \\ q & , x = 0 \\ \frac{\sqrt{x+x^2} - \sqrt{x}}{x^{3/2}} & , x > 0 \end{cases} \text{ is continuous at } x = 0,$$

Therefore, $f(0) = f(0) = f(0^+) \dots (1)$

$$f(0^-) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{\sin(p+1)(-h) + \sin(-h)}{-h}$$

$$= \lim_{h \rightarrow 0} \left[\frac{-\sin(p+1)h}{-h} + \frac{\sin h}{h} \right]$$

$$= \lim_{h \rightarrow 0} \frac{\sin(p+1)h}{h(p+1)} \times (p+1) + \lim_{h \rightarrow 0} \frac{\sin h}{h}$$

$$= (p+1) + 1 = p+2 \dots (2)$$

$$\text{And } f(0^+) = \lim_{h \rightarrow 0} f(0+h) = \frac{\sqrt{h^2+h} - \sqrt{h}}{h^{3/2}}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{h^2} [\sqrt{h+1} - 1]}{\frac{1}{h} \left(\frac{1}{h^2}\right)}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{h+1} - 1}{h} \times \frac{\sqrt{h+1} + 1}{\sqrt{h+1} + 1} = \lim_{h \rightarrow 0} \frac{h+1-1}{h(\sqrt{h+1} + 1)}$$

$$= \lim_{h \rightarrow 0} \frac{1}{\sqrt{h+1} + 1} = \frac{1}{1+1} = \frac{1}{2} \dots (3)$$

Now, from equation (1),

$$f(0^-) = f(0) = f(0^+) \Rightarrow p+2 = q = \frac{1}{2}$$

$$\Rightarrow q = \frac{1}{2} \text{ and } p = -\frac{3}{2}$$

$$\therefore (p, q) = \left(-\frac{3}{2}, \frac{1}{2}\right)$$

Question 153

Let $f(x) = \log_e(\sin x)$, ($0 < x < \pi$) and $g(x) = \sin^{-1}(e^{-x})$, ($x \geq 0$).

If α is a positive real number such that $a = (f \circ g)'(\alpha)$ and $b = (f \circ g)(\alpha)$, then:

[April 10, 2019 (II)]

Options:

A. $a\alpha^2 + b\alpha + a = 0$

B. $a\alpha^2 - b\alpha - a = 1$

C. $a\alpha^2 - b\alpha - a = 0$

D. $a\alpha^2 + b\alpha - a = -2a^2$

Answer: B

Solution:

$$\begin{aligned}f(x) &= \ln(\sin x), g(x) = \sin^{-1}(e^{-x}) \\ \Rightarrow f(g(x)) &= \ln(\sin(\sin^{-1}e^{-x})) = -x \\ \Rightarrow f(g(x)) &= -\alpha \\ \text{But given that } (f \circ g)(\alpha) &= b \\ \therefore -\alpha &= b \text{ and } f'(g(\alpha)) = a, \text{ i.e., } a = -1 \\ \therefore a\alpha^2 - b\alpha - a &= -\alpha^2 + \alpha^2 - (-1) \\ \Rightarrow a\alpha^2 - b\alpha - a &= 1.\end{aligned}$$

Question154

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable at $c \in \mathbb{R}$ and $f(c) = 0$. If $g(x) = |f(x)|$, then at $x = c$, g is:
[April 10, 2019 (I)]

Options:

A. not differentiable if $f'(c) = 0$

B. differentiable if $f''(c) \neq 0$

C. differentiable if $f'(c) = 0$

D. not differentiable

Answer: C

Solution:

$$\begin{aligned}g'(c) &= \lim_{x \rightarrow c} \frac{g(x) - g(c)}{x - c} \\ \Rightarrow g'(c) &= \lim_{x \rightarrow c} \frac{|f(x)| - |f(c)|}{x - c} \\ \text{Since, } f(c) &= 0 \text{ Then, } g'(c) = \lim_{x \rightarrow c} \frac{|f(x)|}{x - c} \\ \Rightarrow g'(c) &= \lim_{x \rightarrow c} \frac{f(x)}{x - c}; \text{ if } f(x) > 0 \\ \text{and } g'(c) &= \lim_{x \rightarrow c} \frac{-f(x)}{x - c}; \text{ if } f(x) < 0 \\ \Rightarrow g'(c) &= f'(c) = -f'(c) \\ \Rightarrow 2f'(c) &= 0 \Rightarrow f'(c) = 0 \\ \text{Hence, } g(x) &\text{ is differentiable if } f'(c) = 0\end{aligned}$$

Question155

Let $f(x) = 15 - |x - 10|$; $x \in \mathbb{R}$. Then the set of all values of x at which the function, $g(x) = f(f(x))$ is not differentiable, is:

[April 09, 2019 (I)]

Options:

- A. {5, 10, 15}
- B. {10, 15}
- C. {5, 10, 15, 20}
- D. {10}

Answer: A

Solution:

Since, $f(x) = 15 - |10 - x|$
 $\therefore g(x) = f(f(x)) = 15 - |10 - [15 - |10 - x|]|$
 $= 15 - ||10 - x| - 5|$
 $\therefore$ Then, the points where function $g(x)$ is Non-differentiable are
 $10 - x = 0$ and $|10 - x| = 5$
 $\Rightarrow x = 10$ and $x - 10 = \pm 5$
 $\Rightarrow x = 10$ and $x = 15, 5$

Question 156

If $f(1) = 1$, $f'(1) = 3$, then the derivative of $f(f(f(x))) + (f(x))^2$ at $x = 1$ is :
[April 08, 2019 (II)]

Options:

- A. 33
- B. 12
- C. 15
- D. 9

Answer: A

Solution:

Let $g(x) = f(f(f(x))) + (f(x))^2$
Differentiating both sides w.r.t. x , we get
 $g'(x) = f'(f(f(x)))f'(f(x))f'(x) + 2f(x)f'(x)$
 $g'(1) = f'(f(f(1)))f'(f(1))f'(1) + 2f(1)f'(1)$
 $= f'(f(1))f'(1)f'(1) + 2f(1)f'(1)$
 $= 3 \times 3 \times 3 + 2 \times 1 \times 3 = 27 + 6 = 33$

Question 157

If $e^y + xy = e$, the ordered pair $\left(\frac{dy}{dx}, \frac{d^2y}{dx^2}\right)$ at $x = 0$ is equal to :

[April 12, 2019 (I)]

Options:

- A. $\left(\frac{1}{e}, -\frac{1}{e^2}\right)$
- B. $\left(-\frac{1}{e}, \frac{1}{e^2}\right)$
- C. $\left(\frac{1}{e}, \frac{1}{e^2}\right)$

D. $\left(-\frac{1}{e}, -\frac{1}{e^2}\right)$

Answer: B

Solution:

Given, $e^y + xy = e \dots (i)$

Putting $x = 0$ in (i), $\Rightarrow e^y = e \Rightarrow y = 1$

On differentiating (i) w. r. to x

$$e^y \frac{dy}{dx} + x \frac{dy}{dx} + y = 0 \dots (ii)$$

Putting $y = 1$ and $x = 0$ in (ii),

$$e \frac{dy}{dx} + 0 + 1 = 0 \Rightarrow \frac{dy}{dx} = -\frac{1}{e}$$

On differentiating (ii) w. r. to x ,

$$e^y \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot e^y + x \frac{d^2y}{dx^2} + \frac{dy}{dx} + \frac{dy}{dx} = 0$$

Putting $y = 1$, $x = 0$ and $\frac{dy}{dx} = -\frac{1}{e}$ in (iii),

$$e \frac{d^2y}{dx^2} + \frac{1}{e} - \frac{2}{e} = 0 \Rightarrow \frac{d^2y}{dx^2} = \frac{1}{e^2}$$

Hence, $\left(\frac{dy}{dx}, \frac{d^2y}{dx^2}\right) = \left(-\frac{1}{e}, \frac{1}{e^2}\right)$

Question158

The derivative of $\tan^{-1}\left(\frac{\sin x - \cos x}{\sin x + \cos x}\right)$, with respect to $\frac{x}{2}$ where left $\left(x \in \left(0, \frac{\pi}{2}\right)\right)$ is:

[April 12, 2019 (II)]

Options:

A. 1

B. $\frac{2}{3}$

C. $\frac{1}{2}$

D. 2

Answer: D

Solution:

$$f(x) = \tan^{-1}\left(\frac{\tan x - 1}{\tan x + 1}\right)$$

$$= -\tan^{-1}\left(\tan\left(\frac{\pi}{4} - x\right)\right) \quad \left[\because \frac{\pi}{4} - x \in \left(-\frac{\pi}{4}, \frac{\pi}{4}\right) \right]$$

$$\text{So, } f(x) = -\left(\frac{\pi}{4} - x\right) = x - \frac{\pi}{4}$$

$$\text{Let } y \Rightarrow f(y) = 2y - \frac{\pi}{4}$$

$$\text{Now, differentiate w.r.t. } y, \frac{df(y)}{dy} = 2.$$

Question159

If $2y = \left(\cot^{-1}\left(\frac{\sqrt{3}\cos x + \sin x}{\cos x - \sqrt{3}\sin x}\right)\right)^2$, $x \in \left(0, \frac{\pi}{2}\right)$ then $\frac{dy}{dx}$ is equal to :

[April 08, 2019 (I)]

Options:

A. $\frac{\pi}{6} - x$

B. $x - \frac{\pi}{6}$

C. $\frac{\pi}{3} - x$

D. None

Answer: D

Solution:

$$\begin{aligned}
 2y &= \left[\cot^{-1} \left(\frac{\frac{\sqrt{3}}{2} \cos x + \frac{1}{2} \sin x}{\frac{1}{2} \cos x - \frac{\sqrt{3}}{2} \sin x} \right) \right]^2 \\
 \Rightarrow 2y &= \left[\cot^{-1} \left(\frac{\cos \left(\frac{\pi}{6} - x \right)}{\sin \left(\frac{\pi}{6} - x \right)} \right) \right]^2 \\
 \Rightarrow 2y &= \left[\cot^{-1} \left(\cot \left(\frac{\pi}{6} - x \right) \right) \right]^2 \because \frac{\pi}{6} - x \in \left(-\frac{\pi}{3}, \frac{\pi}{6} \right) \\
 \Rightarrow 2y &= \begin{cases} \left(\frac{7\pi}{6} - x \right)^2, & \text{if } \frac{\pi}{6} - x \in \left(-\frac{\pi}{3}, 0 \right) \\ \left(\frac{\pi}{6} - x \right)^2, & \text{if } \frac{\pi}{6} - x \in \left(0, \frac{\pi}{6} \right) \end{cases} \\
 \Rightarrow \frac{dy}{dx} &= \begin{cases} x - \frac{7\pi}{6} & \text{if } x \in \left(\frac{\pi}{6}, \frac{\pi}{2} \right) \\ x - \frac{\pi}{6} & \text{if } x \in \left(0, \frac{\pi}{6} \right) \end{cases}
 \end{aligned}$$

Question 160

If the function f defined as

$$f(x) = \frac{1}{x} - \frac{k-1}{e^{2x}-1}$$

$x \neq 0$, is continuous at $x = 0$,
then the ordered pair $(k, f(0))$ is equal to?
[Online April 16, 2018]

Options:

A. (3,1)

B. (3,2)

C. $\left(\frac{1}{3}, 2 \right)$

D. (2,1)

Answer: A

Solution:

If the function is continuous at $x = 0$, then

$$\lim_{x \rightarrow 0} f(x) \text{ will exist and } f(0) = \lim_{x \rightarrow 0} f(x)$$

$$\text{Now, } \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{k-1}{e^{2x}-1} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{e^{2x}-1-kx+x}{(x)(e^{2x}-1)} \right)$$

$$= \lim_{x \rightarrow 0} \left[\frac{\left(1 + 2x + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \dots\right) - 1 - kx + x}{(x) \left(\left(1 + 2x + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \dots\right) - 1 \right)} \right]$$

$$= \lim_{x \rightarrow 0} \left[\frac{(3-k)x + \frac{4x^2}{2!} + \frac{8x^3}{3!} + \dots}{\left(2x^2 + \frac{4x^3}{2!} + \frac{8x^3}{3!} + \dots\right)} \right]$$

For the limit to exist, power of x in the numerator should be greater than or equal to the power of x in the denominator. Therefore, coefficient of x in numerator is equal to zero

$$\Rightarrow 3 - k = 0$$

$$\Rightarrow k = 3$$

So the limit reduces to

$$\lim_{x \rightarrow 0} \frac{(x^2) \left(\frac{4}{2!} + \frac{8x}{3!} + \dots \right)}{(x^2) \left(2 + 4x + \frac{8x^2}{3!} + \dots \right)}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{4}{2!} + \frac{8x}{3!} + \dots}{2 + 4x + \frac{8x^2}{3!} + \dots} = 1$$

Hence $f(0) = 1$

Question 161

Let $f(x) = \begin{cases} (x-1)^{\frac{1}{2-x}}, & x > 1, x \neq 2 \\ k, & x = 2 \end{cases}$ The value of k for which f is continuous at $x = 2$ is

[Online April 15, 2018]

Options:

A. e^{-2}

B. e

C. e^{-1}

D. 1

Answer: C

Solution:

Since $f(x)$ is continuous at $x = 2$.

$$\therefore \lim_{x \rightarrow 2} f(x) = f(2)$$

$$\Rightarrow \lim_{x \rightarrow 2} (x-1)^{\frac{1}{2-x}} = k$$

$$\therefore e^1 = k$$

$$\text{where } 1 = \lim_{x \rightarrow 2} (x-1-1) \times \frac{1}{2-x} = \lim_{x \rightarrow 2} \frac{x-2}{2-x}$$

$$= \lim_{x \rightarrow 2} \left(\frac{x-2}{x-2} \right)$$

$$\Rightarrow k = e^{-1}$$

Question 162

Let $S = \{t \in \mathbb{R} : f(x) = |x - \pi| (e^{|x|} - 1) \sin |x| \text{ is not differentiable at } t\}$. Then the set S is equal to :
[2018]

Options:

A. $\{0\}$

- B. $\{\pi\}$
- C. $\{0, \pi\}$
- D. ϕ (an empty set)

Answer: D

Solution:

$$f(x) = |x - \pi| (e^{|x|} - 1) \sin |x|$$

Check differentiability of $f(x)$ at $x = \pi$ and $x = 0$
at $x = \pi$:

$$R.H.D. = \lim_{h \rightarrow 0} \frac{|\pi + h - \pi|(e^{|\pi + h|} - 1) \sin |\pi + h| - 0}{h}$$

$$L.H.D. = \lim_{h \rightarrow 0} \frac{|\pi - h - \pi|(e^{|\pi - h|} - 1) \sin |\pi - h| - 0}{-h} = 0$$

$$\therefore RHD = LHD$$

Therefore, function is differentiable at $x = \pi$

at $x = 0$:

$$R.H.D. = \lim_{h \rightarrow 0} \frac{|h - \pi|(e^{|h|} - 1) \sin |h| - 0}{h} = 0$$

$$L.H.D. = \lim_{h \rightarrow 0} \frac{|-h - \pi|(e^{-|h|} - 1) \sin |-h| - 0}{-h} = 0$$

$$\therefore RHD = LHD$$

Therefore, function is differentiable.

at $x = 0$.

Since, the function $f(x)$ is differentiable at all the points including π and 0 .

i.e., $f(x)$ is every where differentiable.

Therefore, there is no element in the set S .

$\Rightarrow S = \phi$ (an empty set)

Question 163

Let $S = \{(\lambda, \mu) \in \mathbb{R} \times \mathbb{R} : f(t) = (|\lambda|e^{|t|} - \mu) \cdot \sin(2|t|), t \in \mathbb{R}, \cdot \text{ is a differentiable function} \}$. Then S is a subset of?

[Online April 15, 2018]

Options:

- A. $\mathbb{R} \times [0, \infty)$
- B. $(-\infty, 0) \times \mathbb{R}$
- C. $[0, \infty) \times \mathbb{R}$
- D. $\mathbb{R} \times (-\infty, 0)$

Answer: A

Solution:

$$S = \{(\lambda, \mu) \in \mathbb{R} \times \mathbb{R} : f(t) = (|\lambda|e^{|t|} - \mu) \sin(2|t|), t \in \mathbb{R}$$

$$f(t) = (|\lambda|e^{|t|} - \mu) \sin(2|t|)$$

$$= \begin{cases} (|\lambda|e^t - \mu) \sin 2t, & t > 0 \\ (|\lambda|e^{-t} - \mu)(-\sin 2t), & t < 0 \end{cases}$$

$$f'(t) = \begin{cases} (|\lambda|e^t) \sin 2t + (|\lambda|e^t - \mu)(2 \cos 2t), & t > 0 \\ |\lambda|e^{-t} \sin 2t + (|\lambda|e^{-t} - \mu)(-2 \cos 2t), & t < 0 \end{cases}$$

As, $f(t)$ is differentiable

$\therefore LHD = RHD$ at $t = 0$

$$\Rightarrow |\lambda| \cdot \sin 2(0) + (|\lambda|e^0 - \mu)2 \cos(0)$$

$$= |\lambda|e^0 \cdot \sin 2(0) - 2 \cos(0) (|\lambda|e^0 - \mu)$$

$$\Rightarrow 0 + (|\lambda| - \mu)2 = 0 - 2(|\lambda| - \mu)$$

$$\Rightarrow 4(|\lambda| - \mu) = 0$$

$$\Rightarrow |\lambda| = \mu$$

So, $S \equiv (\lambda, \mu) = \{\lambda \in \mathbb{R} \text{ \& } \mu \in [0, \infty)\}$
 Therefore set S is subset of $\mathbb{R} \times [0, \infty)$

Question164

If $x = \sqrt{2^{\operatorname{cosec}^{-1}t}}$ and $y = \sqrt{2^{\sec^{-1}t}}$ ($|t| \geq 1$), then $\frac{dy}{dx}$ is equal to.
 [Online April 16, 2018]

Options:

A. $\frac{y}{x}$

B. $-\frac{y}{x}$

C. $-\frac{x}{y}$

D. $\frac{x}{y}$

Answer: B

Solution:

$$\text{Here, } \frac{dx}{dt} = \frac{1}{2\sqrt{2^{\operatorname{cosec}^{-1}t}}} 2^{\operatorname{cosec}^{-1}t} \log 2 \cdot \frac{-1}{x\sqrt{x^2-1}}$$

$$\frac{dy}{dt} = \frac{1}{2\sqrt{2^{\sec^{-1}t}}} 2^{\sec^{-1}t} \log 2 \cdot \frac{1}{x\sqrt{x^2-1}}$$

$$\therefore \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{-\sqrt{2^{\operatorname{cosec}^{-1}t}}}{\sqrt{2^{\sec^{-1}t}}} \frac{2^{\sec^{-1}t}}{2^{\operatorname{cosec}^{-1}t}}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = -\sqrt{\frac{2^{\sec^{-1}t}}{2^{\operatorname{cosec}^{-1}t}}} = -\frac{y}{x}$$

Question165

If $f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2 \sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$, then $\lim_{x \rightarrow 0} \frac{f'(x)}{x}$

[Online April 15, 2018]

Options:

A. Exists and is equal to -2

B. Does not exist

C. Exist and is equal to 0

D. Exists and is equal to 2

Answer: A

Solution:

$$\begin{aligned}
 f(x) &= \begin{vmatrix} \cos x & x & 1 \\ 2 \sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix} \\
 &= \cos x(x^2 - 2x^2) - x(2 \sin x - 2x \tan x) + 1(2x \sin x - x^2 \tan x) \\
 &= -x^2 \cos x - 2x \sin x + 2x^2 \tan x + 2x \sin x - x^2 \tan x \\
 &= x^2 \tan x - x^2 \cos x = x^2(\tan x - \cos x) \\
 \Rightarrow f'(x) &= 2x(\tan x - \cos x) + x^2(\sec^2 x + \sin x) \\
 \therefore \lim_{x \rightarrow 0} \frac{f'(x)}{x} &= \lim_{x \rightarrow 0} \frac{2x(\tan x - \cos x) + x^2(\sec^2 x + \sin x)}{x} \\
 &= \lim_{x \rightarrow 0} (\tan x - \cos x) + x(\sec^2 x + \sin x) \\
 &= 2(0 - 1) + 0 = -2 \\
 \text{So, } \lim_{x \rightarrow 0} \frac{f'(x)}{x} &= -2
 \end{aligned}$$

Question 166

If $f(x) = \sin^{-1}\left(\frac{2 \times 3^x}{1 + 9^x}\right)$, then $f'\left(-\frac{1}{2}\right)$ equals.

[Online April 15, 2018]

Options:

- A. $\sqrt{3} \log_e \sqrt{3}$
- B. $-\sqrt{3} \log_e \sqrt{3}$
- C. $-\sqrt{3} \log_e 3$
- D. $\sqrt{3} \log_e 3$

Answer: A

Solution:

$$\begin{aligned}
 \text{Since } f(x) &= \sin^{-1}\left(\frac{2 \times 3^x}{1 + 9^x}\right) \\
 \text{Suppose } 3^x &= \tan t \\
 \Rightarrow f(x) &= \sin^{-1}\left(\frac{2 \tan t}{1 + \tan^2 t}\right) = \sin^{-1}(\sin 2t) = 2t \\
 &= 2 \tan^{-1}(3x) \\
 \text{So, } f'(x) &= \frac{2}{1 + (3^x)^2} \times 3^x \cdot \log_e 3 \\
 \therefore f'\left(-\frac{1}{2}\right) &= \frac{2}{1 + \left(3^{-\frac{1}{2}}\right)^2} \times 3^{-\frac{1}{2}} \cdot \log_e 3 \\
 &= \frac{1}{2} \times \sqrt{3} \times \log_e 3 = \sqrt{3} \times \log_e \sqrt{3}
 \end{aligned}$$

Question 167

If $x^2 + y^2 + \sin y = 4$, then the value of $\frac{d^2y}{dx^2}$ at the point $(-2, 0)$ is

[Online April 15, 2018]

Options:

- A. -34
- B. -32
- C. -2

D. 4

Answer: A

Solution:

Given, $x^2 + y^2 + \sin y = 4$

After differentiating the above equation w. r. t. x we get

$$2x + 2y \frac{dy}{dx} + \cos y \frac{dy}{dx} = 0 \dots\dots(1)$$

$$\Rightarrow 2x + (2y + \cos y) \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2x}{2y + \cos y}$$

$$\text{At } (-2, 0), \left(\frac{dy}{dx} \right)_{(-2, 0)} = \frac{-2 \times -2}{2 \times 0 + \cos 0}$$

$$\Rightarrow \left(\frac{dy}{dx} \right)_{(-2, 0)} = \frac{4}{0+1}$$

$$\Rightarrow \left(\frac{dy}{dx} \right)_{(-2, 0)} = 4 \dots\dots (2)$$

Again differentiating equation (1) w. r. t to x, we get

$$2 + 2 \left(\frac{dy}{dx} \right)^2 + 2y \frac{d^2y}{dx^2} - \sin y \left(\frac{dy}{dx} \right)^2 + \cos y \frac{d^2y}{dx^2} = 0$$

$$\Rightarrow 2 + (2 - \sin y) \left(\frac{dy}{dx} \right)^2 + (2y + \cos y) \frac{d^2y}{dx^2} = 0$$

$$\Rightarrow (2y + \cos y) \frac{d^2y}{dx^2} = -2 - (2 - \sin y) \left(\frac{dy}{dx} \right)^2$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{-2 - (2 - \sin y) \left(\frac{dy}{dx} \right)^2}{2y + \cos y}$$

So, at $(-2, 0)$,

$$\frac{d^2y}{dx^2} = \frac{-2 - (2 - 0) \times 4^2}{2 \times 0 + 1}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{-2 - 2 \times 16}{1}$$

$$\Rightarrow \frac{d^2y}{dx^2} = -34$$

Question168

The value of k for which the function

$$f(x) = \begin{cases} \left(\frac{4}{5}\right)^{\frac{\tan 4x}{\tan 5x}}, & 0 < x < \frac{\pi}{2} \\ k + \frac{2}{5}, & x = \frac{\pi}{2} \end{cases}$$

is continuous at $x = \frac{\pi}{2}$, is:

[Online April 9, 2017]

Options:

A. $\frac{17}{20}$

B. $\frac{2}{5}$

C. $\frac{3}{5}$

D. $-\frac{2}{5}$

Answer: C

Solution:

$$\lim_{x \rightarrow \pi/2} f(x) = f(\pi/2)$$

$$\Rightarrow k + 2/5 = 1 \Rightarrow k = 1 - \frac{2}{5} \Rightarrow k = \frac{3}{5}$$

Question169

If for $x \in \left(0, \frac{1}{4}\right)$, the derivative of $\tan^{-1}\left(\frac{6x\sqrt{x}}{1-9x^3}\right)$ is $\sqrt{x} \cdot g(x)$, then $g(x)$ equals:

[2017]

Options:

A. $\frac{3}{1+9x^3}$

B. $\frac{9}{1+9x^3}$

C. $\frac{3x\sqrt{x}}{1-9x^3}$

D. $\frac{3x}{1-9x^3}$

Answer: B

Solution:

Let $F(x) = \tan^{-1}\left(\frac{6x\sqrt{x}}{1-9x^3}\right)$ where $x \in \left(0, \frac{1}{4}\right)$

$$= \tan^{-1}\left(\frac{2 \cdot (3x^{3/2})}{1 - (3x^{3/2})^2}\right) = 2\tan^{-1}(3x^{3/2})$$

As $3x^{3/2} \in \left(0, \frac{3}{8}\right)$

$$\left[\because 0 < x < \frac{1}{4} \Rightarrow 0 < x^{3/2} < \frac{1}{8} \Rightarrow 0 < 3x^{3/2} < \frac{3}{8} \right]$$

So $\frac{dF(x)}{dx} = 2 \times \frac{1}{1+9x^3} \times 3 \times \frac{3}{2} \times x^{1/2} = \frac{9}{1+9x^3} \sqrt{x}$

On comparing

$$\therefore g(x) = \frac{9}{1+9x^3}$$

Question170

If $2x = y^{\frac{1}{5}} + y^{-\frac{1}{5}}$ and $(x^2 - 1)\frac{d^2y}{dx^2} + \lambda x \frac{dy}{dx} + ky = 0$ then $\lambda + k$ is equal to

[Online April 9, 2017]

Options:

A. - 23

B. - 24

C. 26

D. - 26

Answer: B

Solution:

$$\begin{aligned}
y^{1/5} + y^{-1/5} &= 2x \\
\Rightarrow \left(\frac{1}{5}y^{-4/5} - \frac{1}{5}y^{-6/5} \right) \frac{dy}{dx} &= 2 \\
\Rightarrow y'(y^{1/5} - y^{-1/5}) &= 10y \\
\Rightarrow y^{1/5} + y^{-1/5} &= \frac{2x}{y^{1/5} - y^{-1/5}} \\
\Rightarrow y^{1/5} - y^{-1/5} &= \sqrt{4x^2 - 4} \\
\Rightarrow y'(2\sqrt{x^2 - 1}) &= 10y \\
\Rightarrow y''(2\sqrt{x^2 - 1}) + y'2 \frac{2x}{2\sqrt{x^2 - 1}} &= 10y' \\
\Rightarrow y''(x^2 - 1) + xy' &= 5\sqrt{x^2 - 1}(y') \\
\Rightarrow y''(x^2 - 1) + xy' - 25y &= 0 \\
\lambda = 1, k = -25
\end{aligned}$$

Question171

Let f be a polynomial function such that $f(3x) = f'(x)$, $f''(x)$, for all $x \in \mathbb{R}$. Then :
[Online April 9, 2017]

Options:

- A. $f(b) + f'(b) = 28$
- B. $f''(b) - f'(b) = 0$
- C. $f''(b) - f'(b) = 4$
- D. $f(b) - f'(b) + f''(b) = 10$

Answer: B

Solution:

$$\begin{aligned}
\text{Let } f(x) &= ax^3 + bx^2 + cx + d \\
\Rightarrow f(3x) &= 27ax^3 + 9bx^2 + 3cx + d \\
\Rightarrow f'(x) &= 3ax^2 + 2bx + c \\
\Rightarrow f''(x) &= 6ax + 2b \\
\Rightarrow f(3x) &= f'(x)f''(x) \\
\Rightarrow 27a &= 18a^2 \\
\Rightarrow a &= \frac{3}{2}, b = 0, c = 0, d = 0 \\
\Rightarrow f(x) &= \frac{3}{2}x^3, \\
f'(x) &= \frac{9}{2}x^2, f''(x) = 9x \\
\Rightarrow f'(2) &= 18 \\
\text{and } f''(2) &= 18 \\
\Rightarrow f''(b) - f'(b) &= 0
\end{aligned}$$

Question172

If $y = \left[x + \sqrt{x^2 - 1} \right]^{15} + \left[x - \sqrt{x^2 - 1} \right]^{15}$, then $(x^2 - 1) \frac{d^2y}{dx^2} + x \frac{dy}{dx}$ is equal to
[Online April 8, 2017]

Options:

- A. $12y$
- B. $224y^2$
- C. $225y^2$
- D. $225y$

Answer: D

Solution:

$$y = \{x + \sqrt{x^2 - 1}\}^{15} + \{x - \sqrt{x^2 - 1}\}^{15}$$

Differentiate w.r.t. 'x'

$$\frac{dy}{dx} = 15(x + \sqrt{x^2 - 1})^{14} \left[1 + \frac{x}{\sqrt{x^2 - 1}} \right] + 15(x - \sqrt{x^2 - 1})^{14} \left(1 - \frac{x}{\sqrt{x^2 - 1}} \right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{15}{\sqrt{x^2 - 1}} \cdot y$$

$$\Rightarrow \sqrt{x^2 - 1} \cdot \frac{dy}{dx} = 15y$$

Again differentiating both sides w.r.t. x

$$\frac{x}{\sqrt{x^2 - 1}} \cdot \frac{dy}{dx} + \sqrt{x^2 - 1} \frac{d^2y}{dx^2} = 15 \frac{dy}{dx}$$

$$\Rightarrow x \frac{dy}{dx} + (x^2 - 1) \frac{d^2y}{dx^2}$$

$$= 15 \sqrt{x^2 - 1} \cdot \frac{15}{\sqrt{x^2 - 1}} \cdot y = 225y$$

Question 173

Let $a, b \in \mathbb{R}$, ($a \neq 0$). if the function f defined as

$$f(x) = \begin{cases} \frac{2x^2}{a} & , \quad 0 \leq x < 1 \\ a & , \quad 1 \leq x < \sqrt{2} \\ \frac{2b^2 - 4b}{x^3} & , \quad \sqrt{2} \leq x \leq \infty \end{cases}$$

is continuous in the interval $[0, \infty)$, then an ordered pair (a, b) is:
[Online April 10, 2016]

Options:

A. $(-\sqrt{2}, 1 - \sqrt{3})$

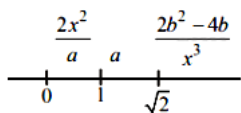
B. $(\sqrt{2}, -1 + \sqrt{3})$

C. $(\sqrt{2}, 1 - \sqrt{3})$

D. $(-\sqrt{2}, 1 + \sqrt{3})$

Answer: C

Solution:



Continuity at $x = 1$

$$\frac{2}{a} = a \Rightarrow a = \pm\sqrt{2}$$

Continuity at $x = \sqrt{2}$ $a = \sqrt{2}$

$$a = \frac{2b^2 - 4b}{2\sqrt{2}}$$

$$\text{Put } a = \sqrt{2}$$

$$2 = b^2 - 2b \Rightarrow b^2 - 2b - 2 = 0$$

$$b = \frac{2 \pm \sqrt{4 + 4 \cdot 2}}{2} = 1 \pm \sqrt{3}$$

$$\text{So, } (a, b) = (\sqrt{2}, 1 - \sqrt{3})$$

Question174

If the function

$$f(x) = \begin{cases} -x, & x < 1 \\ a + \cos^{-1}(x+b), & 1 \leq x \leq 2 \end{cases}$$

is differentiable at $x = 1$, then $\frac{a}{b}$ is equal to:

[Online April 9, 2016]

Options:

A. $\frac{\pi+2}{2}$

B. $\frac{\pi-2}{2}$

C. $\frac{-\pi-2}{2}$

D. $-1 - \cos^{-1}(2)$

Answer: A

Solution:

$$f(x) = \begin{cases} -x, & x < 1 \\ a + \cos^{-1}(x+b), & 1 \leq x \leq 2 \end{cases}$$

$f(x)$ is continuous

$$\Rightarrow \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} a + \cos^{-1}(x+b) = f(x)$$

$$\Rightarrow -1 = a + \cos^{-1}(1+b)$$

$$\cos^{-1}(1+b) = -1 - a \dots\dots(a)$$

$f(x)$ is differentiate

$$\Rightarrow \text{LHD} = \text{RHD}$$

$$\Rightarrow -1 = \frac{-1}{\sqrt{1-(1+b)^2}}$$

$$\Rightarrow 1 - (1+b)^2 = 1 \Rightarrow b = -1 \dots\dots(b)$$

$$\text{From (a)} \Rightarrow \cos^{-1}(0) = -1 - a$$

$$\therefore -1 - a = \frac{\pi}{2}$$

$$a = -1 - \frac{\pi}{2} \Rightarrow a = \frac{-\pi-2}{2} \dots\dots(c)$$

$$\therefore \frac{a}{b} = \frac{\pi+2}{2}$$

Question175

For $x \in \mathbb{R}$, $f(x) = |\log 2 - \sin x|$ and $g(x) = f(f(x))$, then
[2016]

Options:

A. $g'(0) = -\cos(\log 2)$

B. g is differentiable at $x = 0$ and $g'(0) = -\sin(\log 2)$

C. g is not differentiable at $x = 0$

D. $g'(0) = \cos(\log 2)$

Answer: D

Solution:

(d) $g(x) = f(f(x))$
 In the neighbourhood of $x = 0$
 $f(x) = |\log 2 - \sin x| = (\log 2 - \sin x)$
 $\therefore g(x) = |\log 2 - \sin |\log 2 - \sin x||$
 $= (\log 2 - \sin(\log 2 - \sin x))$
 $\therefore g(x)$ is differentiable
 and $g'(x) = -\cos(\log 2 - \sin x) (-\cos x)$
 $\Rightarrow g'(0) = \cos(\log 2)$

Question176

Let k be a non-zero real number

$$\text{If } f(x) = \begin{cases} \frac{(e^x - 1)}{\sin\left(\frac{x}{k}\right) \log\left(1 + \frac{x}{4}\right)} & , x \neq 0 \\ 12 & , x = 0 \end{cases}$$

is a continuous function then the value of k is:
 [Online April 11, 2015]

Options:

- A. 4
- B. 1
- C. 3
- D. 2

Answer: C

Solution:

Since $f(x)$ is a continuous function therefore limit of $f(x)$ at $x \rightarrow 0 =$ value of $f(x)$ at 0

$$\begin{aligned} \therefore \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{(e^x - 1)^2}{\sin\left(\frac{x}{k}\right) \log\left(1 + \frac{x}{4}\right)} \\ &= \lim_{x \rightarrow 0} \frac{x^2 \left(\frac{e^x - 1}{x}\right)^2}{\frac{x}{k} \left[\frac{\sin\left(\frac{x}{k}\right)}{\frac{x}{k}} \right] \cdot \frac{\log\left(1 + \frac{x}{4}\right)}{\left(\frac{x}{4}\right)}} \times \left(\frac{x}{4}\right) \\ &= \lim_{x \rightarrow 0} \frac{x^2 \left(\frac{e^x - 1}{x}\right)^2 4k}{\frac{x}{k} \cdot \frac{x}{4}} \end{aligned}$$

on applying limit we get
 $4k = 12 \Rightarrow k = 3$

Question177

If the function.

$$g(x) = \begin{cases} k\sqrt{x+1}, & 0 \leq x \leq 3 \\ mx + 2, & 3 < x \leq 5 \end{cases}$$

is differentiable, then the value of $k + m$ is:
 [2015]

Options:

A. $\frac{10}{3}$

B. 4

C. 2

D. $\frac{16}{5}$

Answer: C

Solution:

Since g (x) is differentiable, it will be continuous at x = 3

$$\therefore \lim_{x \rightarrow 3^-} g(x) = \lim_{x \rightarrow 3^+} g(x)$$

$$2k = 3m + 2 \dots\dots\dots(1)$$

Also g(x) is differentiable at x = 0

$$\therefore \lim_{x \rightarrow 3^-} g'(x) = \lim_{x \rightarrow 3^+} g'(x)$$

$$\frac{k}{2\sqrt{3+1}} = m$$

$$k = 4m \dots\dots\dots(2)$$

(1) and (2), we get Solving

$$m = \frac{2}{5}, k = \frac{8}{5}$$

$$k + m = 2$$

Question178

If Rolle's theorem holds for the function $f(x)2x^3 + bx^2 + cx$, $x \in [-1, 1]$, at the point $x = \frac{1}{2}$, then $2b + c$ equals:

[Online April 10, 2015]

Options:

A. -3

B. -1

C. 2

D. 1

Answer: B

Solution:

Conduction for Rolle's theorem

$$f(1) = f(-1)$$

$$\text{and } f'\left(\frac{1}{2}\right) = 0$$

$$c = -2 \text{ and } b = \frac{1}{2}$$

$$2b + c = -1$$

Question179

If the function

$$f(x) = \begin{cases} \frac{\sqrt{2+\cos x}-1}{(\pi-x)^2}, & x \neq \pi \\ k, & x = \pi \end{cases}$$

is continuous at $x = \pi$, then k equals:
[Online April 19, 2014]

Options:

A. 0

B. $\frac{1}{2}$

C. 2

D. $\frac{1}{4}$

Answer: D

Solution:

$$\begin{aligned} \text{Since } f(x) &= \frac{\sqrt{2 + \cos x} - 1}{(\pi - x)^2} \text{ is} \\ \text{Continuous at } x &= \pi \\ \therefore L \cdot H \cdot L &= R \cdot H \cdot L = f(\pi) \\ \text{Let } (\pi - x) &= \theta, \theta \rightarrow 0 \text{ when } x \rightarrow \pi \\ \therefore \lim_{\theta \rightarrow 0} \frac{\sqrt{2 - \cos \theta} - 1}{\theta^2} \\ &= \lim_{\theta \rightarrow 0} \frac{(2 - \cos \theta) - 1}{\theta^2} \times \frac{1}{\sqrt{2 - \cos \theta} + 1} \\ &= \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2} \cdot \frac{1}{2} (\because \cos 0 = 1) \\ &= \frac{1}{2} \lim_{\theta \rightarrow 0} \frac{2 \sin^2 \theta / 2}{\theta^2} = \frac{2}{2} \lim_{\theta \rightarrow 0} \frac{\sin^2 \theta / 2}{\frac{\theta^2}{4} \cdot 4} \\ &= \frac{1}{4} \left(\because \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right) \end{aligned}$$

Question 180

If $f(x)$ is continuous and $f\left(\frac{9}{2}\right) = \frac{2}{9}$, then $\lim_{x \rightarrow 0} f\left(\frac{1 - \cos 3x}{x^2}\right)$ is equal to:

[Online April 9, 2014]

Options:

A. $\frac{9}{2}$

B. $\frac{2}{9}$

C. 0

D. $\frac{8}{9}$

Answer: B

Solution:

$$\begin{aligned} \text{Given that } f\left(\frac{9}{2}\right) &= \frac{2}{9} \\ \lim_{x \rightarrow 0} f\left(\frac{1 - \cos 3x}{x^2}\right) &= \lim_{x \rightarrow 0} f\left(\frac{x^2}{1 - \cos 3x}\right) \\ &= \lim_{x \rightarrow 0} f\left(\frac{x^2}{2 \sin^2 \frac{3x}{2}}\right) = \frac{1}{2} \lim_{x \rightarrow 0} f\left(\frac{\frac{9}{4} \cdot x^2 \cdot \frac{4}{9}}{\sin^2 \frac{3x}{2}}\right) \end{aligned}$$

$$\begin{aligned}
&= \frac{4}{9 \times 2} \lim_{x \rightarrow 0} \left(\frac{\frac{1}{\sin^2 \frac{3x}{2}}}{\left(\frac{3x}{2}\right)^2} \right) \\
&= \frac{2}{9} \left[\frac{\lim_{x \rightarrow 0} 1}{\lim_{x \rightarrow 0} \frac{\sin^2 \frac{3x}{2}}{\left(\frac{3x}{2}\right)^2}} \right] \left\{ \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right\} \\
&= \frac{2}{9} \cdot \left[\frac{1}{1} \right] = \frac{2}{9}
\end{aligned}$$

Question181

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $|f(x)| \leq x^2$, for all $x \in \mathbb{R}$. Then, at $x = 0$, f is :
[Online April 19, 2014]

Options:

- A. continuous but not differentiable.
- B. continuous as well as differentiable.
- C. neither continuous nor differentiable.
- D. differentiable but not continuous.

Answer: B

Solution:

Let $|f(x)| \leq x^2, \forall x \in \mathbb{R}$
Now, at $x = 0, |f(0)| \leq 0$
 $\Rightarrow f(0) = 0$
 $\therefore f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h - 0} = \lim_{h \rightarrow 0} \frac{f(h)}{h} \dots (1)$
Now, $\left| \frac{f(h)}{h} \right| \leq |h| (\because |f(x)| \leq x^2)$
 $\Rightarrow -|h| \leq \frac{f(h)}{h} \leq |h|$
 $\Rightarrow \lim_{h \rightarrow 0} \frac{f(h)}{h} \rightarrow 0 \dots (2)$
(using sandwich Theorem)
 $\therefore$ from (1) and (2), we get $f'(0) = 0$,
i.e. $f(x)$ is differentiable, at $x = 0$
Since, differentiability $\Rightarrow$ Continuity
 $\therefore |f(x)| \leq x^2$, for all $x \in \mathbb{R}$ is continuous as well as differentiable at $x = 0$

Question182

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be two functions defined by $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$, and $g(x) = xf(x)$

Statement I: f is a continuous function at $x = 0$.

Statement II: g is a differentiable function at $x = 0$.

[Online April 12, 2014]

Options:

- A. Both statement I and II are false.
- B. Both statement I and II are true.

C. Statement I is true, statement II is false.

D. Statement I is false, statement II is true.

Answer: B

Solution:

$$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

and $g(x) = xf(x)$

For $f(x)$

$$\begin{aligned} \text{LH L} &= \lim_{h \rightarrow 0^-} \left\{ -h \sin\left(-\frac{1}{h}\right) \right\} \\ &= 0 \times \text{a finite quantity between -1 and 1} = 0 \end{aligned}$$

$$\text{RH L} = \lim_{h \rightarrow 0^+} h \sin \frac{1}{h} = 0$$

Also, $f(0) = 0$

Thus $\text{LH L} = \text{RH L} = f(0)$

$\therefore f(x)$ is continuous at $x = 0$

$$g(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

For $g(x)$

$$\begin{aligned} \text{LH L} &= \lim_{h \rightarrow 0^-} \left\{ -h^2 \sin\left(\frac{1}{h}\right) \right\} \\ &= 0^2 \times \text{a finite quantity between -1 and 1} = 0 \end{aligned}$$

$$\text{RH L} = \lim_{h \rightarrow 0^+} h^2 \sin\left(\frac{1}{h}\right) = 0$$

Also $g(0) = 0$

$\therefore g(x)$ is continuous at $x = 0$

Question 183

If $f(x) = x^2 - x + 5$, $x > \frac{1}{2}$, and $g(x)$ is its inverse function, then $g'(7)$ equals:

[Online April 12, 2014]

Options:

A. $-\frac{1}{3}$

B. $\frac{1}{13}$

C. $\frac{1}{3}$

D. $-\frac{1}{13}$

Answer: C

Solution:

$$f(x) = y = x^2 - x + 5$$

$$x^2 - x + \frac{1}{4} - \frac{1}{4} + 5 = y$$

$$\left(x - \frac{1}{2}\right)^2 + \frac{19}{4} = y$$

$$\left(x - \frac{1}{2}\right)^2 = y - \frac{19}{4}$$

$$x - \frac{1}{2} = \pm \sqrt{y - \frac{19}{4}}$$

$$x = \frac{1}{2} \pm \sqrt{y - \frac{19}{4}}$$

$$\begin{aligned} \text{As } x &> \frac{1}{2} \\ x &= \frac{1}{2} + \sqrt{y - \frac{19}{4}} \\ g(x) &= \frac{1}{2} + \sqrt{x - \frac{19}{4}} \\ g'(x) &= \frac{1}{2\sqrt{x - \frac{19}{4}}} \\ g'(7) &= \frac{1}{2\sqrt{7 - \frac{19}{4}}} = \frac{1}{2\sqrt{\frac{28-19}{4}}} = \frac{1}{3} \end{aligned}$$

Question 184

If f and g are differentiable functions in $[0, 1]$ satisfying $f(0) = 2 = g(1)$, $g(0) = 0$ and $f(1) = 6$, then for some $c \in [0, 1]$ [2014]

Options:

- A. $f(c) = g'(c)$
- B. $f(c) = 2g'(c)$
- C. $2f(c) = g'(c)$
- D. $2f(c) = 3g'(c)$

Answer: B

Solution:

Since, f and g both are continuous function on $[0, 1]$ and differentiable on $(0, 1)$ then $\exists c \in (0, 1)$ such that

$$f'(c) = \frac{f(1) - f(0)}{1 - 0} = \frac{6 - 2}{1} = 4$$

$$\text{and } g'(c) = \frac{g(1) - g(0)}{1 - 0} = \frac{2 - 0}{1} = 2$$

Thus, we get $f'(c) = 2g'(c)$

Question 185

Let $f(x) = x|x|$, $g(x) = \sin x$ and $h(x) = (g \circ f)(x)$. Then [Online April 11, 2014]

Options:

- A. $h(x)$ is not differentiable at $x = 0$.
- B. $h(x)$ is differentiable at $x = 0$, but $h'(x)$ is not continuous at $x = 0$
- C. $h'(x)$ is continuous at $x = 0$ but it is not differentiable at $x = 0$
- D. $h'(x)$ is differentiable at $x = 0$

Answer: C

Solution:

Let $f(x) = xx|x| = xx|x|$, $g(x) = \sin x$
and $h(x) = g \circ f(x) = g[f(x)]$

$$\therefore h(x) = \begin{cases} \sin x^2, & x \geq 0 \\ -\sin x^2, & x < 0 \end{cases}$$

$$\text{Now, } h'(x) = \begin{cases} 2x \cos x^2, & x \geq 0 \\ -2x \cos x^2, & x < 0 \end{cases}$$

Since, L.H.L and R.H.L at $x = 0$ of $h'(x)$ is equal to 0 therefore $h'(x)$ is continuous at $x = 0$

Now, suppose $h'(x)$ is differentiable

$$\therefore h''(x) = \begin{cases} 2(\cos x^2 + 2x^2(-\sin x^2)), & x \geq 0 \\ 2(-\cos x^2 + 2x^2 \sin x^2), & x < 0 \end{cases}$$

Since, L.H.L and R.H.L at $x = 0$ of $h''(x)$ are different therefore $h''(x)$ is not continuous.

$\Rightarrow h''(x)$ is not differentiable

$\Rightarrow$ our assumption is wrong

Hence $h'(x)$ is not differentiable at $x = 0$.

Question186

Let for $i = 1, 2, 3$, $p_i(x)$ be a polynomial of degree 2 in x , $p_i'(x)$ and $p_i''(x)$ be the first and second order derivatives of $p_i(x)$ respectively. Let,

$$A(x) = \begin{bmatrix} p_1(x) & p_1'(x) & p_1''(x) \\ p_2(x) & p_2'(x) & p_2''(x) \\ p_3(x) & p_3'(x) & p_3''(x) \end{bmatrix}$$

and $B(x) = [A(x)]^T A(x)$. Then determinant of $B(x)$
[Online April 11, 2014]

Options:

- A. is a polynomial of degree 6 in x .
- B. is a polynomial of degree 3 in x .
- C. is a polynomial of degree 2 in x .
- D. does not depend on x .

Answer: A

Solution:

$$\text{Let } p_1(x) = a_1x^2 + b_1x + c_1$$

$$p_2(x) = a_2x^2 + b_2x + c_2$$

$$\text{and } p_3(x) = a_3x^2 + b_3x + c_3$$

where $a_1, a_2, a_3, b_1, b_2, b_3, c_1, c_2, c_3$ are real numbers.

$$\therefore A(x) = \begin{bmatrix} a_1x^2 + b_1x + c_1 & 2a_1x + b_1 & 2a_1 \\ a_2x^2 + b_2x + c_2 & 2a_2x + b_2 & 2a_2 \\ a_3x^2 + b_3x + c_3 & 2a_3x + b_3 & 2a_3 \end{bmatrix}$$

$$B(x) = \begin{bmatrix} a_1x^2 + b_1x + c_1 & a_2x^2 + b_2x + c_2 & a_3x^2 + b_3x + c_3 \\ 2a_1x + b_1 & 2a_2x + b_2 & 2a_3x + b_3 \\ 2a_1 & 2a_2 & 2a_3 \end{bmatrix}$$

$$\times \begin{bmatrix} a_1x^2 + b_1x + c_1 & 2a_1x + b_1 & 2a_1 \\ a_2x^2 + b_2x + c_2 & 2a_2x + b_2 & 2a_2 \\ a_3x^2 + b_3x + c_3 & 2a_3x + b_3 & 2a_3 \end{bmatrix}$$

It is clear from the above multiplication, the degree of determinant of $B(x)$ can not be less than 4.

Question187

If the Rolle's theorem holds for the function $f(x) = 2x^3 + ax^2 + bx$ in the interval $[-1,1]$ for the point $c = \frac{1}{2}$, then the value of $2a + b$ is:

[Online April 9, 2014]

Options:

- A. 1
- B. -1
- C. 2
- D. -2

Answer: B

Solution:

$$f(x) = 2x^3 + ax^2 + bx$$

$$\text{let, } a = -1, b = 1$$

Given that $f(x)$ satisfy Rolle's theorem in interval $[-1,1]$

$f(x)$ must satisfy two conditions.

$$(1) f(a) = f(b)$$

$$(2) f'(c) = 0 \text{ (c should be between a and b)}$$

$$f(a) = f(1) = 2(1)^3 + a(1)^2 + b(1) = 2 + a + b$$

$$f(b) = f(-1) = 2(-1)^3 + a(-1)^2 + b(-1)$$

$$= -2 + a - b$$

$$f(a) = f(b)$$

$$2 + a + b = -2 + a - b$$

$$2b = -4$$

$$b = -2$$

$$\text{(given that } c = \frac{1}{2} \text{)}$$

$$f'(x) = 6x^2 + 2ax + b$$

$$\text{at } x = \frac{1}{2}, f'(x) = 0$$

$$\frac{3}{2} + a + b = 0$$

$$\frac{3}{2} + a - 2 = 0$$

$$a = 2 - \frac{3}{2} = \frac{1}{2}$$

$$2a + b = 2 \times \frac{1}{2} - 2 = 1 - 2 = -1$$

Question 188

Consider the function :

$f(x) = [x] + |1 - x|$, $-1 \leq x \leq 3$ where $[x]$ is the greatest integer function.

Statement 1: f is not continuous at $x = 0, 1, 2$ and 3 .

Statement 2 :

$$f(x) = \begin{cases} -x, & -1 \leq x < 0 \\ 1 - x, & 0 \leq x < 1 \\ 1 + x, & 1 \leq x < 2 \\ 2 + x, & 2 \leq x \leq 3 \end{cases}$$

[Online April 25, 2013]

Options:

- A. Statement 1 is true ; Statement 2 is false,
- B. Statement 1 is true; Statement 2 is true; Statement 2 is not correct explanation for Statement 1.
- C. Statement 1 is true; Statement 2 is true; Statement It is a correct explanation for Statement 1.

D. Statement 1 is false; Statement 2 is true.

Answer: A

Solution:

Let $f(x) = [x] + |1 - x|$, $-1 \leq x \leq 3$

where $[x]$ = greatest integer function.

f is not continuous at $x = 0, 1, 2, 3$

But in statement- 2 $f(x)$ is continuous at $x = 3$.

Hence, statement- 1 is true and 2 is false.

Question 189

Let f be a composite function of x defined by

$$f(u) = \frac{1}{u^2 + u - 2}, \quad u(x) = \frac{1}{x-1}.$$

Then the number of points x where f is discontinuous is :
[Online April 23, 2013]

Options:

A. 4

B. 3

C. 2

D. 1

Answer: B

Solution:

$u(x) = \frac{1}{x-1}$, which is discontinuous at $x = 1$

$$f(u) = \frac{1}{u^2 + u - 2} = \frac{1}{(u+2)(u-1)}$$

which is discontinuous at $u = -2, 1$

when $u = -2$, then $\frac{1}{x-1} = -2 \Rightarrow x = \frac{1}{2}$

when $u = 1$, then $\frac{1}{x-1} = 1 \Rightarrow x = 2$

Hence given composite function is discontinuous at three points, $x = 1, \frac{1}{2}$ and 2 .

Question 190

Let $f(x) = -1 + |x - 2|$, and $g(x) = 1 - |x|$; then the set of all points where $f \circ g$ is discontinuous is :
[Online April 22, 2013]

Options:

A. $\{0, 2\}$

B. $\{0, 1, 2\}$

C. $\{0\}$

D. an empty set

Answer: D

Solution:

$$\begin{aligned}
 f \circ g &= f(g(x)) = f(1 - |x|) \\
 &= -1 + |1 - |x|| - 2| \\
 &= -1 + |-|x| - 1| = -1 + ||x| + 1| \\
 \text{Let } f \circ g &= y \\
 \therefore y &= -1 + ||x| + 1|
 \end{aligned}$$

$$\Rightarrow y = \begin{cases} -1 + x + 1, & x \geq 0 \\ -1 - x + 1, & x < 0 \end{cases}$$

$$\Rightarrow y = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$\text{LHL at } (x = 0) = \lim_{x \rightarrow 0} (-x) = 0$$

$$\text{RHL at } (x = 0) = \lim_{x \rightarrow 0} (x) = 0$$

When $x = 0$, then $y = 0$

Hence, LHL at $(x = 0) = \text{RHL at } (x = 0)$

= value of y at $(x = 0)$

Hence y is continuous at $x = 0$.

Clearly at all other point y continuous. Therefore, the set of all points where $f \circ g$ is discontinuous is an empty set.

Question 191

If $y = \sec(\tan^{-1}x)$, then $\frac{dy}{dx}$ at $x = 1$ is equal to :
[2013]

Options:

A. $\frac{1}{\sqrt{2}}$

B. $\frac{1}{2}$

C. 1

D. $\sqrt{2}$

Answer: A

Solution:

$$\text{Let } y = \sec(\tan^{-1}x) = \sec(\sec^{-1}\sqrt{1+x^2})$$

$$\Rightarrow y = \sqrt{1+x^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2\sqrt{1+x^2}} \cdot 2x$$

$$\text{At } x = 1$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{2}}$$

Question 192

If the curves $\frac{x^2}{a} + \frac{y^2}{4} = 1$ and $y^3 = 16x$ intersect at right angles, then a value of a is :
[Online April 23, 2013]

Options:

A. 2

B. $\frac{4}{3}$

C. $\frac{1}{2}$

D. $\frac{3}{4}$

Answer: B

Solution:

$$\frac{x^2}{a} + \frac{y^2}{4} = 1 \Rightarrow \frac{2x}{a} + \frac{2y}{4} \cdot \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{-4x}{ay} \dots\dots(i)$$

$$y^3 = 16x \Rightarrow 3y^2 \cdot \frac{dy}{dx} = 16 \Rightarrow \frac{dy}{dx} = \frac{16}{3y^2} \dots\dots(ii)$$

Since curves intersect at right angles

$$\therefore \frac{-4x}{ay} \times \frac{16}{3y^2} = -1 \Rightarrow 3ay^3 = 64x$$

$$\Rightarrow a = \frac{64x}{3 \times 16x} = \frac{4}{3}$$

Question193

For $a > 0, t \in \left(0, \frac{\pi}{2}\right)$, let $x = \sqrt{a^{\sin^{-1}t}}$ and $y = \sqrt{a^{\cos^{-1}t}}$ Then, $1 + \left(\frac{dy}{dx}\right)^2$ equals:

[Online April 22, 2013]

Options:

A. $\frac{x^2}{y^2}$

B. $\frac{y^2}{x^2}$

C. $\frac{x^2 + y^2}{y^2}$

D. $\frac{x^2 + y^2}{x^2}$

Answer: D

Solution:

$$\text{Let } x = \sqrt{a^{\sin^{-1}t}}$$

$$\Rightarrow x^2 = a^{\sin^{-1}t}$$

$$\Rightarrow 2 \log x = \sin^{-1}t \cdot \log a$$

$$\Rightarrow \frac{2}{x} = \frac{\log a}{\sqrt{1-t^2}} \cdot \frac{dt}{dx}$$

$$\Rightarrow \frac{2\sqrt{1-t^2}}{x \log a} = \frac{dt}{dx} \dots\dots(1)$$

$$\text{Now, let } y = \sqrt{a^{\cos^{-1}t}}$$

$$\Rightarrow 2 \log y = \cos^{-1}t \cdot \log a$$

$$\Rightarrow \frac{2}{y} \cdot \frac{dy}{dx} = \frac{-\log a}{\sqrt{1-t^2}} \cdot \frac{dt}{dx}$$

$$\Rightarrow \frac{2}{y} \cdot \frac{dy}{dx} = \frac{-\log a}{\sqrt{1-t^2}} \times \frac{2\sqrt{1-t^2}}{x \log a} \text{ (from (1))}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{y}{x}$$

$$\text{Hence, } 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \left(\frac{-y}{x}\right)^2 = \frac{x^2 + y^2}{x^2}$$

Question194

Let $f(x) = \frac{x^2 - x}{x^2 + 2x} x \neq 0, -2$. Then $\frac{d}{dx}[f^{-1}(x)]$ (wherever it is defined) is equal to :

[Online April 9, 2013]

Options:

A. $\frac{-1}{(1-x)^2}$

B. $\frac{3}{(1-x)^2}$

C. $\frac{1}{(1-x)^2}$

D. $\frac{-3}{(1-x)^2}$

Answer: B

Solution:

$$\begin{aligned}\text{Let } y &= \frac{x^2 - x}{x^2 + 2x} \\ \Rightarrow (x^2 + 2x)y &= x^2 - x \\ \Rightarrow x(x + 2)y &= x(x - 1) \\ \Rightarrow x[(x + 2)y - (x - 1)] &= 0 \\ \because x \neq 0, \therefore (x + 2)y - (x - 1) &= 0 \\ \Rightarrow xy + 2y - x + 1 &= 0 \\ \Rightarrow x(y - 1) &= -(2y + 1) \\ \therefore x &= \frac{2y + 1}{1 - y} \Rightarrow f^{-1}(x) = \frac{2x + 1}{1 - x} \\ \frac{d}{dx}(f^{-1}(x)) &= \frac{2(1 - x) - (2x + 1)(-1)}{(1 - x)^2} \\ &= \frac{2 - 2x + 2x + 1}{(1 - x)^2} = \frac{3}{(1 - x)^2}\end{aligned}$$

Question195

If $f(x) = \sin(\sin x)$ and $f''(x) + \tan x f'(x) + g(x) = 0$, then $g(x)$ is :
[Online April 23, 2013]

Options:

A. $\cos^2 x \cos(\sin x)$

B. $\sin^2 x \cos(\cos x)$

C. $\sin^2 x \sin(\cos x)$

D. $\cos^2 x \sin(\sin x)$

Answer: D

Solution:

$$\begin{aligned}f(x) &= \sin(\sin x) \\ \Rightarrow f'(x) &= \cos(\sin x) \cdot \cos x \\ \Rightarrow f''(x) &= -\sin(\sin x) \cdot \cos^2 x + \cos(\sin x) \cdot (-\sin x) = -\cos^2 x \cdot \sin(\sin x) - \sin x \cdot \cos(\sin x) \\ \text{Now } f''(x) + \tan x \cdot f'(x) + g(x) &= 0 \\ \Rightarrow g(x) &= \cos^2 x \cdot \sin(\sin x) + \sin x \cdot \cos(\sin x) - \tan x \cdot \cos x \cdot \cos(\sin x) \\ \Rightarrow g(x) &= \cos^2 x \cdot \sin(\sin x)\end{aligned}$$

Question196

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function defined by $f(x) = [x] \cos\left(\frac{2x-1}{2}\right) \pi$, where $[x]$ denotes the greatest integer function, then f is.
[2012]

Options:

- A. continuous for every real x .
- B. discontinuous only at $x = 0$
- C. discontinuous only at non-zero integral values of x .
- D. continuous only at $x = 0$.

Answer: A

Solution:

$$\text{Let } f(x) = [x] \cos\left(\frac{2x-1}{2}\right)$$

We know that $[x]$ is discontinuous at all integral points and $\cos x$ is continuous at $x \in \mathbb{R}$

So, check at $x = n, n \in \mathbb{I}$

$$\text{L.H.L.} = \lim_{x \rightarrow n^-} [x] \cos\left(\frac{2x-1}{2}\right) \pi$$

$$= (n-1) \cos\left(\frac{2n-1}{2}\right) \pi = 0$$

($\because [x]$ is the greatest integer function)

$$\text{R.H.L.} = \lim_{x \rightarrow n^+} [x] \cos\left(\frac{2x-1}{2}\right) \pi$$

$$= n \cos\left(\frac{2n-1}{2}\right) \pi = 0$$

Now, value of the function at $x = n$ is

$$f(n) = 0$$

Since, L.H.L. = R.H.L. = $f(n)$

therefore $f(x) = [x] \cos\left(\frac{2x-1}{2}\right)$ is continuous for every real x .

Question 197

Let $f : [1, 3] \rightarrow \mathbb{R}$ be a function satisfying $\frac{x}{[x]} \leq f(x) \leq \sqrt{6-x}$, for all $x \neq 2$ and $f(2) = 1$, where $\mathbb{R}$ is the set of all real numbers and $[x]$ denotes the largest integer less than or equal to x .

Statement 1: $\lim_{x \rightarrow 2^-} f(x)$ exists.

Statement 2: f is continuous at $x = 2$.

[Online May 19, 2012]

Options:

- A. Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation for Statement 1.
- B. Statement 1 is false, Statement 2 is true.
- C. Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation for Statement 1.
- D. Statement 1 is true, Statement 2 is false.

Answer: D

Solution:

$$\text{Consider } \frac{x}{[x]} \leq f(x) \leq \sqrt{6-x}$$

$$\Rightarrow \lim_{x \rightarrow 2^-} \frac{x}{[x]} = \frac{2}{1} = 2$$

$$\Rightarrow \lim_{x \rightarrow 2^-} \sqrt{6-x} = 2$$

therefore $\lim_{x \rightarrow 2^-} f(x) = 2$ [By Sandwich theorem]

$$\text{Now } \lim_{x \rightarrow 2^+} \frac{x}{[x]} = 1, \lim_{x \rightarrow 2^+} \sqrt{6-x} = 2$$

Hence by Sandwich theorem $\lim_{x \rightarrow 2^+} f(x)$ does not exist.

Therefore f is not continuous at $x = 2$. Thus statement-1 is true but statement-2 is not true

Question 198

Statement 1: A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at x_0 if and only if $\lim_{x \rightarrow x_0} f(x)$ exists and $\lim_{x \rightarrow x_0} f(x) = f(x_0)$

Statement 2: A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is discontinuous at x_0 if and only if, $\lim_{x \rightarrow x_0} f(x)$ exists and

$$\lim_{x \rightarrow x_0} f(x) \neq f(x_0).$$

[Online May 12, 2012]

Options:

- A. Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.
- B. Statement 1 is false, Statement 2 is true.
- C. Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation of Statement 1.
- D. Statement 1 is true, Statement 2 is false.

Answer: D

Solution:

Statement - 1 is true.
It is the definition of continuity.
Statement - 2 is false

Question 199

Consider the function, $f(x) = |x - 2| + |x - 5|, x \in \mathbb{R}$.

Statement- 1: $f'(4) = 0$

Statement - 2: f is continuous in $[2, 5]$, differentiable in $(2, 5)$ and $f(2) = f(5)$.

[2012]

Options:

- A. Statement-1 is false, Statement-2 is true.
- B. Statement-1 is true, statement-2 is true; statement-2 is a correct explanation for Statement-1.
- C. Statement-1 is true, statement-2 is true; statement-2 is not a correct explanation for Statement-1.
- D. Statement-1 is true, statement-2 is false.

Answer: C

Solution:

$$f(x) = |x - 2| = \begin{cases} x - 2, & x - 2 \geq 0 \\ 2 - x, & x - 2 \leq 0 \end{cases}$$

$$= \begin{cases} x - 2, & x \geq 2 \\ 2 - x, & x \leq 2 \end{cases}$$

Similarly,

$$f(x) = |x - 5| = \begin{cases} x - 5, & x \geq 5 \\ 5 - x, & x \leq 5 \end{cases}$$

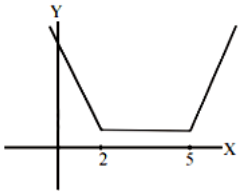
$$\therefore f(x) = |x - 2| + |x - 5| \\ = \{x - 2 + 5 - x = 3, 2 \leq x \leq 5\}$$

Thus $f(x) = 3, 2 \leq x \leq 5$

$$f'(x) = 0, 2 < x < 5$$

$$f'(4) = 0$$

$\therefore$ Statement-1 is true



Since $f(x) = 3, 2 \leq x \leq 5$ is constant function.

So, it continuous in 2,5 and differentiable in (2,5)

$$\therefore f(2) = 0 + |2 - 5| = 3$$

$$\text{and } f(5) = |5 - 2| + 0 = 3$$

statement- 2 is also true.

Question200

If $f(x) = a |\sin x| + be^{|x|} + cx^3$, where $a, b, c \in \mathbb{R}$, is differentiable at $x = 0$, then
[Online May 26, 2012]

Options:

- A. $a = 0$, b and c are any real numbers
- B. $c = 0$, $a = 0$, b is any real number
- C. $b = 0$, $c = 0$, a is any real number
- D. $a = 0$, $b = 0$, c is any real number

Answer: D

Solution:

$|\sin x|$ and $e^{|x|}$ are not differentiable at $x = 0$ and $|x|^3$ is differentiable at $x = 0$

$\therefore$ for $f(x)$ to be differentiable at $x = 0$, we must have $a = 0$, $b = 0$ and c is any real number.

Question201

If $x + |y| = 2y$, then y as a function of x , at $x = 0$ is
[Online May 7, 2012]

Options:

- A. differentiable but not continuous
- B. continuous but not differentiable
- C. continuous as well as differentiable
- D. neither continuous nor differentiable

Answer: B

Solution:

$$\text{Given } x + |y| = 2y$$

$$\Rightarrow x + y = 2y \text{ or } x - y = 2y$$

$$\Rightarrow x = y \text{ or } x = 3y$$

This represent a straight line which passes through origin.

Hence, $x + |y| = 2y$ is continuous at $x = 0$.

Now, we check differentiability at $x = 0$

$$x + |y| = 2y \Rightarrow x + y = 2y, y \geq 0$$

$$x - y = 2y, y < 0$$

$$\text{Thus, } f(x) = \begin{cases} x, & y < 0 \\ x/3, & y \geq 0 \end{cases}$$

$$\text{Now, L.H.D.} = \lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{-h}$$

$$= \lim_{h \rightarrow 0^-} \frac{x+h-x}{-h} = -1$$

$$\text{R.H.D} = \lim_{h \rightarrow 0^+} \frac{f(x+h)-f(x)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{\frac{x+h}{3} - \frac{x}{3}}{h} = \lim_{h \rightarrow 0^+} \frac{1}{3} = \frac{1}{3}$$

Since, L.H.D $\neq$ R.H.D. at $x = 0$

$\therefore$ given function is not differentiable at $x = 0$

Question202

If $f'(x) = \sin(\log x)$ and $y = f\left(\frac{2x+3}{3-2x}\right)$, then $\frac{dy}{dx}$ equals

[Online May 12, 2012]

Options:

A. $\sin\left[\log\left(\frac{2x+3}{3-2x}\right)\right]$

B. $\frac{12}{(3-2x)^2}$

C. $\frac{12}{(3-2x)^2} \sin\left[\log\left(\frac{2x+3}{3-2x}\right)\right]$

D. $\frac{12}{(3-2x)^2} \cos\left[\log\left(\frac{2x+3}{3-2x}\right)\right]$

Answer: C

Solution:

Let $f'(x) = \sin[\log x]$ and $y = f\left(\frac{2x+3}{3-2x}\right)$

$$\begin{aligned} \text{Now, } \frac{dy}{dx} &= f'\left(\frac{2x+3}{3-2x}\right) \cdot \frac{d}{dx}\left(\frac{2x+3}{3-2x}\right) \\ &= \sin\left[\log\left(\frac{2x+3}{3-2x}\right)\right] \cdot \frac{[(6-4x)-(-4x-6)]}{(3-2x)^2} \\ &= \frac{12}{(3-2x)^2} \cdot \sin\left[\log\left(\frac{2x+3}{3-2x}\right)\right] \end{aligned}$$

Question203

Consider a quadratic equation $ax^2 + bx + c = 0$, where $2a + 3b + 6c = 0$ and let $g(x) = a\frac{x^3}{3} + b\frac{x^2}{2} + cx$.

Statement 1: The quadratic equation has at least one root in the interval $(0, 1)$.

Statement 2: The Rolle's theorem is applicable to function $g(x)$ on the interval $[0, 1]$.

[Online May 19, 2012]

Options:

A. Statement 1 is false, Statement 2 is true.

B. Statement 1 is true, Statement 2 is false.

C. Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation for Statement 1.

D. Statement 1 is true, Statement 2 is true, , Statement 2 is a correct explanation for Statement 1.

Answer: D

Solution:

Let $g(x) = \frac{ax^3}{3} + b \cdot \frac{x^2}{2} + cx$

$g'(x) = ax^2 + bx + c$

Given: $ax^2 + bx + c = 0$ and $2a + 3b + 6c = 0$

Statement-2:

(i) $g(0) = 0$ and $g(1) = \frac{a}{3} + \frac{b}{2} + c = \frac{2a + 3b + 6c}{6}$

$= \frac{0}{6} = 0$

$\Rightarrow g(0) = g(1)$

(ii) g is continuous on $[0, 1]$ and differentiable on $(0, 1)$

$\therefore$ By Rolle's theorem $\exists k \in (0, 1)$ such that $g'(k) = 0$

This holds the statement 2. Also, from statement-2, we can say $ax^2 + bx + c = 0$ has at least one root in $(0, 1)$.

Thus statement- 1 and 2 both are true and statement- 2 is a correct explanation for statement-1.

Question204

Define $f(x)$ as the product of two real functions

$$f_1(x) = x, x \in \mathbb{R}, \text{ and } f_2(x) = \begin{cases} \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

as follows:

$$f(x) = \begin{cases} f_1(x) \cdot f_2(x), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

Statement -1: $f(x)$ is continuous on $\mathbb{R}$

Statement -2: $f_1(x)$ and $f_2(x)$ are continuous on $\mathbb{R}$

[2011RS]

Options:

A. Statement -1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.

B. Statement-1 is true, Statement-2 is true; Statement-2 is NOT a correct explanation for Statement-1

C. Statement-1 is true, Statement-2 is false

D. Statement-1 is false, Statement-2 is true

Answer: C

Solution:

Given that $f(x) = \begin{cases} x \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$

At $x = 0$

LHL = $\lim_{h \rightarrow 0^-} \left\{ -h \sin\left(-\frac{1}{h}\right) \right\}$

$= 0 \times \text{a finite quantity between } -1 \text{ and } 1 = 0$

RHL = $\lim_{h \rightarrow 0^+} h \sin \frac{1}{h} = 0$

Also, $f(0) = 0$ Thus LHL = RHL = $f(0)$

$\therefore f(x)$ is continuous on $\mathbb{R}$.

but $f_2(x)$ is not continuous at $x = 0$

Question205

The values of p and q for which the function

$$f(x) = \begin{cases} \frac{\sin(p+1)x + \sin x}{x} & , x < 0 \\ q & , x = 0 \\ \frac{\sqrt{x+x^2} - \sqrt{x}}{x^{3/2}} & , x > 0 \end{cases}$$

is continuous for all x in R, are
[2011]

Options:

A. $p = \frac{5}{2}, q = \frac{1}{2}$

B. $p = -\frac{3}{2}, q = \frac{1}{2}$

C. $p = \frac{1}{2}, q = \frac{3}{2}$

D. $p = \frac{1}{2}, q = -\frac{3}{2}$

Answer: B

Solution:

$$\begin{aligned} L.H.L &= \lim_{x \rightarrow 0^-} f(x) \\ &= \lim_{h \rightarrow 0^+} \frac{\sin\{(p+1)(-h)\} - \sin h}{-h} = p + 1 + 1 = p + 2 \end{aligned}$$

$$\begin{aligned} R.H.L &= \lim_{x \rightarrow 0^+} f(x) \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{x+x^2} - \sqrt{x}}{x^{3/2}} \times \frac{\sqrt{x+x^2} + \sqrt{x}}{\sqrt{x+x^2} + \sqrt{x}} = \frac{1}{1+1} = \frac{1}{2} \end{aligned}$$

$$f(0) = 2$$

Given that f(x) is continuous at x = 0

$$\therefore p + 2 = q = \frac{1}{2}$$

$$\Rightarrow p = -\frac{3}{2}, q = \frac{1}{2}$$

Question 206

If function f(x) is differentiable at x = a,

then $\lim_{x \rightarrow a} \frac{x^2 f(a) - a^2 f(x)}{x - a}$ is :

[2011RS]

Options:

A. $-a^2 f'(a)$

B. $af(a) - a^2 f'(a)$

C. $2af(a) - a^2 f'(a)$

D. $2af(a) + a^2 f'(a)$

Answer: C

Solution:

$$\lim_{x \rightarrow a} \frac{x^2 f(a) - a^2 f(x)}{x - a}$$

Applying L-Hospital rule

$$= \lim_{x \rightarrow a} \frac{2xf(a) - a^2 f'(x)}{1} = 2af(a) - a^2 f'(a)$$

Question 207

$\frac{d^2 x}{dy^2}$ equals:

[2011]

Options:

A. $-\left(\frac{d^2 y}{dx^2}\right)^{-1} \left(\frac{dy}{dx}\right)^{-3}$

B. $\left(\frac{d^2 y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-2}$

C. $-\left(\frac{d^2 y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-3}$

D. $\left(\frac{d^2 y}{dx^2}\right)^{-1}$

Answer: C

Solution:

$$\begin{aligned} \frac{d^2 x}{dy^2} &= \frac{d}{dy} \left(\frac{dx}{dy} \right) = \frac{d}{dx} \left(\frac{dx}{dy} \right) \frac{dx}{dy} = \frac{d}{dx} \left(\frac{1}{dy/dx} \right) \frac{dx}{dy} \\ &= -\frac{1}{\left(\frac{dy}{dx}\right)^2} \cdot \frac{d^2 y}{dx^2} \cdot \frac{1}{\frac{dy}{dx}} \left[\because \frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2} \right] \\ &= -\frac{1}{\left(\frac{dy}{dx}\right)^3} \frac{d^2 y}{dx^2} \end{aligned}$$

Question 208

Let $f : (-1, 1) \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = -1$ and $f'(0) = 1$. Let $g(x) = [f(2f(x) + 2)]^2$.

Then $g'(0) =$

[2010]

Options:

A. -4

B. 0

C. -2

D. 4

Answer: A

Solution:

Given that $g(x) = [f(2f(x)) + 2]^2$
 $\therefore g'(x) = 2(f(2f(x)) + 2) \left(\frac{d}{dx}(f(2f(x)) + 2) \right)$
 $= 2f(2f(x)) + 2 \cdot f'(2f(x)) \cdot (2f'(x))$
 $\Rightarrow g'(0) = 2f(2f(0)) + 2 \cdot f'(2f(0)) \cdot 2$
 $2f'(0) = 4f(0)(f'(0))^2 = 4(-1)(1)^2 = -4$

Question209

Let y be an implicit function of x defined by $x^{2x} - 2x^x \cot y - 1 = 0$. Then $y'(1)$ equals [2009]

Options:

- A. 1
- B. $\log 2$
- C. $-\log 2$
- D. -1

Answer: D

Solution:

$$x^{2x} - 2x^x \cot y - 1 = 0$$

$$\Rightarrow 2 \cot y = x^x - x^{-x}$$

Let $u = x^x$

$$\Rightarrow 2 \cot y = u - \frac{1}{u}$$

Differentiating both sides with respect to x , we get

$$-2 \operatorname{cosec}^2 y \frac{dy}{dx} = \left(1 + \frac{1}{u^2} \right) \frac{du}{dx}$$

Now $u = x^x$ Taking log both sides

$$\Rightarrow \log u = x \log x$$

$$\Rightarrow \frac{1}{u} \frac{du}{dx} = 1 + \log x$$

$$\Rightarrow \frac{du}{dx} = x^x (1 + \log x)$$

$\therefore$ We get

$$-2 \operatorname{cosec}^2 y \frac{dy}{dx} = (1 + x^{-2x}) \cdot x^x (1 + \log x)$$

$$\Rightarrow \frac{dy}{dx} = \frac{(x^x + x^{-x})(1 + \log x)}{-2(1 + \cot^2 y)} \dots (i)$$

Put $x = 1$ in eqn. $x^{2x} - 2x^x \cot y - 1 = 0$, gives

$$1 - 2 \cot y - 1 = 0$$

$$\Rightarrow \cot y = 0$$

$\therefore$ Putting $x = 1$ and $\cot y = 0$ in eqn. (i), we get

$$y'(1) = \frac{(1+1)(1+0)}{-2(1+0)} = -1$$

Question210

Let $f(x) = x |x|$ and $g(x) = \sin x$.

Statement-1 : g is differentiable at $x = 0$ and its derivative is continuous at that point.

Statement-2 : g is twice differentiable at $x = 0$.

[2009]

Options:

- A. Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.
- B. Statement-1 is true, Statement-2 is false.
- C. Statement-1 is false, Statement-2 is true.

D. Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.

Answer: B

Solution:

Given that $f(x) = xx|$ and $g(x) = \sin x$

So that

$$\text{gof}(x) = g(f(x)) = g(x|x|) = \sin x|x|$$

$$= \begin{cases} \sin(-x^2), & \text{if } x < 0 \\ \sin(x^2) & \text{if } x \geq 0 \end{cases} = \begin{cases} -\sin x^2, & \text{if } x < 0 \\ \sin(x^2) & \text{if } x \geq 0 \end{cases}$$

$$\therefore (\text{gof})'(x) = \begin{cases} -2x \cos x^2, & \text{if } x < 0 \\ 2x \cos x^2, & \text{if } x \geq 0 \end{cases}$$

Here we observe $L(\text{gof})'(0) = 0 = R(\text{gof})'(0)$

$\Rightarrow$ go f is differentiable at $x = 0$

and $(\text{gof})'$ is continuous at $x = 0$

$$\text{Now } (\text{gof})''(x) = \begin{cases} -2 \cos x^2 + 4x^2 \sin x^2, & x < 0 \\ 2 \cos x^2 - 4x^2 \sin x^2, & x \geq 0 \end{cases}$$

Here

$$L(\text{gof})''(0) = -2 \text{ and } R(\text{gof})''(0) = 2$$

$$\therefore L(\text{gof})''(0) \neq R(\text{gof})''(0)$$

$\Rightarrow$ go f (x) is not twice differentiable at $x = 0$.

$\therefore$ Statement - 1 is true but statement -2 is false.

Question 211

$$\text{Let } f(x) = \begin{cases} (x-1) \sin \frac{1}{x-1} & \text{if } x \neq 1 \\ 0 & \text{if } x = 1 \end{cases}$$

Then which one of the following is true?
[2008]

Options:

A. f is neither differentiable at $x = 0$ nor at $x = 1$

B. f is differentiable at $x = 0$ and at $x = 1$

C. f is differentiable at $x = 0$ but not at $x = 1$

D. f is differentiable at $x = 1$ but not at $x = 0$

Answer: C

Solution:

Given that,

$$f(x) = \begin{cases} (x-1) \sin \frac{1}{x-1} & , \text{ if } x \neq 1 \\ 0 & , \text{ if } x = 1 \end{cases}$$

At $x = 1$

$$\text{R.H.D.} = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h \sin \frac{1}{h} - 0}{h} = \lim_{h \rightarrow 0} \sin \frac{1}{h} = \text{a finite number}$$

Let this finite number be l

$$\text{L.H.D.} = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h}$$

$$= \lim_{h \rightarrow 0} -h \sin \left(\frac{1}{-h} \right) - 0 = \lim_{h \rightarrow 0} \sin \left(\frac{1}{-h} \right)$$

$$= - \lim_{h \rightarrow 0} \sin \left(\frac{1}{h} \right) = -(\text{a finite number}) = -l$$

Thus $R.H.D. \neq L.H.D.$

∴ f is not differentiable at x = 1

$$\text{At } x = 0 \quad f'(0) = \sin \frac{1}{(x-1)} - \frac{x-1}{(x-1)^2} \cos\left(\frac{1}{x-1}\right) \Big|_{x=0}$$

$$= -\sin 1 + \cos 1$$

∴ f is differentiable at x = 0

Question 212

The function $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ given by

$$f(x) = \frac{1}{x} - \frac{2}{e^{2x} - 1}$$

can be made continuous at x = 0 by defining f(0) as [2007]

Options:

A. 0

B. 1

C. 2

D. -1

Answer: B

Solution:

Given, $f(x) = \frac{1}{x} - \frac{2}{e^{2x} - 1}$ is continuous at x = 0

$$\Rightarrow f(0) = \lim_{x \rightarrow 0} \frac{1}{x} - \frac{2}{e^{2x} - 1}$$
$$= \lim_{x \rightarrow 0} \frac{(e^{2x} - 1) - 2x}{x(e^{2x} - 1)} \quad \left[\frac{0}{0} \text{ form} \right]$$

∴ Applying, L'Hospital rule

Differentiate two times, we get

$$f(0) = \lim_{x \rightarrow 0} \frac{4e^{2x}}{2(xe^{2x} + e^{2x} \cdot 1) + e^{2x} \cdot 2}$$
$$= \lim_{x \rightarrow 0} \frac{4e^{2x}}{4xe^{2x} + 2e^{2x} + 2e^{2x}} \quad \left[\frac{0}{0} \text{ form} \right]$$
$$= \lim_{x \rightarrow 0} \frac{4e^{2x}}{4(xe^{2x} + e^{2x})} = \frac{4 \cdot e^0}{4(0 + e^0)} = 1$$

Question 213

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \min\{x + 1, |x| + 1\}$, Then which of the following is true? [2007]

Options:

A. f(x) is differentiable everywhere

B. f(x) is not differentiable at x = 0

C. $f(x) \geq 1$ for all $x \in \mathbb{R}$

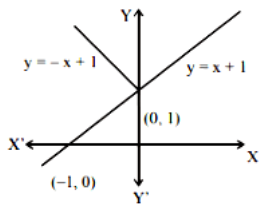
D. f(x) is not differentiable at x = 1

Answer: A

Solution:

$$f(x) = \min\{x + 1, |x| + 1\}$$

$$\Rightarrow f(x) = x + 1 \quad \forall x \in \mathbb{R}$$



Since $f(x) = x + 1$ is polynomial function
Hence, $f(x)$ is differentiable everywhere for all $x \in \mathbb{R}$.

Question214

A value of c for which conclusion of Mean Value Theorem holds for the function $f(x) = \log_e x$ on the interval $[1, 3]$ is
[2007]

Options:

- A. $\log_3 e$
- B. $\log_e 3$
- C. $2\log_3 e$
- D. $\frac{1}{2}\log_3 e$

Answer: C

Solution:

Using Lagrange's Mean Value Theorem

Let $f(x)$ be a function defined on $[a, b]$

then, $f'(c) = \frac{f(b) - f(a)}{b - a}$ (i)

$c \in [a, b]$

$\therefore$ Given $f(x) = \log_e x \therefore f'(x) = \frac{1}{x}$

$\therefore$ equation (i) become

$$\frac{1}{c} = \frac{f(3) - f(1)}{3 - 1}$$

$$\Rightarrow \frac{1}{c} = \frac{\log_e 3 - \log_e 1}{2} = \frac{\log_e 3}{2}$$

$$\Rightarrow c = \frac{2}{\log_e 3} \Rightarrow c = 2\log_3 e$$

Question215

The set of points where $f(x) = \frac{x}{1 + |x|}$ is differentiable is
[2006]

Options:

- A. $(-\infty, 0) \cup (0, \infty)$
- B. $(-\infty, -1) \cup (-1, \infty)$
- C. $(-\infty, \infty)$
- D. $(0, \infty)$

Answer: C

Solution:

$$f(x) = \begin{cases} \frac{x}{1-x}, & x < 0 \\ \frac{x}{1+x}, & x \geq 0 \end{cases}$$

$f(x) = x/(1-x)$ is not define at $x \neq 1$ but here $x < 0$ and $f(x) = \frac{x}{1+x}$ is not define at $x = -1$ but here $x > 0$. So, $f(x)$ is continuous for $x \in \mathbb{R}$.

$$\text{and } f'(x) = \begin{cases} \frac{x}{(1-x)^2}, & x < 0 \\ \frac{x}{(1+x)^2}, & x \geq 0 \end{cases}$$

$\therefore f'(x)$ exist at everywhere.

Question216

If $x^m \cdot y^n = (x+y)^{m+n}$, then $\frac{dy}{dx}$ is
[2006]

Options:

- A. $\frac{y}{x}$
- B. $\frac{x+y}{xy}$
- C. xy
- D. $\frac{x}{y}$

Answer: A

Solution:

$$\begin{aligned} x^m \cdot y^n &= (x+y)^{m+n} \\ \text{taking log both sides} \\ \Rightarrow m \ln x + n \ln y &= (m+n) \ln(x+y) \\ \text{Differentiating both sides, we get} \\ \therefore \frac{m}{x} + \frac{n}{y} \frac{dy}{dx} &= \frac{m+n}{x+y} \left(1 + \frac{dy}{dx} \right) \\ \Rightarrow \left(\frac{m}{x} - \frac{m+n}{x+y} \right) &= \left(\frac{m+n}{x+y} - \frac{n}{y} \right) \frac{dy}{dx} \\ \Rightarrow \frac{my - nx}{x(x+y)} &= \left(\frac{my - nx}{y(x+y)} \right) \frac{dy}{dx} \\ \Rightarrow \frac{dy}{dx} &= \frac{y}{x} \end{aligned}$$

Question217

If f is a real valued differentiable function satisfying $|f(x) - f(y)| \leq (x-y)^2$, $x, y \in \mathbb{R}$ and $f(0) = 0$, then $f'(1)$ equals
[2005]

Options:

- A. -1
- B. 0
- C. 2
- D. 1

Answer: B

Solution:

Given that $|f(x) - f(y)| \leq (x - y)^2$, $x, y \in \mathbb{R}$ (i) and $f(0) = 0$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$|f'(x)| = \lim_{h \rightarrow 0} \left| \frac{f(x+h) - f(x)}{h} \right| \leq \lim_{h \rightarrow 0} \left| \frac{(h)^2}{h} \right|$$

$$\Rightarrow |f'(x)| \leq 0 \Rightarrow f'(x) = 0$$

$$\Rightarrow f(x) = \text{constant}$$

$$\text{As } f(0) = 0$$

$$\Rightarrow f(1) = 0$$

Question218

Suppose $f(x)$ is differentiable at $x = 1$ and $\lim_{h \rightarrow 0} \frac{1}{h} f(1+h) = 5$, then $f'(1)$ equals
[2005]

Options:

A. 3

B. 4

C. 5

D. 6

Answer: C

Solution:

$$(c) f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h};$$

Given that function is differentiable so it is continuous also

$$\text{and } \lim_{h \rightarrow 0} \frac{f(1+h)}{h} = 5 \text{ and hence } f(1) = 0$$

$$\text{Hence, } f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h)}{h} = 5$$

Question219

Let f be differentiable for all x . If $f(1) = -2$ and $f'(x) \geq 2$ for $x \in [1, 6]$, then
[2005]

Options:

A. $f(6) \geq 8$

B. $f(6) < 8$

C. $f(6) < 5$

D. $f(6) = 5$

Answer: A

Solution:

As $f(1) = -2$ & $f'(x) \geq 2 \forall x \in [1, 6]$

Applying Lagrange's mean value theorem

$$\frac{f(6) - f(1)}{5} = f'(c) \geq 2$$

$$\Rightarrow f(6) \geq 10 + f(1)$$

$$\Rightarrow f(6) \geq 10 - 2 \Rightarrow f(6) \geq 8.$$

Question 220

If the equation $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x = 0$, $a_1 \neq 0$, $n \geq 2$, has a positive root $x = \alpha$, then the equation $na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \dots + a_1 = 0$ has a positive root, which is [2005]

Options:

- A. greater than α
- B. smaller than α
- C. greater than or equal to α
- D. equal to α

Answer: B

Solution:

Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x = 0$

The other given equation,

$na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \dots + a_1 = 0 = f'(x)$

Given $a_1 \neq 0 \Rightarrow f(0) = 0$

Again $f(x)$ has root α , $\Rightarrow f(\alpha) = 0$

$\therefore f(0) = f(\alpha)$

$\therefore$ By Rolle's theorem $f'(x) = 0$ has root between $(0, \alpha)$

Hence $f'(x)$ has a positive root smaller than α .

Question 221

Let $f(x) = \frac{1 - \tan x}{4x - \pi}$, $x \neq \frac{\pi}{4}$, $x \in \left[0, \frac{\pi}{2}\right]$.

If $f(x)$ is continuous in $\left[0, \frac{\pi}{2}\right]$, then $f\left(\frac{\pi}{4}\right)$ is [2004]

Options:

- A. -1
- B. $\frac{1}{2}$
- C. $-\frac{1}{2}$
- D. 1

Answer: C

Solution:

Given that $f(x) = \frac{1 - \tan x}{4x - \pi}$ is continuous in $\left[0, \frac{\pi}{2}\right]$

$\therefore f\left(\frac{\pi}{4}\right) = \lim_{x \rightarrow \frac{\pi}{4}^-} f(x) = \lim_{x \rightarrow \frac{\pi}{4}^+} f(x)$

$\lim_{x \rightarrow \frac{\pi}{4}^-} f(x) = \lim_{h \rightarrow 0} f\left(\frac{\pi}{4} + h\right)$

$$= \lim_{h \rightarrow 0} \frac{1 - \tan\left(\frac{\pi}{4} + h\right)}{4\left(\frac{\pi}{4} + h\right) - \pi}, h > 0 = \lim_{h \rightarrow 0} \frac{1 - \frac{1 + \tan h}{1 - \tan h}}{4h}$$

$$= \lim_{h \rightarrow 0} \frac{-2}{1 - \tan h} \cdot \frac{\tan h}{4h} = \frac{-2}{4} = -\frac{1}{2} \left[\because \lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1 \right]$$

Question222

If $x = e^{y + e^y + \dots + t^0 \infty}$, $x > 0$, then $\frac{dy}{dx}$ is
[2004]

Options:

A. $\frac{1+x}{x}$

B. $\frac{1}{x}$

C. $\frac{1-x}{x}$

D. $\frac{x}{1+x}$

Answer: C

Solution:

Given that $x = e^{y + e^y + \dots + t^0 \infty} \Rightarrow x = e^{y+x}$.

Taking log both sides.

$\log x = y + x$ differentiating both side $\Rightarrow \frac{1}{x} = \frac{dy}{dx} + 1$

$$\therefore \frac{dy}{dx} = \frac{1}{x} - 1 = \frac{1-x}{x}$$

Question223

If $2a + 3b + 6c = 0$, then at least one root of the equation $ax^2 + bx + c = 0$ lies in the interval
[2004]

Options:

A. (1, 3)

B. (1, 2)

C. (2, 3)

D. (0, 1)

Answer: D

Solution:

Let us define a function

$$f(x) = \frac{ax^3}{3} + \frac{bx^2}{2} + cx$$

Being polynomial, it is continuous and differentiable, also,

$$f(0) = 0 \text{ and } f(1) = \frac{a}{3} + \frac{b}{2} + c$$

$$\Rightarrow f(1) = \frac{2a + 3b + 6c}{6} = 0 \text{ (given)}$$

$$\therefore f(0) = f(1)$$

$\therefore f(x)$ satisfies all conditions of Rolle's theorem therefore $f'(x) = 0$ has a root in (0,1)

i.e. $ax^2 + bx + c = 0$ has at lease one root in (0,1)

Question 224

$$\text{If } f(x) = \begin{cases} xe^{-(\frac{1}{|x|} + \frac{1}{x})}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

then $f(x)$ is
[2003]

Options:

- A. discontinuous every where
- B. continuous as well as differentiable for all x
- C. continuous for all x but not differentiable at $x = 0$
- D. neither differentiable nor continuous at $x = 0$

Answer: C

Solution:

$$\text{Given that } f(0) = 0; f(x) = xe^{-(\frac{1}{|x|} + \frac{1}{x})}$$
$$R.H.L = \lim_{h \rightarrow 0} (0+h)e^{-2/h} = \lim_{h \rightarrow 0} \frac{h}{e^{2/h}} = 0$$

$$L.H.L = \lim_{h \rightarrow 0} (0-h)e^{-(\frac{1}{h} - \frac{1}{h})} = 0$$

therefore, $f(x)$ is continuous at $x = 0$.

$$\text{Now, } R.H.D = \lim_{h \rightarrow 0} \frac{(0+h)e^{-(\frac{1}{h} + \frac{1}{h})} - 0}{h} = 0$$

$$L.H.D. = \lim_{h \rightarrow 0} \frac{(0-h)e^{-(\frac{1}{h} - \frac{1}{h})} - 0}{-h} = 1$$

therefore, $L.H.D. \neq R.H.D.$

$f(x)$ is not differentiable at $x = 0$

Question 225

Let $f(x)$ be a polynomial function of second degree. If $f(1) = f(-1)$ and a, b, c are in A.P., then $f'(a), f'(b), f'(c)$ are in
[2003]

Options:

- A. Arithmetic -Geometric Progression
- B. A.P
- C. G..P
- D. H.P

Answer: B

Solution:

$$f(x) = ax^2 + bx + c$$

$$f(1) = f(-1)$$

$$\Rightarrow a + b + c = a - b + c \text{ or } b = 0$$

$$\therefore f(x) = ax^2 + c \text{ or } f'(x) = 2ax$$

Now $f'(a); f'(b)$ and $f'(c)$

are $2a(a)$; $2a(b)$; $2a(c)$

i.e. $2a^2$, $2ab$, $2ac$

$\Rightarrow$ If a, b, c are in A.P. then $f'(a)$; $f'(b)$ and $f'(c)$ are also in A.P.

Question 226

If $f(x) = x^n$, then the value of $f(1) - f'(1) + \frac{f''(1)}{2!} - \frac{f'''(1)}{3!} + \dots + \frac{(-1)^n f^{(n)}(1)}{n!}$ is
[2003]

Options:

A. 1

B. 2^n

C. $2^n - 1$

D. 0.

Answer: D

Solution:

Given that $f(x) = x^n \Rightarrow f(1) = 1$

$f'(x) = nx^{n-1} \Rightarrow f'(1) = n$

$f''(x) = n(n-1)x^{n-2} \Rightarrow f''(1) = n(n-1)$

.....

$f^{(n)}(x) = n! \Rightarrow f^{(n)}(1) = n!$

$f(1) - \frac{f'(1)}{1!} + \frac{f''(1)}{2!} - \frac{f'''(1)}{3!} + \dots + \frac{(-1)^n f^{(n)}(1)}{n!}$

$= 1 - \frac{n}{1!} + \frac{n(n-1)}{2!} - \frac{n(n-1)(n-2)}{3!} + \dots + (-1)^n \frac{n!}{n!}$

$= {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \dots + (-1)^n {}^nC_n$

$= (1-1)^n = 0$

Question 227

Let $f(a) = g(a) = k$ and their n th derivatives $f^{(n)}(a)$, $g^{(n)}(a)$ exist and are not equal for some n . Further if

$$\lim_{x \rightarrow a} \frac{f(a)g(x) - f(x)g(a) + f(x) - g(x)}{g(x) - f(x)} = 4$$

then the value of k is

[2003]

Options:

A. 0

B. 4

C. 2

D. 1

Answer: B

Solution:

$$\lim_{x \rightarrow a} \frac{f(a)g'(x) - g(a)f'(x)}{g'(x) - f'(x)} = 4 \text{ (By Applying L' Hospital rule)}$$

$$\lim_{x \rightarrow a} \frac{k g'(x) - k f'(x)}{g'(x) - f'(x)} = 4$$

$$\therefore k = 4$$

Question 228

f is defined in $[-5, 5]$ as
 $f(x) = x$ if x is rational
 $= -x$ if x is irrational. Then
[2002]

Options:

- A. $f(x)$ is continuous at every x , except $x = 0$
- B. $f(x)$ is discontinuous at every x , except $x = 0$
- C. $f(x)$ is continuous everywhere
- D. $f(x)$ is discontinuous everywhere

Answer: B

Solution:

Let a is a rational number other than 0, in $[-5, 5]$, then $f(a) = a$ and $\lim_{x \rightarrow a} f(x) = -a$

$\therefore x \rightarrow a^-$ and $x \rightarrow a^+$ is tends to irrational number

$\therefore f(x)$ is discontinuous at any rational number

If a is irrational number, then

$f(a) = -a$ and $\lim_{x \rightarrow a} f(x) = a$

$\therefore f(x)$ is not continuous at any irrational number. For $x = 0$, $\lim_{x \rightarrow 0} f(x) = f(0) = 0$

$\therefore f(x)$ is continuous at $x = 0$

Question 229

If $f(x + y) = f(x) \cdot f(y) \forall x, y$ and $f(5) = 2$, $f'(0) = 3$, then $f'(5)$ is
[2002]

Options:

- A. 0
- B. 1
- C. 6
- D. 2

Answer: C

Solution:

Given that $f(x + y) = f(x) \times f(y)$

Differentiate with respect to x , treating y as constant

$f'(x + y) = f'(x)f(y)$

Putting $x = 0$ and $y = x$, we get $f'(x) = f'(0)f(x)$;

$\Rightarrow f'(5) = 3f(5) = 3 \times 2 = 6$

Question 230

If $y = (x + \sqrt{1 + x^2})^n$, then $(1 + x^2)^{\frac{d^2 y}{dx^2}} + x \frac{dy}{dx}$ is
[2002]

Options:

- A. n^2y
- B. $-n^2y$
- C. $-y$
- D. $2x^2y$

Answer: A

Solution:

Given that $y = (x + \sqrt{1+x^2})^n$ (i)

Differentiating both sides w.r. to x

$$\frac{dy}{dx} = n(x + \sqrt{1+x^2})^{n-1} \left(1 + \frac{1}{2}(1+x^2)^{-1/2} \cdot 2x \right)$$

$$\begin{aligned} \frac{dy}{dx} &= n(x + \sqrt{1+x^2})^{n-1} \frac{(\sqrt{1+x^2} + x)}{\sqrt{1+x^2}} \\ &= \frac{n(\sqrt{1+x^2} + x)^n}{\sqrt{1+x^2}} \end{aligned}$$

$$\text{or } \sqrt{1+x^2} \frac{dy}{dx} = ny \text{ [from (i)]}$$

$$\Rightarrow \sqrt{1+x^2} y_1 = ny \left(\because y_1 = \frac{dy}{dx} \right) \text{ Squaring both sides,}$$

$$\text{we get } (1+x^2)y_1^2 = n^2y^2$$

Differentiating it w.r. to x,

$$(1+x^2)2y_1y_2 + y_1^2 \cdot 2x = n^2 \cdot 2yy_1$$

$$\Rightarrow (1+x^2)y_2 + xy_1 = n^2y$$

Question 231

If $2a + 3b + 6c = 0$, ($a, b, c \in \mathbb{R}$) then the quadratic equation $ax^2 + bx + c = 0$ has [2002]

Options:

- A. at least one root in $[0, 1]$
- B. at least one root in $[2, 3]$
- C. at least one root in $[4, 5]$
- D. None of these

Answer: A

Solution:

$$\text{Let } f(x) = \frac{ax^3}{3} + \frac{bx^2}{2} + cx$$

$$\Rightarrow f(0) = 0 \text{ and } f(1) = \frac{a}{3} + \frac{b}{2} + c = \frac{2a+3b+6c}{6} = 0$$

Also $f(x)$ is continuous and differentiable in $[0, 1]$ and $[0, 1]$ So by Rolle's theorem, $f'(x) = 0$.

i.e $ax^2 + bx + c = 0$ has at least one root in $[0, 1]$